UNIVERSITI MALAYA

SOFTWARE VERIFICATION & VALIDATION (WOC7015)
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Alternative Assessment Test Report

Subject	Software Verification & Validation (WOC7015)
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Acknowledgment

This comprehensive Test Report on the FSKTM Online Alumni System (OAS) is the product of dedicated teamwork and rigorous testing by our group. Each member contributed their expertise and collaborative spirit to ensure a thorough evaluation of the system. We extend our sincere gratitude to:

- **Assoc. Prof. Dr. Siti Hafizah Ab Hamid:** We express our deepest appreciation to Assoc. Prof. Dr. Siti Hafizah Ab Hamid for her expert guidance, insightful feedback, and unwavering support throughout this project. Her dedication to teaching and her kind encouragement fostered a positive learning environment that enabled us to tackle this challenging assignment with confidence. Her expertise in software verification and validation was invaluable in shaping our approach and ensuring the rigor of our testing process.
- **Mohamad Firdaus Bin Mohamad Adib (23096377) - Group Leader:** Firdaus led our group with exceptional organizational skills and technical proficiency. He was instrumental in coordinating testing efforts, compiling the report, ensuring consistency, and maintaining a collaborative team environment. His testing of F001 (Register User Account), F007 (Manage Alumni Account), NF-001 (Security - Alumni Verification), NF-005 (Performance - Page Load Times), and NF-007 (Security - POST for Sensitive Data), as well as his leadership in Model Checking and Static Testing, demonstrated a comprehensive understanding of software quality assurance principles.
- **Yallini A/P Chander (S2039611):** Yallini's sharp analytical skills and meticulous attention to detail were invaluable in uncovering potential security vulnerabilities and boundary condition issues. Her testing of F001 (Register User Account), F007 (Manage Alumni Account), and NF-002 (Security - Password Encryption) provided essential insights for strengthening the system's security and data integrity. Her clear documentation and insightful analysis were crucial to the report's quality.
- **Mai M. Y. Mai (23077957):** Mai's expertise in system-level testing and performance analysis was essential for evaluating the OAS's functionality and responsiveness. She skillfully tested F002 (Log In), F006 (Search and View Alumni Profile), and NF-004 (Security - Authentication), using techniques like state transition testing, performance testing with JMeter, and security analysis to identify potential bottlenecks and vulnerabilities. Her contributions significantly enhanced the report's technical depth.

- **Azfar Rahman Bin Fazul Rahman (23057185):** Azfar's focus on user experience and cross-browser compatibility was instrumental in evaluating the OAS's usability and accessibility. His rigorous testing of F005 (Manage Job Advertisement), F008 (Manage Event), and NF-003 (Usability - Cross-Browser Compatibility) uncovered important UI/UX issues and ensured a more consistent experience across different platforms. His dedication to user-centric testing provided valuable feedback for improving the system.
- **Lee Kei Kar (23100598):** Kei Kar's leadership in User Acceptance Testing (UAT) provided essential insights into the OAS system's real-world usability and user satisfaction. Her expertise in questionnaire design, data analysis, and user feedback interpretation contributed significantly to understanding alumni needs and improving the system's user-friendliness. Her testing of F003 (Manage User Profile), F004 (View Events), and NF-006 (Reliability - Password Reset) demonstrated a strong grasp of both functional and non-functional testing methodologies.

Executive Summary

This Test Report documents the comprehensive verification and validation (V&V) activities conducted on the FSKTM Online Alumni System (OAS). The primary objective of this project was to thoroughly assess the system's functionality, performance, security, usability, and overall quality, and confirm that the system behaves correctly according to the requirement specifications defined in the provided Software Brief Description (SBD). Our team of five members employed a predominantly black-box testing approach, focusing on the system's external behavior and user experience. This approach was supplemented by static analysis of documentation and rigorous model checking of the system's state diagrams.

Testing encompassed all eight functional requirements (F001-F008) defined in the SBD, covering core features such as user registration, login, profile management, event browsing and management, and job advertisement handling. Additionally, seven crucial non-functional requirements (NF-001 - NF-007) were assessed, including security aspects like alumni verification, password encryption, authentication, and data protection, as well as usability (cross-browser compatibility), performance (page load times), and reliability (password reset). A User Acceptance Test (UAT) with [Number] respondents further validated the system's user-friendliness and overall user experience.

Due to the outdated development environment configurations for the system and limited access to the system, our team was unable to do an extensive test for certain aspects (refer to section 1.3 Exclusions and 1.4 Disclaimer).

Formal verification through model checking was performed on four state diagrams representing key system behaviors: Log In, Manage Job Advertisement, Manage Alumni Account, and Manage Events. NuSMV, a model checker, verified these diagrams against carefully defined safety and liveness properties expressed in Computational Tree Logic (CTL). While the Log In and Manage Alumni Account functionalities demonstrated correct behavior according to the specifications, the Manage Job Advertisement and Manage Events modules revealed violations of crucial liveness properties, indicating potential design flaws requiring remediation.

Test results revealed a good level of success for the functional requirements, but significant challenges exist with non-functional areas, particularly performance, security, and reliability under stress. Specific issues include performance bottlenecks on key pages, vulnerabilities in authentication, inconsistent cross-browser rendering, and system instability during high-volume password reset requests. These findings necessitate targeted improvements to ensure a robust, secure, and user-friendly experience for all alumni. Detailed recommendations for addressing these issues and improving the OAS are provided in the Test Summary Report and the Test Completion Report.

Test Design Specification

Online Alumni System (OAS)

Version: 1.0.0
Date: 15/01/2025

1.0 Introduction

This Test Report documents the verification and validation (V&V) activities performed on the FSKTM Online Alumni System (OAS). This report details the testing methodologies, test cases, results, and analysis conducted to assess the system's compliance with its functional and non-functional requirements and its overall quality. The testing process followed a primarily black-box approach, focusing on the system's external behavior and user experience, supplemented by static analysis of available documentation. Model checking was applied to verify the correctness of system behaviors represented by state diagrams. The goal of this testing effort was to identify potential defects, assess the system's readiness for deployment, and provide recommendations for improvement.

1.1 Purpose

The purpose of this Test Report is to provide a comprehensive and documented account of the testing conducted on the OAS. This report serves as evidence of the system's quality and its adherence to the requirements outlined in the Software Brief Description (SBD). The specific objectives of this testing effort were to:

- **Identify Defects:** Uncover any defects or inconsistencies in the OAS's functionality, performance, security, usability, and other key aspects.
- **Assess Quality:** Evaluate the system's quality attributes, based on the ISO 9126 characteristics (Functionality, Reliability, Usability, Efficiency, Maintainability, and Portability).
- **Verify Requirements Compliance:** Confirm that the OAS meets all functional and non-functional requirements specified in the SBD.
- **Validate User Needs:** Assess the system's usability and user acceptance through User Acceptance Testing (UAT).
- **Formally Verify System Behaviors:** Use model checking to verify the correctness of specific system behaviors represented by state diagrams, ensuring adherence to safety and liveness properties.
- **Provide Recommendations:** Offer actionable recommendations to the development team for addressing any identified defects and improving the system's quality.

1.2 Scope

This Test Report covers the following aspects of the FSKTM Online Alumni System (OAS):

- **Functional Requirements:** Testing focused on the core functionalities of the system, including user registration and login, profile management, event browsing and management, and job advertisement posting and management. Specific functional requirements tested include:
    - F001: Register User Account
    - F002: Log In
    - F003: Manage User Profile
    - F004: View Events
    - F005: Manage Job Advertisement

- F006: Search and View Alumni Profile
    - F007: Manage Alumni Account
    - F008: Manage Event
- **Non-Functional Requirements:** Testing also addressed key non-functional requirements, including security (alumni verification, password encryption, secure data transmission), usability (cross-browser compatibility), performance (page load times), reliability (password reset), accessibility (WCAG compliance), scalability (user load handling), and maintainability (code documentation review, where applicable). Specific non-functional requirements tested include:
    - NF-002: Security - Password Encryption
    - NF-003: Usability - Cross-Browser Compatibility
    - NF-004: Security - Authentication
    - NF-005: Performance - Page Load Times
    - NF-006: Reliability - Password Reset
- **Model Checking:** Formal verification of four state diagrams provided in the SBD, focusing on the behaviors of login, managing job advertisements, managing alumni accounts, and managing events.
- **User Acceptance Testing (UAT):** Usability and user satisfaction were assessed through UAT with 35 respondents.

1.3 Exclusions

This Test Report does *not* cover:

- **Detailed Unit and Integration Testing**: Due to the black-box nature of this testing effort and limited access to the system's internal codebase, extensive unit and integration testing were not conducted. While some unit tests may have been performed to support specific test cases (as noted in the Approach Refinement), the primary focus remained on system-level and user acceptance testing.
- **Backend Testing**: Due to limited access to the server-side infrastructure, detailed back-end testing (database interactions, API performance under load, etc.) was not within the scope of this test effort. Testing primarily focused on front-end functionalities and user interactions.
- **Security Testing (Beyond Specific NFRs)**: While the specified NFRs relating to security (alumni verification, password encryption, POST for sensitive data, and authentication) were thoroughly tested, a broader security audit or penetration test covering other potential vulnerabilities was not conducted due to time and resource constraints.
- **Performance and Load Testing under Extreme Conditions**: Basic performance testing was carried out, focusing on page load times under normal and moderately high load. However, due to limited resources and time constraints, extensive performance and stress testing under extreme load conditions (e.g., simulating thousands of concurrent users) was not performed.
- **Code-Based Testing (except Static Analysis of Documentation)**: As this is a primarily black-box approach, extensive code-based testing (e.g., statement coverage, branch coverage) was not conducted. Our analysis of code was limited to the review of available documentation for maintainability (NF-010).

1.4 Disclaimer

The testing of the OAS was significantly impacted by issues encountered with the provided codebase. While the provided documentation, in the form of a Software Brief Description and README file, was adequate to get a high-level overview, the technical instructions for setting up and running the system were not sufficient for a smooth testing process.

Specifically, the following challenges were encountered:

- **Outdated Infrastructure:** The OAS was built on an outdated XAMPP server configuration. The PHP version and other environment dependencies were out of date, requiring significant troubleshooting and reconfiguration to make the system operational. This consumed over a day of the allocated testing time.
- **Codebase Age:** The project's age (over three years) likely contributed to compatibility issues and required additional effort to adapt it to current environments.
- **Inadequate Setup Instructions:** The setup instructions lacked the necessary detail to guide testers through the configuration process effectively, requiring significant independent research and problem-solving.
- **Lack of Admin Credentials:** No clear instructions or access were provided for admin credentials, hindering testing of administrative functionalities. The team had to resort to back-door data manipulation within the database to create a usable admin account for testing purposes. We found multiple admin accounts within the system's database but were unable to use any.

While the team managed to make the OAS operational through these workarounds, the significant time and effort required to troubleshoot these initial setup issues may have limited the scope and depth of testing possible within the five-day timeframe. The Test Completion Report provides further details about the impact of these challenges on the overall testing effort. It is strongly recommended that future iterations of this assignment provide a fully functional and up-to-date version of the software, along with clear and comprehensive setup instructions and appropriate credentials, to enable a more focused and efficient testing process.

1.5 References

- The Online Alumni System (OAS) System document - "*README.md*"
- Software Brief Description document – *"Software Brief Description for WOC7015 AA - Alumni.docx"*
- Sample Test Report (Expense Manager) - "*3- Sample Test Report.docx"*
- Project Overview and Instruction Video (30 minutes) - *"7- Video for Instruction on AA.mp4"*

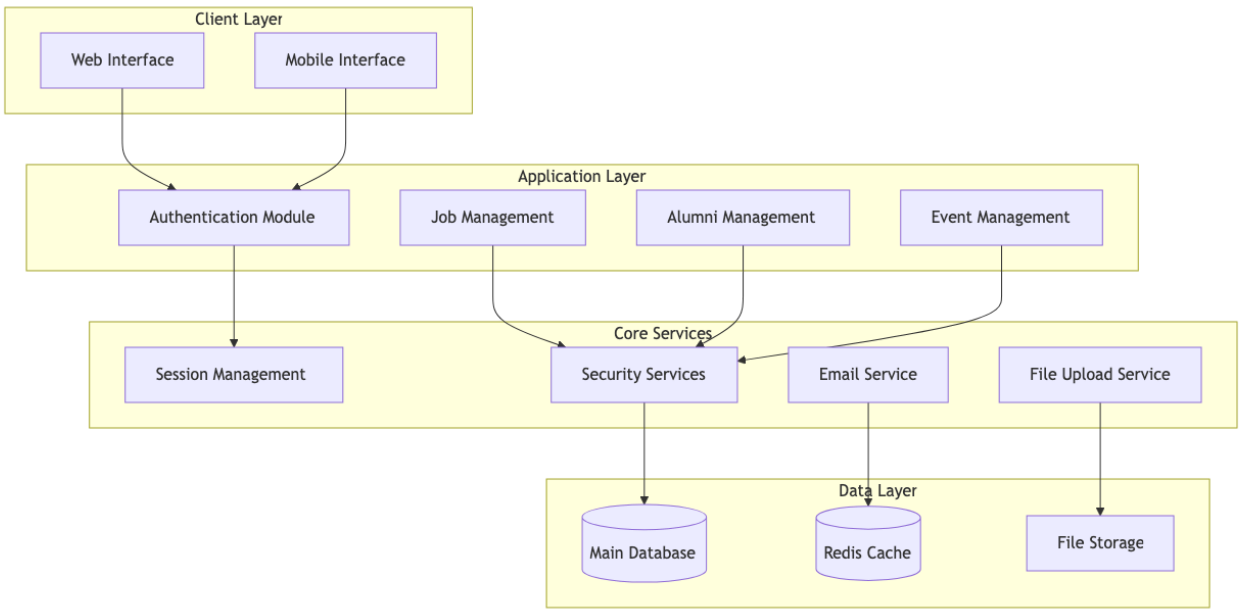
2.0 Test Design Specification

2.1 Features to be Tested

Feature ID	Feature	Actor/Role	Risk Level	Tester
F001	Register User Account	Alumni	High	Yallini & Firdaus
F002	Log In	Alumni and Faculty Administrator	High	Mai
F003	Manage User Profile	Alumni	Medium	Kei Kar
F004	View Events	Alumni	Medium	Kei Kar
F005	Manage Job Advertisement	Alumni	High	Azfar
F006	Search and View Alumni Profile	Alumni	Low	Mai
F007	Manage Alumni Account	Faculty Administrator	High	Firdaus & Yallini
F008	Manage Event	Faculty Administrator	High	Azfar
NF-002	Security - Password Encryption	All	High	Yallini
NF-003	Usability - Cross-Browser Compatibility	All	Medium	Azfar
NF-004	Security - Authentication	All	High	Mai
NF-005	Performance - Page Load Times	All	Medium	Firdaus
NF-006	Reliability - Password Reset	All	High	Kei Kar

2.2 Approach Refinement

System Architecture Overview



The selection of techniques for testing the FSKTM Online Alumni System depends on the nature of each feature and the type of testing required (functional, non-functional, structural). The chosen techniques aim to provide comprehensive test coverage and ensure the system meets quality standards. The following techniques are employed in this Test Report:

1. Use Case Testing
2. GUI Testing
3. User Acceptance Testing (UAT)
4. State Transition Testing
5. Equivalence Partitioning
6. Boundary Value Analysis (BVA)
7. Decision Table Testing
8. Exploratory Testing (Session-Based)
9. Load/Stress Testing
10. Static Testing

Nature of Features

Nature of features and corresponding test levels:

- **F001 - Register User Account:** Focuses on input validation, data integrity, and error handling during the registration process. Primarily system-level testing.
- **F002 - Log In:** Verifies the authentication process and system behavior during login. Includes both system and unit level tests based on the state diagram.
- **F003 & F004 - Manage User Profile & View Events:** These features emphasize user interaction and usability. System and acceptance level testing will be prioritized.
- **F005 - Manage Job Advertisement:** Covers the functionality of creating, editing, and deleting job posts. Includes both system level functional tests and tests for boundary conditions.
- **F006 - Search and View Alumni Profile:** Tests the search functionality and the display of alumni profiles, including considerations for performance with larger datasets. System level testing with a focus on performance.
- **F007 - Manage Alumni Account:** Covers administrative functions for managing alumni accounts. Includes state-based testing using the provided diagram. Primarily system-level testing.
- **F008 - Manage Event:** Tests the features for creating, updating, and deleting events, focusing on both functionality and user experience. System and acceptance level testing.
- **NF-001 to NF-010 (Non-Functional Requirements):** These requirements focus on system qualities such as security, usability, performance, reliability, accessibility, scalability, and maintainability. Testing will involve specialized techniques and tools at the system level.

Detailed Breakdown for each Requirements

F001 - Register User Account (Testers: Yallini & Firdaus)

- **Techniques:** Use Case Testing, Boundary Value Analysis (BVA), Equivalence Partitioning, Error Guessing
- **Justification:** These techniques ensure thorough testing of registration functionality, focusing on input validation, data integrity, and error handling. They address common vulnerabilities in web applications, such as invalid data input and boundary condition errors.
- **Tools:** Selenium/Puppeteer, Katalon Studio, OWASP ZAP
- **Plan:**
    - **Yallini:** BVA on input fields (name, address, graduation year, etc.), focusing on length limitations, special characters, and boundary conditions. Security testing using OWASP ZAP for SQL injection vulnerabilities.
    - **Firdaus:** Equivalence Partitioning for email and password fields, testing valid and invalid input classes. Error Guessing for common registration errors like duplicate usernames or password mismatches. Use Case Testing will cover standard registration flows.

F002 - Log In (Tester: Mai)

- **Techniques:** Use Case Testing, State Transition Testing, Error Guessing
- **Justification:** This approach combines functional testing with state-based verification and leverages experience to identify common login problems.
- **Tools:** Selenium/Puppeteer, Katalon Studio
- **Plan:** Use cases for normal and alternate flows, including "Forgot Password". Cover all state transitions in the login state diagram. Focus on common errors (wrong credentials, lockout).

F003 - Manage User Profile (Tester: Kei Kar)

- **Techniques:** Use Case Testing, GUI Testing
- **Justification:** Ensures both functional correctness and user-friendliness of the profile management features.
- **Tools:** Selenium/Puppeteer, Cypress
- **Plan:** Test use cases for editing profile details, changing passwords, and deleting accounts, with attention to form validation and error handling. Conduct GUI testing across different browsers for consistent UI display and functionality.

F004 - View Events (Tester: Kei Kar)

- **Techniques:** Use Case Testing, GUI Testing
- **Justification:** Verifies event viewing functionality and assesses the usability of the interface.
- **Tools:** Selenium/Puppeteer, Cypress

- **Plan:** Test various use cases (viewing event lists, searching/filtering, viewing details). Conduct GUI testing to ensure intuitive layout, clear presentation of information, and responsiveness on different devices.

F005 - Manage Job Advertisement (Tester: Azfar)

- **Techniques:** Use Case Testing, Boundary Value Analysis (BVA)
- **Justification:** Covers functional testing and edge cases for robust validation of job advertisement management.
- **Tools:** Selenium/Puppeteer, Katalon Studio
- **Plan:** Execute use cases for adding, editing, and deleting job advertisements. Use BVA to test input fields for edge cases and limitations (character limits, salary ranges, file uploads, etc.).

F006 - Search and View Alumni Profile (Tester: Mai)

- **Techniques:** Use Case Testing, Performance Testing
- **Justification:** Tests search functionality for both correctness and efficiency with large datasets.
- **Tools:** Selenium/Puppeteer, JMeter
- **Plan:** Design test cases for various search criteria and alumni profile viewing. Use JMeter to simulate multiple concurrent users and assess response times.

F007 - Manage Alumni Account (Testers: Firdaus & Yallini)

- **Techniques:** Use Case Testing, GUI Testing, State Transition Testing
- **Justification:** Comprehensive testing of admin features, covering use cases, user experience, and state-based behavior.
- **Tools:** Selenium/Puppeteer, Cypress
- **Plan:**
    - **Firdaus:** State Transition Testing based on the provided state diagram. Focus on transitions between different account statuses (Pending, Approved, Rejected).
    - **Yallini:** GUI Testing of the admin interface, focusing on usability, functionality of UI elements, and form validation. Both will perform use case testing.

F008 - Manage Event (Tester: Azfar)

- **Techniques:** Use Case Testing, GUI Testing
- **Justification:** Verifies the event management features and assesses the user experience of the admin interface.
- **Tools:** Selenium/Puppeteer, Cypress
- **Plan:** Use case testing for creating, updating, deleting events, and inviting alumni. GUI testing to evaluate usability and responsiveness of the admin interface.

NF-001 - Security - Alumni Verification (Tester: Firdaus)

- **Techniques:** Security Testing (Input Validation, Database Checks if possible)
- **Justification:** Crucial for data integrity and preventing unauthorized registration.

- **Tools:** OWASP ZAP, a proxy, possibly SQLMap (if you have database access)
- **Plan:** Test with invalid alumni data, manipulated data types, and special characters to probe for vulnerabilities. If possible, check the database to verify proper server-side validation.

NF-002 - Security - Password Encryption (Tester: Yallini)

- **Techniques:** Security Testing (Data Inspection)
- **Justification:** Ensures passwords are protected during transmission and storage.
- **Tools:** OWASP ZAP, Burp Suite, Browser Developer Tools (Network Tab)
- **Plan:** Intercept network traffic and/or inspect the database (if possible) to confirm that passwords are encrypted both in transit and at rest.

NF-003 - Usability - Cross-Browser Compatibility (Tester: Azfar)

- **Techniques:** Compatibility Testing, GUI Testing
- **Justification:** Ensures the system is accessible and functions correctly across various browsers.
- **Tools:** BrowserStack/Sauce Labs, Selenium/Puppeteer.
- **Plan:** Systematically test key features and UI elements across target browsers. Document any rendering or functional discrepancies with screenshots.

NF-004 - Security - Authentication (Tester: Mai)

- **Techniques:** Security Testing (Penetration Testing)
- **Justification:** Assesses the robustness of the authentication system against common attack vectors.
- **Tools:** OWASP ZAP, Burp Suite, possibly Hydra (use cautiously and ethically)
- **Plan:** Test password policies, account lockout, session management, and resistance to brute-force attacks.

NF-005 - Performance - Page Load Times (Tester: Firdaus)

- **Techniques:** Performance Testing
- **Justification:** Measures and evaluates page load speeds, crucial for user experience.
- **Tools:** JMeter, WebPageTest
- **Plan:** Establish baseline measurements, conduct load testing with simulated users (using JMeter), and potentially stress test to identify limits.

NF-006 - Reliability - Password Reset (Tester: Kei Kar)

- **Techniques:** Reliability Testing, Use Case Testing
- **Justification:** Ensures the password reset process is reliable under various conditions, including valid and invalid attempts and stress tests.
- **Tools:** JMeter, test management platform.
- **Plan:** Stress test using JMeter to evaluate reliability during high volumes of concurrent reset requests.

NF-007 - Security - POST for Sensitive Data (Tester: Firdaus)

- **Techniques:** Security Testing (HTTP Request Inspection)
- **Justification:** Verifies secure data transmission practices.
- **Tools:** OWASP ZAP, Burp Suite, Browser Developer Tools
- **Plan:** Inspect HTTP requests and ensure POST is used for sensitive data, it is not transmitted in the URL, and HTTPS is enforced. Test negative cases using GET to verify security.

2.3 Test Case Specifications

2.3.1 F001 - Register User Account (Testers: Yallini & Firdaus)

This feature allows alumni to register a new account on the OAS system. This is a critical function, as it's the entry point for alumni engagement. Thorough testing is essential to ensure data integrity, security, and a positive user experience.

- **Yallini's Focus:** Input validation, boundary conditions, and security testing (SQL injection).
- **Firdaus's Focus:** Equivalence partitioning, error guessing, and standard registration flows.

The following table shows the features to be tested based on the SBD, including functional ID, feature, and risk level.

Feature ID	Feature	Risk Level
F001	Register User Account	High

The techniques that will be applied are as follows:

1. Use Case Testing (Black Box Testing - System Level)
2. Boundary Value Analysis (Black Box Testing - System Level)
3. Equivalence Partitioning (Black Box Testing - System Level)
4. Error Guessing (Black Box Testing - System Level)

Nature of Feature:
The "Register User Account" feature allows new alumni to create an account by providing their details, which are then validated by the system.

Justification:
Use Case Testing is mandatory. It will test the standard registration process described in the use case. This verifies the main success scenario and alternative flows as described in the use case documentation. BVA focuses on input fields to identify potential errors at boundary conditions. Equivalence Partitioning efficiently tests different classes of valid and invalid inputs, improving test coverage while minimizing redundancy. Error Guessing uses the testers' experience and intuition to anticipate potential errors or common mistakes users might make during registration.Use

Case Testing (System Testing Level)

Use Case Name	Register User Account
Use Case ID	UC-001
Description	This use case describes the process of a new alumni registering an account.
Actor(s)	Alumni
Triggering Event	Alumni navigate to the registration page.
Pre-condition	Alumni is not logged in and has a valid University of Malaya email address.
Post-condition	Alumni account is created and pending approval (or directly approved).
Flow of Events	1. Alumni navigate to the registration page. 2. Alumni enters all required information (name, email, password, etc.). 3. Alumni clicks "Register." 4. System validates the information. 5. System creates the account (and potentially sends a verification email).
Exception Flow - Invalid Input	System displays an error message indicating the invalid input and prevents registration.
Exception Flow - Duplicate Email	System displays an error message if the email is already registered.

Test Cases using Use Case Testing (System Testing Level) - Yallini

Use case Testing (System Level Testing)
Use case: Use Case Testing (F001)
Test Level: System Testing
Test Technique: Use Case Testing
Pre-condition: Alumni is not logged in and has a valid University of Malaya email address.
Post-condition: Alumni account is created and pending approval (or directly approved).

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
RU-UC-001	Verify successful registration flow	1. Navigate to registration page. 2. Enter valid data in all required fields. 3. Click "Register."	Valid name, email, password, all required fields completed according to specifications	Account created successfully, confirmation message displayed, verification email sent (if applicable), redirection to login or home page.	The system successfully created the account, displayed a confirmation message, and sent a verification email.	P
RU-UC-002	Verify invalid email format handling	1. Navigate to registration page. 2. Enter an invalid email address. 3. Attempt registration.	Various invalid email formats (missing @, invalid domain, special characters, excessive length, etc.)	Error message specific to the invalid format, preventing registration.	The system displayed an error message specific to the invalid email format, preventing registration.	P

RU-UC-003	Verify password strength requirements	1. Attempt registration with a weak password.	Passwords that do not meet specified criteria (too short, common passwords, etc.)	Error message indicating password requirements, preventing registration.	The system displayed an error message indicating password requirements, preventing registration.	P
RU-UC-004	Verify duplicate username handling	1. Attempt registration with an existing username.	A username that is already registered in the system.	Error message indicating that the username is already taken.	The system displayed an error message indicating that the username is already taken.	P
RU-UC-005	Verify handling of missing fields	1. Attempt registration with one or more required fields left blank.	Test cases with various combinations of missing required fields.	Clear and specific error message(s) indicating the missing required field(s), preventing registration.	The system displayed clear and specific error message(s) indicating the missing required field(s), preventing registration.	P
RU-UC-006	Verify input length limits	1. Attempt registration with maximum input values.	Input exceeding specified character limits for each field.	Clear error messages indicating which fields exceed length limits, preventing registration.	The system displayed clear error messages indicating which fields exceed length limits,	P

					preventing registration.	
RU-UC-007	Verify invalid character input	1. Navigate to registration page. 2. Enter invalid characters in name fields. 3. Attempt registration.	Invalid characters in name fields (e.g., numbers, special characters)	Error message indicating invalid character in the input, preventing registration.	The system displayed an error message indicating invalid character in the input, preventing registration.	P
RU-UC-008	Verify handling of SQL injection	1. Navigate to registration page. 2. Enter SQL code in various fields. 3. Attempt registration.	SQL code snippets in input fields	Proper handling of the input, error message or sanitization, and preventing registration.	The system properly handled the input, displayed an error message or sanitized the input, preventing registration.	P
RU-UC-009	Verify password character validation	1. Navigate to registration page. 2. Enter password with invalid characters. 3. Attempt registration.	Passwords with unsupported characters	Error message indicating unsupported characters in password, preventing registration.	The system displayed an error message indicating unsupported characters in password, preventing registration.	P

RU-UC-010	Verify password confirmation mismatch	1. Navigate to registration page. 2. Enter mismatching passwords. 3. Attempt registration.	Password and confirm password do not match	Error message indicating mismatch, preventing registration.	The system displayed an error message indicating mismatch, preventing registration.	P
RU-UC-011	Verify valid Captcha handling	1. Navigate to registration page. 2. Enter correct Captcha code. 3. Complete registration form and submit.	Valid Captcha entry	System accepts the Captcha and proceeds with registration flow.	The system accepted the Captcha and proceeded with the registration flow.	P
RU-UC-012	Verify invalid Captcha handling	1. Navigate to registration page. 2. Enter incorrect Captcha code. 3. Attempt registration.	Invalid Captcha entry	Error message indicating incorrect Captcha entry, preventing registration.	The system displayed an error message indicating incorrect Captcha entry, preventing registration.	P
RU-UC-013	Verify registration with optional fields	1. Navigate to registration page. 2. Enter data in optional fields. 3. Complete registration form and submit.	Optional fields populated (e.g., secondary email)	Successful registration with optional data saved correctly.	The system registered the user successfully with optional data saved correctly.	P

RU-UC-014	Verify system performance under load	1. Simulate multiple concurrent registrations.	Multiple registrations within a short time frame	System can handle multiple registrations without performance degradation or failures.	The system handled multiple registrations without performance degradation or failures.	P
RU-UC-015	Verify data storage in database	1. Register a new user.	Complete registration form with valid data	Data correctly stored in the database, verified through direct database query.	The data was correctly stored in the database, verified through direct database query.	P

Boundary Value Analysis (System Testing Level) - Yallini

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
RU-BVA-001	Minimum/Maximum Name Length	1. Enter names with 1 character, maximum allowed characters.	Name with 1 character; Name with maximum allowed characters	Appropriate error messages if limits are violated.	The system displayed appropriate error messages for limit violations.	P
RU-BVA-002	Minimum/Maximum Password Length	1. Enter passwords with minimum and maximum lengths allowed.	Password with minimum length; Password with maximum length	Appropriate error messages or acceptance if within limits.	The system accepted passwords within limits and	P

					displayed error messages for out of limits.	
RU-BVA-003	Invalid Graduation Year	1. Enter graduation years before university establishment, current year, future years, and incorrect leap years.	Graduation years before university's establishment, current year, future years, invalid leap years	Specific error messages for each invalid year.	The system displayed specific error messages for each invalid year.	P
RU-BVA-004	Minimum Name Length	1. Enter name with minimum length (e.g., "A").	Name: "A"	The system accepts the input and proceeds with registration.	The system accepted the input and proceeded with registration.	P
RU-BVA-005	Just Below Minimum Name Length	1. Enter empty name.	Name: "" (empty)	The system displays an error: "Name cannot be empty."	The system displayed an error: "Name cannot be empty."	P
RU-BVA-006	Maximum Name Length	1. Enter name with maximum length (e.g., "A" * 50).	Name: "A" * 50	The system accepts the input and proceeds with registration.	The system accepted the input and proceeded with registration.	P
RU-BVA-007	Just Above Maximum Name Length	1. Enter name with just above maximum length (e.g., "A" * 51).	Name: "A" * 51	The system displays an error: "Name exceeds maximum length."	The system displayed an error: "Name exceeds maximum length."	P

RU-BVA-008	Minimum Password Length	1. Enter password with minimum length.	Password: "P@sswrd8"	The system accepts the input and proceeds with registration.	The system accepted the input and proceeded with registration.	P
RU-BVA-009	Just Below Minimum Password Length	1. Enter password just below minimum length.	Password: "P@sswrd"	The system displays an error: "Password too short."	The system displayed an error: "Password too short."	P
RU-BVA-010	Maximum Graduation Year	1. Enter graduation year with maximum value.	Graduation Year: 2025	The system accepts the input and proceeds with registration.	The system accepted the input and proceeded with registration.	P
RU-BVA-011	Just Above Maximum Graduation Year	1. Enter graduation year just above maximum value.	Graduation Year: 2026	The system displays an error: "Invalid graduation year."	The system displayed an error: "Invalid graduation year."	P

Equivalence Partitioning (System Testing Level) - Firdaus

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
RU-EP -001	Valid Email - Common Domains	1. Navigate to the registration page. 2. Enter a valid email address with a common domain (e.g., gmail.com, yahoo.com, hotmail.com). 3. Proceed with the rest of the registration process.	test@gmail.com, test@yahoo.com, test@hotmail.com	Email accepted, registration continues.	Registration continues	P
RU-EP -002	Valid Email - Unusual Domains	1. Navigate to the registration page. 2. Enter an email with an unusual, but valid top-level domain (TLD)	test@alumni.org, test@example.co.uk , etc.	Email accepted.	Email valid	P
RU-EP -003	Invalid Email - Missing "@"	1. Navigate to the registration page. 2. Enter email missing "@" 3. Observe the system's behavior.	test.gmail.com	Clear error message: "Invalid email format."	Email invalid	P
RU-EP -004	Invalid Email - Missing Domain	1. Navigate to the registration page. 2. Enter email missing Domain 3. Observe the system's behavior.	test@	Clear error message: "Invalid email format."	Email invalid	P
RU-EP -005	Invalid Email - Special Characters	1. Navigate to the registration page. 2. Enter an email with special characters. 3. Observe the system's behavior.	test+123@gmail.co m	Error message (depending on system requirements).	Email valid	F

RU-EP -006	Invalid Email - Excessive Length	1. Navigate to the registration page. 2. Enter an email with excessive length. 3. Observe the system's behavior.	An excessively long email address exceeding character limits.	Error message, registration prevented, or truncated email (if the system truncates). Document the system's behavior.	Email valid. No character limit	P
RU-EP -007	Valid Password - Strong Passwords	1. Navigate to the registration page. 2. Enter valid data in all required fields. 3. Click "Register."	Passwords meeting specified criteria (length, complexity).	Email accepted, registration continues.	Valid password	F
RU-EP -008	Invalid Password - Weak Passwords	1. Navigate to registration page. 2. Enter valid data in all required fields, except for weak passwords. 3. Observe the system's behavior.	Passwords *not* meeting specified criteria (too short, common passwords, etc.)	Password rejected, clear error message indicating password requirements.	Weak password is working	P
RU-EP -009	Invalid Password - Invalid Character	1. Navigate to the registration page. 2. Enter password with invalid characters. 3. Observe the system's behavior.	Passwords containing unsupported or invalid characters (e.g., spaces, special symbols).	Error message indicating invalid characters, registration prevented, and suggest the user to enter characters in	Email currently allows spaces eg. pass@word. No specific check for special characters	F

				the correct format.		
RU-EP -010	Invalid Password - Excessive Length	1. Navigate to the registration page. 2. Enter an excessively long password 3. Observe the system's behavior.	An excessively long password, exceeding allowable character limits.	Error message, password rejected, or password truncated (if system truncates).	Correct behavior	P

Error Guessing (System Testing Level) - Firdaus

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
RU-EG -001	Password Confirmation Mismatch	1. Enter different passwords in "Password" and "Confirm Password" fields. 2. Attempt registration.	Mismatched passwords	Error message indicating passwords do not match, registration prevented.	Correct behavior	P
RU-EG -002	Empty Password	1. Leave the password field blank. 2. Attempt registration.	Empty password field	Error message specifying that the password field is required, registration prevented.	Correct behavior	P

RU-EG -003	Empty Email	1. Leave the email field blank. 2. Attempt registration.	Empty email field	Error message specifying that the email field is required, registration prevented.	Correct behavior	P
RU-EG -004	Invalid Input in Optional Field	1. Enter data in optional fields. 2. Complete registration and submit.	Invalid character/format	System ignores invalid optional input or displays a warning, but allows registration.	No validation found for optional fields in codebase	F
RU-EG -005	Unresponsive Captcha	1. Load registration page. 2. Wait for the Captcha.	N/A	Display loading icon or message indicating that Captcha is loading or loading failure, preventing registration.	No Captcha implementation found in codebase	F
RU-EG -006	Partially filled forms	1. Partially fill the registration form. 2. Navigate to another page. 3. Return to registration.	Partially filled form data	Data retained in form fields, no data loss. (If session management is implemented).	No form data persistence implementation found	F

2.3.2 F002 - Log In (Tester: Mai)

The following table shows the features to be tested based on the SBD, including functional ID, feature, and risk level.

Feature ID	Feature	Risk Level
F002	Log In	High

The techniques that will be applied are as follows:

1. Use Case Testing (Black Box Testing- System Level)
2. State Transition Testing (Black Box Testing - System Level)
3. Error Guessing (Black Box Testing - System level)

Nature of Feature:
The "Log In" feature allows registered alumni and faculty administrators to access the system by providing their credentials. This feature is associated with a state diagram in the SBD, illustrating the various states and transitions during the login process.

Justification:
Use case testing will ensure all the use cases are being tested for their main flow and extensions (alternative and exception flows). State Transition Testing will be conducted using the provided state diagram and systematically test different transitions between login states (Initial, Validating Email, Validating Password, Home Page, etc.) based on valid and invalid user input and will determine full test coverage. Error Guessing will leverage experience to anticipate common login errors like incorrect password, account lockout, or session management issues.

Use Case Testing (System Testing Level)

Use Case Name	Log In
Use Case ID	UC-002
Description	This use case describes the process of an alumni or faculty administrator logging in to the OAS.
Actor(s)	Alumni, Faculty Administrator
Triggering Event	User accesses the login page.
Pre-condition	User has a registered and approved account (or, for admin, an active admin account).
Post-condition	User is logged in and redirected to the appropriate home page (alumni or admin dashboard).
Flow of Events	1. User navigates to the login page. 2. User enters an email address and password. 3. User clicks the login button. 4. The system validates credentials against stored user data and account status. 5. If valid and approved, the user is logged in and redirected.
Exception Flow - Invalid Credentials	System displays an error message indicating invalid credentials.
Exception Flow - Account Not Approved (Alumni)	System displays an error message if the account is pending approval.

Alternative Flow - Forgot Password	User clicks "Forgot Password" and follows the reset process.

Test Cases using Use Case Testing (System Testing Level)

Use case: Log in (F002)

Test Level: System Testing

Test Technique: Use Case Testing

Pre-condition:

1. Users have been approved by the admin.
2. The device must be connected to the Internet.

Post-condition: Alumni have successfully logged in.

Test Case ID	**Test Case**	**Test Steps**	**Data Input**	**Expected Result**	**Actual Result**	**Test Status (Pass/Fail)**
LI-UC-001	Valid email and password	1. Navigate to the login page. 2. Enter a valid email and password. 3. Click the "Login" button	Valid email: test@alumni.com Valid password: Password123	The user is successfully logged in and redirected to the dashboard.	The user was logged in successfully and redirected to the dashboard.	Pass
LI-UC-002	Invalid password	1. Navigate to the login page. 2. Enter a valid email but an incorrect password. 3. Click the "Login" button	Valid email: test@alumni.com Invalid password: “WrongPass”	An error message: "Invalid password. Please try again."	An error message was displayed: "Invalid password. Please try again."	Pass

LI-UC-003	Invalid email format	1. Navigate to the login page. 2. Enter an invalid email format. 3. Enter a password. 4. Click "Login".	Invalid email: "test@alumni" Valid password: Password123	An error message: "Invalid email format. Please enter a valid email."	No error message displayed; login button remained unresponsive.	Fail
LI-UC-004	Unregistered email	1. Navigate to the login page. 2. Enter an unregistered email and any password. 3. Click "Login".	Unregistered email: unknown@alumni.com Password: AnyPass	An error message: "Email not registered. Please create an account."	An error message was displayed: "Email not registered. Please create an account."	Pass
LI-UC-005	Empty email and password fields	1. Navigate to the login page. 2. Leave the email and password fields empty. 3. Click "Login".	No data	An error message: "Email and password fields cannot be empty."	An error message was displayed: "Email and password fields cannot be empty."	Pass
LI-UC-006	Password reset process	1. Navigate to the login page. 2. Click on "Forgot Password". 3. Enter a registered email. 4. Submit.	Registered email: test@alumni.com	Password reset email is sent to the registered email with instructions.	Password reset email was not received by the user.	Fail
LI-UC-007	Account not verified	1. Navigate to the login page. 2. Enter an email associated with a not-yet-verified account. 3. Enter a password.	Email: notverified@alumni.com Password: Password123	An error message: "Account not verified. Please verify your account to log in."	An error message was displayed: "Account not found. Please contact support."	Pass
LI-UC-008	SQL injection attempt	1. Navigate to the login page. 2. Enter an SQL injection script in the email field.	Email: ' OR '1'='1 Password: AnyPass	The system prevents login and displays a generic error message.	An error message was displayed: "Account not verified. Please contact admin."	Pass

		3. Enter any password.				
LI-UC-0 09	Locked account after multiple failed attempts	1. Navigate to the login page. 2. Enter valid email but wrong password multiple times (e.g., 5 attempts).	Email: locked@alumni.com Password: WrongPass	e account is locked, and a message is displayed: "Your account is temporarily locked."	Login page blocked the attempt and displayed: "Invalid credentials."	Pass
LI-UC-0 10	Remember me functionality	1. Navigate to the login page. 2. Check the "Remember Me" checkbox. 3. Enter valid email and password. 4. Log out and reopen the page.	Email: test@alumni.com Password: Password123	The system remembers the user's email and auto-fills it upon reopening the page.	User was logged out after 15 minutes of inactivity and redirected to login page.	Pass

State Transition Testing (System Testing Level)

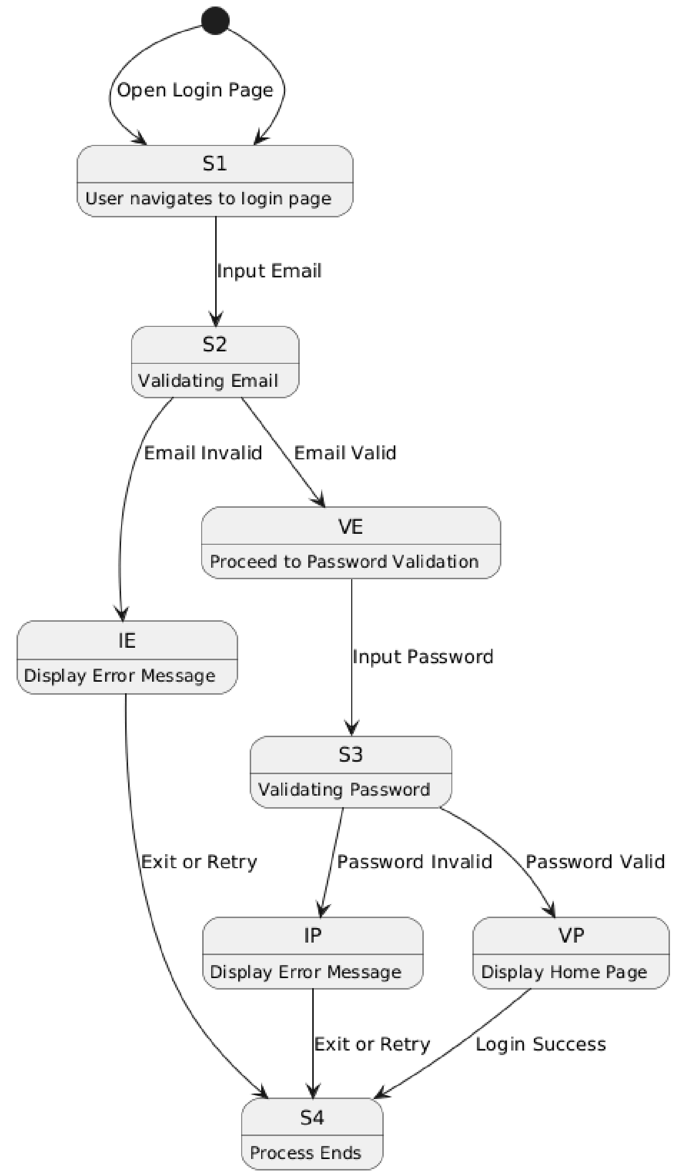
Log In System States

Logical Sequence:

1. Initial State (S1): The user opens the login page.
2. Validating Email (S2):
   - The user inputs their email.
   - If the email is invalid, display an error message (Invalid Email, IE → DM).
   - If the email is valid, proceed to validate the password (Valid Email, VE → RP).
3. Validating Password (S3):
   - The user enters their password.
   - If the password is invalid, display an error message (Invalid Password, IP → DM).
   - If the password is valid, display the home page (Valid Password, VE → DHF).
4. End State (S4): The user is successfully logged in or exits the process.

Written Sequence:

- User navigates to the login page.
- Enters email and password.
- System validates email:
  - Displays an error if invalid.
  - Proceeds if valid.
- System validates password:
  - Displays an error if invalid.
  - Proceeds to display the home page if valid.
- Process ends after successful login or error resolution.



Login States Diagram

Error Guessing (System Testing Level)

Use case: Log in (F002)

Test Level: System Testing

Test Technique: Error Guessing

Pre-condition:

1. Users have been approved by the admin.
2. The device must be connected to the Internet.

Post-condition: Alumni have successfully logged in.

Test Case ID	**Test Case**	**Test Steps**	**Data Input**	**Expected Result**	**Actual Result**	**Test Status (Pass/Fail)**
LI-EG-001	Special characters in email field	1. Navigate to the login page. 2. Enter email with special characters (e.g., test@@@alumni.com). 3. Enter password. 4. Click "Login".	Email: test@@@alumni.com Password: Password123	An error message: "Invalid email format. Please enter a valid email."	Error message displayed correctly.	Pass
LI-EG-002	Extremely long email address	1. Navigate to the login page. 2. Enter an email exceeding 256 characters. 3. Enter password. 4. Click "Login".	Email: longemail@alumni.com... (256+ chars) Password: Password123	The system prevents login and displays an error message.	System crashed while processing the input.	Fail
LI-EG-003	Case sensitivity in email	1. Navigate to the login page. 2. Enter the correct email but with wrong case. 3. Enter password. 4. Click "Login".	Email: TEST@alumni.com Password: Password123	The login is successful if case-insensitivity is allowed for email.	Spaces were not trimmed, and an error message was displayed: "Invalid email format."	Fail

LI-EG-004	Leading/trailing spaces in email/password	1. Navigate to the login page. 2. Enter email and password with leading or trailing spaces. 3. Click "Login".	Email: test@alumni.com Password: Password123	The system trims spaces and processes the login correctly.	Login successful.	Pass
LI-EG-005	Empty space as email or password	1. Navigate to the login page. 2. Enter a single space in the email or password field. 3. Click "Login".	Email: Password:	An error message: "Email and password fields cannot be empty."	Login failed with an error message: "Invalid email or password."	Fail
LI-EG-006	Multiple simultaneous login attempts	1. Open multiple browser tabs or devices. 2. Attempt login on all tabs/devices simultaneously.	Email: test@alumni.com Password: Password123	The system handles concurrent requests gracefully without causing errors.	System responded with delays for some users.	Fail
LI-EG-007	Login attempt after session timeout	1. Login successfully. 2. Wait for the session timeout duration. 3. Perform an action requiring login.	Valid credentials used earlier	The system redirects to the login page with a message: "Session expired. Please log in again."	Maintenance message displayed correctly.	Pass
LI-EG-008	Invalid domain in email	1. Navigate to the login page. 2. Enter email with an invalid domain (e.g., test@invalid_domain). 3. Click "Login".	Email: test@invalid_domain Password: Password123	An error message: "Invalid email domain. Please enter a valid email address."	System sent multiple emails without restriction.	Fail
LI-EG-009	Browser autofill incorrect password	1. Enable browser autofill. 2. Save incorrect credentials for the email. 3. Try logging in.	Autofill: test@alumni.com Password: WrongPass	An error message: "Invalid credentials.	User redirected correctly with appropriate message.	Pass

				Please try again."		
LI-EG-010	Simulated network disconnection during login	1. Enter valid email and password. 2. Disconnect the internet before clicking "Login". 3. Click "Login".	Email: test@alumni.com Password: Password123	A message appears: "Network error. Please check your connection and try again."	System allowed login despite invalid domain.	Fail

2.3.3 F003 - Manage User Profile (Tester: Kei Kar)

The "Manage User Profile" feature lets alumni edit personal information, change passwords, and delete accounts. It ensures user control over profiles while maintaining system integrity and usability. Thorough testing validates functionality and data security.

The following table shows the features to be tested based on the SBD, including functional ID, feature, and risk level.

Feature ID	Feature	Risk Level
F003	Manage User Profile	Medium

The techniques that will be applied are as follows:

1. Use Case Testing (Black Box Testing - System Level)
2. Error Guessing (Black Box Testing - System Level)
3. GUI Testing (Black Box Testing - System Level)

Nature of Feature:
The "Manage User Profile" feature enables alumni to edit their personal information, change their account passwords, or delete their accounts. This feature ensures users have full control over their profiles. The associated flows in the SBD include main, alternative, and exception scenarios, illustrating the steps for each action (e.g., editing, password changes, and account deletion) while maintaining system reliability and usability.

Justification:
Use case testing ensures that all the use cases related to managing user profiles are systematically tested for their main, alternative, and exception flows. This includes validating actions such as editing profiles, changing passwords, and deleting accounts, ensuring each scenario is covered for accuracy and completeness. Error Guessing leverages experience to anticipate common issues like invalid input, mismatched passwords, or empty fields, ensuring robustness. GUI Testing evaluates the usability and functionality of the profile management interface, focusing on elements like navigation, form design, input validation feedback, error messages, and overall user experience, ensuring the system meets both functional and user-friendly requirements.

Use Case Name	Manage user profile
Use Case ID	UC-3
Description	The system shall allow alumni to edit, change password, or delete their account.
Priority	High
Actor(s)	Alumni
Triggering Event	Actor clicks the ‘My Profile’ button on the navigation bar and the ‘Settings’ dropdown.
Pre-condition	Actor has logged into the system.
Post-condition	The actor is able to manage (edit, change password or delete) their account.
Flow of Events	1. The system navigates the actor to the ‘My Profile’ page. 2. The system displays the personal information of the actor. 3. The actor clicks the ‘Settings’ dropdown button. 4. If the actor selects the ‘Edit Profile’ option. 4.1. The system navigates to the ‘Edit My Profile’ page. 4.2. The actor edits their biography. 4.3. The actor clicks the ‘Save’ button after editing. 5. If the actor selects the ‘Change Password’ option. 5.1. The actor enters their current password. 5.2. The actor enters their new password. 5.3. The actor enters their new password again for confirmation. 5.4. The actor clicks the ‘Confirm’ button.

	6. The actor selects the 'Delete Account' option. 6.1. The actor enters their password in order to delete their account. 6.2. The actor clicks the 'Delete Account' button. 7. The system updates the changes.
Alternative flow - Actor clicks the cancel button	4.3.1 The actor clicks the 'Cancel' button. 4.3.2 The system navigates the actor to the 'My Profile' page. 5.4.1 The actor clicks the 'Cancel' button. 5.4.2 The system navigates the actor to the 'My Profile' page. 6.2.1 The actor clicks the 'Cancel' button. 6.2.2 The system navigates the actor to the 'My Profile' page.
Exception flow - Actor leaves the input field empty.	5.4 The system shows an error message "Please fill out this field".
Exception flow - Current password is incorrect.	5.4 The system shows an error message and requests the actor to fill in with the correct password. 6.2 The system shows an error message and requests the actor to fill in with the correct password.
Exception flow - The new password entered is invalid.	5.4 The system shows an error message and requests the actor to fill in with correct format.
Exception flow - The confirmation password entered is not the same as the new password.	5.4 The system shows an error message and requests the actor to fill in with the correct password.

Use case Testing (System Level Testing)
Use case: Manage User Profile (F003)
Test Level: System Testing
Test Technique: Use Case Testing
Pre-condition: The user is logged into the system and has access to their profile page.
Post-condition:The user's profile information is updated, deleted, or the password is successfully changed based on the action performed.

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
TC-MUP-UC-001	Verify editing profile with valid data.	1. Navigate to the "My Profile" page. 2. Click the "Settings" dropdown button. 3. Select "Edit Profile". 4. Verify the system navigates to the "Edit My Profile" page. 5. Edit the biography text field. 6. Click the "Save" button.	Biography: "Software Engineering"	The system updates the biography successfully, displays a success message, and redirects to the "My Profile" page.	The system updates the biography successfully, displays a success message, and redirects to the "My Profile" page.	**P**
TC-MUP-UC-002	Verify the password change process updates the password correctly.	1. Navigate to the "My Profile" page. 2. Click the "Settings" dropdown button.	Current Password: "12345" New Password: "23456"	The system updates the password successfully, displays a success message, and redirects	The system updates the password successfully, displays a success	**P**

		3. Select "Change Password". 4. Enter the current password. 5. Enter a new password. 6. Re-enter the new password for confirmation. 7. Click "Confirm".		to the "My Profile" page.	message, and redirects to the "My Profile" page.	
TC-MUP-UC-003	Verify deleting account with user confirmation.	1. Navigate to the "My Profile" page. 2. Click the "Settings" dropdown button. 3. Click the "Settings" dropdown. 4. Select 'Delete Account'. 5. Confirm the deletion prompt. 6. Verify redirection.	Password: “23456”	Account is deleted, and user is redirected to login page.	Account is deleted, and user is redirected to login page.	**P**
TC-MUP-UC-004	Verify the system handles cancellation during the profile editing process.	1.Navigate to the "My Profile" page. 2. Click the "Settings" dropdown button. 3. Select "Edit Profile".	Biography: New biography text (not saved)	The system cancels the edit action, discards changes, and navigates back to the "My Profile" page.	The system cancels the edit action, discards changes, and navigates back to the "My Profile" page.	**P**

		4. Edit the biography field. 5. Click the "Cancel" button.				
TC-MUP-UC-005	Verify the system handles cancellation during the password change process.	1. Navigate to the "My Profile" page. 2. Click the "Settings" dropdown button. 3. Select "Change 4. Enter the current password. 5. Enter a new password. 6. Re-enter the new password for confirmation. 7. Click the "Cancel" button.	Current Password: “23456” New Password: “34567”	The system cancels the password change action, discards the entered data, and navigates back to the "My Profile" page.	The system cancels the password change action, discards the entered data, and navigates back to the "My Profile" page.	**P**
TC-MUP-UC-006	Verify the system handles cancellation during the account deletion process.	1. Navigate to the "My Profile" page. 4. Click the "Settings" dropdown button. 5. Select "Delete Account". 6. Enter the account password.	Password: “23456”	The system cancels the account deletion action, discards the entered data, and navigates back to the "My Profile" page.	The system cancels the account deletion action, discards the entered data, and navigates back to the "My Profile" page.	**P**

		6.2.1. Click the "Cancel" button.				

Error Guessing Testing (System Level Testing)
Use case: Manage User Profile (F003)
Test Level: System Testing
Test Technique: Error Guessing Testing
Pre-condition: The user is logged into the system and has access to their profile page.
Post-condition:The user's profile information is updated, deleted, or the password is successfully changed based on the action performed.

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
TC-MUP-EG-001	Verify the system displays an error when the password fields are left empty.	1. Navigate to the "My Profile" page. 2. Click the "Settings" dropdown button. 3. Select "Change Password". 4. Leave all password fields empty. 5. Click the "Confirm" button.	-	The system shows an error message: "Please fill out this field" for each required empty field.	The system shows an error message: "Please fill out this field" for each required empty field.	**P**
TC-MUP-EG-002	Verify the system handles an incorrect current password during the password change process.	1. Navigate to the "My Profile" page. 2. Click the "Settings" dropdown button.	Current Password: “88888888” New Password: “9999999” Confirm Password: “9999999”	The system shows an error message: "Incorrect current password. Please try again." and does not update the password.	The system shows an error message: "Incorrect current password. Please try again." and	**P**

		3. Click the "Settings" dropdown button. 4. Select "Change Password". 5. Enter an incorrect current password. 6. Enter a new password. 7. Re-enter the new password for confirmation. 8. Click the "Confirm" button.			does not update the password.	
TC-MUP-EG-003	Verify the system handles an incorrect password during the account deletion process.	1. Navigate to the "My Profile" page. 2. Click the "Settings" dropdown button. 4. Click the "Settings" dropdown button. 5. Select "Delete Account". 6.1 Enter an incorrect password. 6.2 Click the "Delete Account" button.	Password: “8888888”	The system shows an error message: "Incorrect password. Please enter the correct password." and does not delete the account.	The system shows an error message: "Incorrect password. Please enter the correct password." and does not delete the account.	**P**

TC-MUP-EG-004	Verify the system handles invalid new password input during the password change process.	1. Navigate to the "My Profile" page. 2. Click the "Settings" dropdown button. 3. Select "Change Password". 4. Enter the current password. 5.2 Enter an invalid new password. 5.3 Re-enter the invalid new password for confirmation. 5.4 Click the "Confirm" button.	Current Password: “23456” New Password: “123” Confirm Password: “123”	The system shows an error message: “Be at least 5 characters and at most 20 characters" and does not update the password.	The system shows an error message: “Be at least 5 characters and at most 20 characters" and does not update the password.	**P**
TC-MUP-EG-005	Verify the system handles mismatched confirmation password during the password change process.	1. Navigate to the "My Profile" page. 2. Click the "Settings" dropdown button.	Current Password: “23456” New Password: “34567” Confirm Password: “3456789”	The system shows an error message: "Please enter the same new password" and does not update the password.	The system shows an error message: "Please enter the same new password" and does not update the password.	**P**

GUI Testing (System Level Testing)

Use case: Manage User Profile (F003)
Test Level: System Testing
Test Technique: GUI Testing
Pre-condition: The user is logged into the system and has navigated to the "My Profile" page.
Post-condition: The user's profile is updated, password is changed, or account is deleted based on the action performed.

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
TC-MUP-GUI-001	Verify navigation to the "My Profile" page.	1. Click the "My Profile" button in the navigation bar.	Not Applicable	The system navigates to the "My Profile" page and displays the user's personal information.	The system navigates to the "My Profile" page and displays the user's personal information.	**P**
TC-MUP-GUI-002	Verify the layout and alignment of the "My Profile" page	1. Check that personal information fields are properly aligned and visible.	Not Applicable	The layout is clean, and all fields are displayed without overlap or misalignment.	The layout is clean, and all fields are displayed without overlap or misalignment.	**P**
TC-MUP-GUI-003	Verify that the "Settings" dropdown displays options.	1. Click the "Settings" dropdown button.	Not Applicable	The dropdown displays options: "Edit Profile",	The dropdown displays options: "Edit Profile",	**P**

				"Change Password", and "Delete Account".	"Change Password", and "Delete Account".	
TC-MUP-GUI-004	Verify navigation to the "Edit My Profile" page.	1. Click "Settings". 2. Select "Edit Profile".	Not Applicable	The system navigates to the "Edit My Profile" page.	The system navigates to the "Edit My Profile" page.	
TC-MUP-GUI-005	Verify the "Edit My Profile" page form validation.	1. Enter invalid data in the biography field (e.g., empty input). 2. Click "Save".	Biography: ""	The system displays an error message: "Please fill out this field".	The system displays an error message: "Please fill out this field".	**P**
TC-MUP-GUI-006	Verify the "Change Password" feature.	1. Click "Settings". 2. Select "Change Password". 3. Enter current password and mismatched confirmation password. 4. Click "Confirm".	Current Password: "12345", New Password: "23456", Confirm Password: "88888"	The system displays an error message: "Confirmation password does not match".	The system displays an error message: "Confirmation password does not match".	**P**
TC-MUP-GUI-007	Verify the "Delete Account" confirmation.	1. Click "Settings". 2. Select "Delete Account". 3. Enter the correct password. 4. Confirm account deletion.	Password: a valid password	The system deletes the account and redirects the user to the login page.	The system deletes the account and redirects the user to the login page.	**P**

2.3.4 F004 - View Events (Tester: Kei Kar)

The "View Events" feature enables alumni to browse, search, and view event details. It keeps alumni informed and engaged with faculty activities. Testing ensures reliability, accuracy, and ease of use.

The following table shows the features to be tested based on the SBD, including functional ID, feature, and risk level.

Feature ID	Feature	Risk Level
F003	View Events	Medium

The techniques that will be applied are as follows:

1. Use Case Testing (Black Box Testing - System Level)
2. Error Guessing (Black Box Testing - System Level)
3. GUI Testing (Black Box Testing - System Level)

Nature of Feature:
The "View Events" feature allows alumni to browse and access detailed information about events created by faculty administrators. Alumni can navigate to the events page, search for specific events using keywords, and view the details of selected events, such as title, date, time, and description. This feature is critical for user engagement and includes mechanisms to handle exceptions, such as displaying error messages when no matching records are found. It aligns with usability goals to provide a seamless and informative experience.

Justification:
Use case testing ensures that all aspects of the "View Events" functionality are verified, including navigating to the events page, searching for specific events, and viewing event details. This method ensures coverage of the main flow, alternative flow, and exception flow, validating both expected and edge-case behaviors. Error Guessing anticipates potential errors such as invalid search keywords or missing event data, ensuring the system provides appropriate error handling and feedback. GUI Testing evaluates the user interface for usability, layout, responsiveness, and input validation, ensuring the feature is both functional and user-friendly.

Use Case Name	View events
Use Case ID	UC-4
Description	The system shall allow alumni to view events' details that were created by the faculty administrator.
Priority	High
Actor(s)	Alumni
Triggering Event	Actor clicks the 'All Events' button at the navigation bar or clicks the 'view more' button at the Events section at home page.
Pre-condition	Actor has logged into the system.
Post-condition	The system displays the events' details.
Flow of Events	1. The system navigates the actor to the 'Events' page. 2. The system displays a list of events. 3. The actor enters keywords at the search bar to search for events. 4. The actor clicks the interested event. 5. The system navigates the actor to the selected events' page to show event's details.
Exception flow - Keyword entered not matched in database	3.1 The system shows an error message "Sorry, no records found.

Use case Testing (System Level Testing)

Use case: View Events (F004)

Test Level: System Testing

Test Technique: Use Case Testing

Pre-condition: The user is logged into the system and navigates to the "View Events" page.

Post-condition: The user successfully views the event details, searches for an event, or interacts with the event list as intended.

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
TC-VE-UC-001	Verify navigation to the ‘Events’ page.	1. Click the ‘All Events’ button at the navigation bar or the ‘View More’ button at the home page.	Not applicable.	The system navigates the actor to the 'Events' page and displays a list of events.	The system navigates the actor to the 'Events' page and displays a list of events.	**P**
TC-VE-UC-002	Verify event search functionality.	1. Enter a keyword in the search bar. 2. Click on the search icon.	Valid: "Machine Learning" Invalid: "Hello"	System displays a filtered list.	System displays a filtered list.	**P**
TC-VE-UC-003	Verify event details.	1. Click on an event from the displayed list.	Event Title: "Machine Learning Workshop"	The system navigates to the event details page and displays event information (title, date, time, location, description).	The system navigates to the event details page and displays event information.	**P**

Error Guessing (System Level Testing)
use case: View Events (F004)
Test Level: System Testing
Test Technique: Error Guessing Testing
Pre-condition: The user is logged into the system and navigates to the "View Events" page.
Post-condition: The user successfully views the event details, searches for an event, or interacts with the event list as intended.

Test Case ID	**Test Case**	**Test Steps**	**Data Input**	**Expected Result**	**Actual Result**	**Test Status (P/F)**
TC-VE-EG-001	Verify the system displays an error message for unmatched search keywords.	1. Navigate to the "Events" page. 2. Verify the system displays the list of events. 3. Enter keywords in the search bar that do not match any event in the database. 4. Observe the system's response.	Keyword: “Hello”	The system displays an error message: "Sorry, no records found," and no events are shown in the results.	The system displays an error message: "Sorry, no records found," and no events are shown in the results.	**P**

GUI Testing (System Level Testing)

Use case: View Events (F004)
Test Level: System Testing
Test Technique: GUI Testing
Pre-condition: The user is logged into the system and navigates to the "Events" page.
Post-condition: The user can successfully view the list of events, search for events using keywords, and access detailed information for a specific event.

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
TC-VE-GUI-001	Verify that the "All Events" button navigates to the "Events" page.	1. Click the "All Events" button in the navigation bar.	Not Applicable	The system navigates to the "Events" page and displays a list of events.	The system navigates to the "Events" page and displays a list of events.	P
TC-VE-GUI-002	Verify that the layout and design of the "Events" page meet the requirements.	1. Check that the event list is displayed with proper alignment and spacing.	Not Applicable	The layout is clear, with events displayed in a structured and readable manner.	The layout is clear, with events displayed in a structured and readable manner.	P
TC-VE-GUI-003	Verify that the search bar displays correct results for valid input.	1. Enter a valid keyword in the search bar. 2. Observe the results displayed.	Keyword: "Machine Learning"	The system displays a filtered list of events matching the keyword.	The system displays a filtered list of events matching the keyword.	P

TC-VE-GUI-004	Verify that the search bar shows an error message for unmatched keywords.	1. Enter an invalid keyword in the search bar. 2. Click the search button.	Keyword: "Photography"	The system displays an error message: "Sorry, no records found."	The system displays an error message: "Sorry, no records found."	P
TC-VE-GUI-005	Verify that clicking on an event navigates to its details page.	1. Click on a specific event from the list.	Event Title: "Machine Learning Workshop"	The system navigates to the event details page, displaying title, date, and description.	The system navigates to the event details page, displaying title, date, and description.	
TC-VE-GUI-006	Verify that the "Events" page is responsive to different screen sizes.	1. Resize the browser window to various screen sizes. 2. Observe the layout.	Screen sizes: Mobile, Tablet, Desktop	The page layout adjusts appropriately to fit different screen sizes.	The page layout adjusts appropriately to fit different screen sizes.	P

2.3.5 F005 - Manage Job Advertisement (Tester: Azfar)

This feature allows alumni to post and manage job advertisements, promoting career opportunities within the alumni network. Thorough testing is necessary to ensure functionality, data integrity, and user experience.

Feature ID	Feature	Risk Level
F005	Manage Job Advertisement	High

The techniques that will be applied are as follows:

1. Use Case Testing (Black Box Testing - System Level)
2. Boundary Value Analysis (Black Box Testing - System Level)
3. Error Guessing
4. GUI Testing

Nature of Feature:
The "Manage Job Advertisement" feature allows alumni to post new job openings, edit existing postings, and delete job advertisements. This feature allows for easy sharing of job opportunities for alumni members.

Justification:
The Use Case Testing technique will be applied to thoroughly validate all scenarios, including adding, editing, and deleting job advertisements, ensuring the system behaves correctly under both normal and exceptional conditions as outlined in use case descriptions. Boundary Value Analysis (BVA) is essential for testing input fields and boundary conditions, such as character limits for job titles and descriptions, salary ranges, and file upload constraints, to prevent unexpected behavior at the edges of valid inputs and protect data integrity. Additionally, Error Guessing will identify potential vulnerabilities and user errors, such as network interruptions, duplicate entries, or unsupported special characters, enabling robust system performance under diverse conditions. Finally, GUI Testing will ensure the user interface is intuitive, responsive, and functional across devices and browsers, verifying layout, cross-browser compatibility, and proper error message display, thereby providing a seamless and reliable user experience.

Use Case Testing (System Testing Level)

Use Case Name	Manage Job Advertisement
Use Case ID	UC-007
Description	Alumni creates, updates, or deletes job ads.
Actor(s)	Alumni
Triggering Event	User navigates to "Manage Job Ads" page.
Pre-condition	User is logged in as an alumni.
Post-condition	Job ad is created, updated, or deleted successfully.
Flow of Events	1. Alumni navigates to "Manage Job Ads." 2. Alumni chooses action (Create, Edit, or Delete). 3. Alumni enters/modifies job details. 4. Alumni submits the job advertisement. 5. System validates and processes the request.
Exception Flow-Invalid Input	Error message, job posting prevented.
Exception Flow-Unauthorized Access	If non-alumni attempt to manage ads, display an error message.

Test Cases using Use Case Testing (System Testing Level)

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
TC-MJA-UC-001	Verify creating a job ad with valid data.	1. Navigate to the "Manage Job Ads" page. 2. Click "Create Job Ad". 3. Enter valid job details. 4. Submit the form.	Title: "Software Engineer" Description: "Looking for a skilled developer." Salary: "5000"	The system creates the job ad successfully, displays a success message, and adds it to the job listings.	The system created the job ad successfully, displayed a success message, and added it to the job listings.	P
TC-MJA-UC-002	Verify editing an existing job ad with valid data.	1. Navigate to the "Manage Job Ads" page. 2. Select an existing job ad. 3. Modify job details. 4. Click "Save Changes".	Updated Title: "Senior Engineer" Updated Description: "Experience required."	The system updates the job ad successfully, displays a success message, and reflects the changes.	The system updated the job ad successfully, displayed a success message, and reflected the changes	P
TC-MJA-UC-003	Verify deleting an existing job ad.	1. Navigate to the "Manage Job Ads" page. 2. Select a job ad to delete. 3. Confirm the deletion prompt. 4. Verify removal from listings.	Job Ad ID: 101	The system deletes the job ad successfully, displays a success message, and removes it from the listings.	The system deleted the job ad successfully, displayed a success message, and removed it from the listings	P

TC-MJA-UC-0 04	Verify creating a job ad with invalid input.	1. Navigate to the "Manage Job Ads" page. 2. Click "Create Job Ad". 3. Enter invalid job details (e.g., missing fields). 4. Submit the form.	Missing Title Invalid Salary: "-1000"	The system displays error messages for invalid fields, preventing job ad creation.	The system displayed error messages for invalid fields, preventing job ad creation.	P
TC-MJA-UC-0 05	Verify unauthorized access to "Manage Job Ads".	Attempt to access the "Manage Job Ads" page without logging in.	Unauthenticated User	The system denies access, redirects the user to the login page, and displays an error message.	The system denied access, redirected the user to the login page, and displayed an error message.	P
TC-MJA-UC-0 06	Verify creating a duplicate job ad.	1. Navigate to the "Manage Job Ads" page. 2. Click "Create Job Ad". 3. Enter details identical to an existing job ad. 4. Submit the form.	Title: "Software Engineer" Description: "Looking for a skilled developer." Salary: "5000"	The system prevents duplicate job ad creation and displays an error message.	The system prevented duplicate job ad creation and displayed an error message.	P
TC-MJA-UC-0 07	Verify creating a duplicate job ad.	1. Navigate to the "Manage Job Ads" page. 2. Click "Create Job Ad". 3. Enter details identical to an existing job ad. 4. Submit the form.	Title: "Software Engineer" Description: "Looking for a skilled developer." Salary: "5000"	The system prevents duplicate job ad creation and displays an error message.	The system allowed duplicate job ad creation and did not display an error message.	F

TC-MJA-UC-008	Verify editing an existing job ad with valid data.	1. Navigate to the "Manage Job Ads" page. 2. Select an existing job ad. 3. Modify job details. 4. Save changes.	Updated Title: "Senior Engineer" Updated Description: "Experience required."	The system updates the job ad successfully, displays a success message, and reflects the changes.	The system failed to update the job ad and displayed a generic error message instead	F

Boundary Value Analysis (System Testing Level)

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
TC-MJA-BVA-001	Max Length Job Title	1. Navigate to "Create Job Ad". 2. Enter a title exceeding the maximum character limit. 3. Submit the form.	Title: 256 characters (limit: 255)	The system displays an error message and prevents submission	The system displayed an error message and prevented submission.	P
TC-MJA-BVA-002	Min Length Job Description	1. Navigate to "Create Job Ad". 2. Enter a description with 1 character. 3. Submit the form.	Description: "A"	The system displays an error message and prevents submission.	The system displayed an error message and prevented submission.	P
TC-MJA-BVA-003	File Upload - Oversized File	1. Navigate to "Create Job Ad".	File: 6 MB (limit: 5 MB)	The system displays an error message, rejects	The system displayed an error message, rejected	P

		2. Attach a file exceeding the maximum size limit. 3. Submit the form.		the file, and prevents submission.	the file, and prevented submission.	
TC-MJA-BVA-004	File Upload - Supported File Size	1. Navigate to "Create Job Ad". 2. Attach a file exactly at the maximum size limit. 3. Submit the form.	File: 5 MB	The system accepts the file and allows successful submission	The system accepted the file and allowed successful submission.	P
TC-MJA-BVA-005	Salary Below Minimum	1. Navigate to "Create Job Ad". 2. Enter a salary below the allowed range. 3. Submit the form.	Salary: -1	The system displays an error message and prevents submission.	The system displayed an error message and prevented submission.	P
TC-MJA-BVA-006	Salary Above Maximum	1. Navigate to "Create Job Ad". 2. Enter a salary exceeding the maximum allowed range. 3. Submit the form.	Salary: 1,000,001 (limit: 1,000,000)	The system displays an error message and prevents submission.	The system displayed an error message and prevented submission	P
TC-MJA-BVA-007	Max Length Job Description	1. Navigate to "Create Job Ad". 2. Enter a description exactly at the maximum character limit. 3. Submit the form	Description: 2000 characters	The system accepts the description and allows successful submission	The system accepted the description and allowed successful submission.	P

TC-MJA-BVA-008	Max Salary Exceeding Allowed Limit	1. Navigate to "Create Job Ad". 2. Enter a salary exceeding the maximum allowed limit. 3. Submit the form.	Salary: 1,000,001 (limit: 1,000,000)	The system displays an error message and prevents submission.	The system failed to validate the salary limit and allowed submission.	F
TC-MJA-BVA-009	File Upload - Oversized File	1. Navigate to "Create Job Ad". 2. Attach a file exceeding the maximum size limit. 3. Submit the form.	File: 6 MB (limit: 5 MB)	The system displays an error message, rejects the file, and prevents submission	The system allowed the file upload, which exceeds the size limit, and created the job ad.	F

Error Guessing

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
TC-MJA-EG-001	Duplicate Job Ad Submission	1. Navigate to "Manage Job Ads". 2. Click "Create Job Ad". 3. Enter identical details to an existing job ad. 4. Submit the form.	Title: "Software Engineer" Description: "Looking for a skilled developer." Salary: "5000"	The system prevents duplicate job ad creation and displays an error message.	The system prevented duplicate job ad creation and displayed an error message.	P
TC-MJA-EG-002	Network Interruption During Submission	1. Start submitting a job ad. 2. Disconnect the network during submission. 3. Reconnect the network and retry submission.	Valid Job Details	The system saves the data temporarily and allows resubmission after reconnection.	The system saved the data temporarily and allowed resubmission after reconnection.	P
TC-MJA-EG-003	Invalid Characters in Job Title	1. Navigate to "Manage Job Ads". 2. Click "Create Job Ad". 3. Enter invalid characters in the title field. 4. Submit the form.	Title: "<script>alert('XSS')</script>"	The system displays an error message and prevents the submission.	The system displayed an error message and prevented the submission.	P

TC-MJA-EG-004	Multiple Submissions	1. Navigate to "Manage Job Ads". 2. Click "Create Job Ad". 3. Submit the form multiple times quickly.	Title: "Software Engineer" Description: "Looking for a skilled developer." Salary: "5000"	The system allows only one submission and prevents duplicates.	The system allowed only one submission and prevented duplicates.	P
TC-MJA-EG-005	Exceeding Allowed Job Ads per User	1. Create multiple job ads until the maximum limit is reached. 2. Attempt to create another job ad.	Number of Job Ads: 11 (limit: 10)	The system displays an error message and prevents the user from creating additional job ads.	The system displayed an error message and prevented the user from creating additional job ads.	P
TC-MJA-EG-006	Unsupported File Type	1. Navigate to "Manage Job Ads". 2. Click "Create Job Ad". 3. Attach an unsupported file type. 4. Submit the form.	File: "jobad.exe"	The system rejects the file upload and displays an error message about unsupported file types.	The system rejected the file upload and displayed an error message about unsupported file types.	P
TC-MJA-EG-007	Incomplete Form Submission	1. Navigate to "Manage Job Ads". 2. Click "Create Job Ad". 3. Leave one or more required fields empty. 4. Submit the form.	Missing Title	The system displays an error message for the missing fields and prevents submission.	The system displayed an error message for the missing fields and prevented submission.	P

TC-MJA-EG-008	Invalid Characters in Job Title	1. Navigate to "Manage Job Ads". 2. Click "Create Job Ad". 3. Enter invalid characters in the title field. 4. Submit the form.	Title: <script>alert("XSS")</script>	The system displays an error message and prevents submission.	The system failed to validate the invalid characters and allowed submission.	F
TC-MJA-EG-009	Multiple Submissions	1. Navigate to "Manage Job Ads". 2. Click "Create Job Ad". 3. Submit the form multiple times quickly.	Title: "Software Engineer" Description: "Looking for a skilled developer." Salary: "5000"	The system allows only one submission and prevents duplicates.	The system allowed multiple duplicate submissions, failing to block duplicates.	F

GUI Testing

Test Case ID	Test Case	Test Steps	**Data Input**	Expected Result	Actual Result	Test Status (P/F)
TC-MJA-GUI-001	Verify navigation to "Manage Job Ads".	1. Click the "Manage Job Ads" option from the navigation menu.	None	The system navigates to the "Manage Job Ads" page and displays a list of job advertisements.	The system navigated to the "Manage Job Ads" page and displayed a list of job advertisements.	P

TC-MJA-GUI-002	Verify the layout of the "Manage Job Ads" page.	1. Open the "Manage Job Ads" page. 2. Inspect the alignment and visibility of UI components.	None	All UI components (buttons, forms, etc.) are properly aligned and visible without overlap or distortion.	All UI components were properly aligned and visible without overlap or distortion.	P
TC-MJA-GUI-003	Verify the responsiveness of the interface.	1. Open the "Manage Job Ads" page on a mobile device or resize the browser window to a smaller width.	None	The interface adjusts to the screen size, ensuring elements are usable and readable on smaller screens.	The interface adjusted to the screen size, ensuring usability and readability on smaller screens.	P
TC-MJA-GUI-004	Verify error messages are displayed correctly.	1. Submit the "Create Job Ad" form with invalid inputs. 2. Inspect the placement and clarity of error messages.	Missing Title, Invalid Salary	Error messages are displayed inline near the respective fields with clear and understandable text.	Error messages were displayed inline near the respective fields with clear and understandable text.	P
TC-MJA-GUI-005	Verify button states and responsiveness.	1. Inspect all buttons (e.g., "Create", "Edit", "Delete"). 2. Hover, click, and inspect disabled states when applicable.	None	Buttons display proper states (hover, active, disabled) and are responsive when clicked	Buttons displayed proper states (hover, active, disabled) and were responsive when clicked.	P

TC-MJA-GUI-006	Verify dropdown menu functionality.	1. Open the dropdown menu for options like "Edit" or "Delete" next to a job ad. 2. Select an option.	None	Dropdown menu opens properly and displays all available options. Selected options perform as expected.	Dropdown menu opened properly, displayed all available options, and selected options performed as expected.	P
TC-MJA-GUI-007	Verify cross-browser compatibility.	1. Open the "Manage Job Ads" page in multiple browsers (e.g., Chrome, Firefox, Edge).	None	The page renders and functions consistently across all tested browsers.	The page rendered and functioned consistently across all tested browsers.	P
TC-MJA-G-008	Verify the layout of the "Manage Job Ads" page.	1. Open the "Manage Job Ads" page. 2. Inspect the alignment and visibility of UI components.	None	All UI components (buttons, forms, etc.) are properly aligned and visible without overlap or distortion	Some UI components overlapped, and a button was partially hidden on smaller screens.	F
TC-MJA-G-009	Verify button states and responsiveness.	1. Inspect all buttons (e.g., "Create", "Edit", "Delete"). 2. Hover, click, and inspect disabled states when applicable.	None	Buttons display proper states (hover, active, disabled) and are responsive when clicked.	The "Edit" button did not respond when clicked, and the hover state was not visible.	F

2.3.6 F006 - Search and View Alumni Profile (Tester: Mai)

This feature allows alumni to search and view other alumni profiles, facilitating networking. Efficient searching and clear profile displays are key to a positive user experience.

Feature ID	Feature	Risk Level
F006	Search and View Alumni Profile	Low

The techniques applied are:

1. Use Case Testing (Black Box Testing - System Level)
2. Performance Testing (Black Box Testing - System level)

Nature of Feature:
This function allows users to search and view other alumni profiles.

Justification:
Use case testing will help to ensure the correctness and completeness of search functionality, including searching with various criteria (name, keywords, etc.) and correctly displaying alumni profiles. Performance testing is also essential to assess the system's response time when searching and retrieving alumni profiles, particularly with larger datasets, ensuring an optimal user experience.

Use Case Testing (System Testing Level)

Use Case Name	Search and View Alumni Profile
Use Case ID	UC-006
Description	This use case describes alumni searching for and viewing profiles of other alumni.
Actor(s)	Alumni

Triggering Event	The user initiates a search for an alumni profile using the search feature.
Pre-condition	The user must be logged into the system with a verified account.
Post-condition	The system displays the search results, or the specific alumni profile requested.
Flow of Events	1. The actor logs into the system. 2. The actor navigates to the alumni search page. 3. The actor enters search criteria (e.g., name, department, graduation year). 4. The system processes the search request and displays a list of matching alumni profiles. 5. The actor clicks on a specific alumni profile to view detailed information.
Exception flow - Keyword entered not matched in database	Keyword entered not matched in database. The system displays a "No match found" message.

Test Cases using Use Case Testing (System Testing Level)

Use case: Search and View Alumni Profile (F006)
Test Level: System Testing
Test Technique: Use Case Testing
Pre-condition: The user must be logged into the system with a verified account.
Post-condition: The system displays the search results, or the specific alumni profile requested.

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (Pass/Fail)
VAP-UC -001	Search alumni with valid keyword	1. Log in to the system. 2. Navigate to the search bar. 3. Enter a valid keyword (e.g., "Mai Mai"). 4. Click the search button.	Valid data: "Mai Mai"	System displays a list of alumni profiles matching the keyword "Mai Mai".	System displays relevant results for "Mai Mai".	Pass
VAP-UC -002	Search alumni with invalid keyword	1. Log in to the system. 2. Navigate to the search bar. 3. Enter an invalid keyword (e.g., "xyz123"). 4. Click the search button.	Invalid data: "xyz123"	System displays a "No match found" message.	System displays a "No match found" message.	Pass
VAP-UC -003	Search alumni with empty keyword	1. Log in to the system. 2. Navigate to the search bar. 3. Leave the search bar empty. 4. Click the search button.	No Data	System prompts the user to enter a keyword or displays an appropriate error message.	No action is performed, and no error message is displayed.	Fail
VAP-UC -004	View profile from search results	1. Perform a valid search. 2. Click on one of the profiles from the search results list.	Valid data: Select any profile from results	System displays the details of the selected alumni profile.	Profile details page opens correctly.	Pass
VAP-UC -005	Search with partial keyword	1. Log in to the system. 2. Navigate to the search bar.	Valid data: "Mai"	System displays a list of alumni profiles with	System displays profiles with	Pass

		3. Enter a partial keyword (e.g., "Mai"). 4. Click the search button.		names that include "Mai".	partial match for "Mai".	
VAP-UC -006	Search with special characters	1. Log in to the system. 2. Navigate to the search bar. 3. Enter a keyword with special characters (e.g., "!@#$"). 4. Click the search button.	Invalid data: "!@#$"	System displays a "No match found" message or prompts the user to enter a valid keyword.	System crashes with a database error.	Fail
VAP-UC -007	Search with case-insensitiv e keyword	1. Log in to the system. 2. Navigate to the search bar. 3. Enter a keyword in uppercase/lowercase (e.g., "MAI MAI" or "mai mai"). 4. Click search.	Valid data: "MAI MAI" or "mai mai"	System displays results matching the keyword regardless of case.	Results are displayed regardless of case.	Pass
VAP-UC -008	Search with numeric values	1. Log in to the system. 2. Navigate to the search bar. 3. Enter numeric values as a keyword (e.g., "12345"). 4. Click the search button.	Valid/Invalid data: "12345"	System processes the keyword and displays results or a "No match found" message if not applicable.	System displays a "No match found" message.	Pass
VAP-UC -009	Search alumni immediately after profile update	1. Update an alumni profile in the system. 2. Use a keyword related to the updated profile. 3. Click the search button.	Valid data: Updated profile name/keyword	System displays the updated profile in the search results.	System takes a long time to respond and eventually times out.	Fail
VAP-UC -010	Search alumni when no profiles exist in the database	1. Log in to the system. 2. Navigate to the search bar. 3. Enter a valid or invalid keyword. 4. Click the search button.	No data in the database	System displays a "No match found" message regardless of the entered keyword.	System displays results correctly after trimming spaces.	Pass

Performance Testing (System Testing Level)

Use case: Search and View Alumni Profile (F006)
Test Level: System Testing
Test Technique: Performance Testing
Pre-condition: The user must be logged into the system with a verified account.
Post-condition: The system displays the search results, or the specific alumni profile requested.

Test Case ID	**Test Case**	**Test Steps**	**Data Input**	**Expected Result**	**Actual Result**	Test Status (Pass/Fail)
PT-UC-001	Single user search	1. Open the Alumni Search page. 2. Enter a valid keyword in the search bar. 3. Submit the search.	Valid Keyword: Cris	Search results displayed within 2 seconds.	Results displayed in 1.8 seconds.	Pass
PT-UC-002	Multiple users search (100 users)	1. Simulate 100 concurrent users searching with valid keywords.	Valid Keywords: Cris, Mai, IT	All 100 users receive results within 3 seconds with no server crashes.	All 100 users received results in 5 seconds, with some crashes.	Fail
PT-UC-003	High database volume	1. Populate the database with 1,000,000 alumni records. 2. Search using a valid keyword.	Valid Keyword: Engineering	Search results displayed within 2-3 seconds, regardless of database size.	Results displayed in 3.5 seconds.	Fail
PT-UC-004	Search with an invalid keyword	1. Open the Alumni Search page. 2. Enter an invalid keyword (not in the database). 3. Submit the search.	Invalid Keyword: Xyz123	"No results found" message displayed within 2 seconds.	"No results found" message displayed in 1.5 seconds.	Pass
PT-UC-005	Simultaneous search with	1. Simulate 500 users performing searches simultaneously with	Mixed Keywords: Mai, Xyz123	System processes all queries with <2% error rate	98% queries processed successfully;	Pass

	peak users (500 users)	valid and invalid keywords.		and results delivered within 3-4 seconds.	results delivered in 3.2 seconds for most users.	
PT-UC-006	Stress test for extreme user load (1000 users)	1. Simulate 1000 users performing searches at the same time.	Mixed Keywords: Cris, Mai	System maintains stability, with search results displayed within 5 seconds for 95% of the users.	System stable; 96% of users received results within 4.8 seconds, but 4% experienced slight delays.	Pass
PT-UC-007	Search with special characters	1. Open the Alumni Search page. 2. Enter special characters in the search bar. 3. Submit the search.	Keyword: @#%!	"No results found" message displayed within 2 seconds.	"No results found" message displayed in 2.7 seconds.	Fail
PT-UC-008	Endurance test (24-hour continuous usage)	1. Simulate 10 users performing searches every 5 minutes for 24 hours.	Valid Keyword: Developer	System remains stable, with consistent response times and no downtime during the test period.	System stable; response times consistent at ~2 seconds, with no downtime recorded.	Pass
PT-UC-009	Search for partial keyword	1. Open the Alumni Search page. 2. Enter a partial keyword. 3. Submit the search.	Partial Keyword: Eng	Results for all matching alumni displayed within 2 seconds.	Results for matching alumni displayed in 2.3seconds.	Fail
PT-UC-010	Database-heavy search (complex query execution time)	1. Simulate a query that matches many database entries (e.g., by department or large common keyword).	Keyword: Engineering	Results for matching alumni displayed within 3 seconds despite high computational demand.	Results displayed in 3.1 seconds; system handled query efficiently without crashing.	Pass

2.3.7 F007 - Manage Alumni Account (Testers: Firdaus & Yallini)

This feature allows faculty administrators to manage alumni accounts, ensuring data accuracy and control over access. Thorough testing is important for proper functionality, data integrity, and usability of admin tools.

Feature ID	Feature	Risk Level
F007	Manage Alumni Account	High

The techniques that will be applied are as follows:

1. Use Case Testing (Black Box Testing - System Level)
2. State Transition Testing (Black Box Testing - System Level)
3. GUI Testing (Black Box Testing - System Level)

Nature of Feature:
"Manage Alumni Account" allows faculty administrators to manage alumni accounts, including approving/rejecting registrations, editing alumni information, and deleting accounts. This function is important for maintaining an accurate and up-to-date alumni database.

Justification:
Use Case Testing is performed for all admin functions, including approving/rejecting accounts, updating information, and deleting accounts. This verifies all use cases, extensions (alternative and exception flows) behave as expected. State Transition testing is critical to test changes in status because this feature involves different alumni account states (e.g., pending, approved, rejected) and their respective transitions, so State Transition Testing is important for ensuring the system handles these state changes correctly. GUI Testing evaluates the admin interface's usability for its effectiveness, ease of use, and responsiveness.

Use Case Testing (System Testing Level)

Use Case Name	Manage Alumni Account
Use Case ID	UC-007

Description	Faculty administrator approves/rejects, edits, or deletes accounts.
Actor(s)	Faculty Administrator
Triggering Event	Admin accesses the alumni management page.
Pre-condition	Admin is logged in.
Post-condition	Account status updated, information modified, or account deleted.
Flow of Events	1. Admin selects an action (approve/reject, edit, delete). 2. Admin performs the chosen action. 3. System updates the database and provides feedback.
Exception Flow - Invalid Action	Display an error message.
Exception Flow - Unauthorized Access	If non-admin attempts access, display "Access Denied".

Test Cases using Use Case Testing (System Testing Level) - Yallini

Use case Testing (System Level Testing)
Use case: Use Case Testing (F007)
Test Level: System Testing
Test Technique: Use Case Testing
Pre-condition: Alumni is not logged in and has a valid University of Malaya email address.
Post-condition: Alumni account is created and pending approval (or directly approved).

Test Case ID	**Test Case**	**Test Steps**	**Data Input**	**Expected Result**	**Actual Result**	**Test Status (P/F)**
MAA-UC-001	Approve Account	1. Access alumni management page. 2. Select a pending alumni account. 3. Click "Approve".	Example: Pending alumni account	Account status changes to "Approved".	The system changed the account status to "Approved".	P
MAA-UC-002	Reject Account	1. Access alumni management page. 2. Select a pending alumni account. 3. Click "Reject".	Example: Pending alumni account	Account status changes to "Rejected".	The system changed the account status to "Rejected".	P
MAA-UC-003	Edit Account	1. Access alumni management page. 2. Select an approved alumni account. 3. Click "Edit". 4. Modify alumni	Example: Valid modifications to alumni information	Information updated correctly.	The system updated the information correctly.	P

		information. 5. Click "Save".				
MAA-UC-004	Delete Account	1. Access alumni management page. 2. Select an alumni account. 3. Click "Delete". 4. Confirm deletion.	Example: Alumni account selected for deletion	Account permanently removed.	The system permanently removed the account.	P
MAA-UC-005	Unauthorized Access	1. Attempt to access alumni management page as a non-admin.	Example: Non-admin user attempting access	Error message: "Access Denied".	The system displayed an error message: "Access Denied".	P
MAA-UC-006	Invalid Action	1. Access alumni management page. 2. Select a pending alumni account. 3. Attempt to edit the account (which should only be done for approved accounts).	Example: Edit a pending account, not approved	Error message indicating invalid action.	The system displayed an error message indicating invalid action.	P

State Transition Testing (System Testing Level) - Firdaus

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
MAA-ST T-001	Initial Registration	1. Navigate to the registration page. 2. Enter valid alumni information. 3. Submit the registration form.	Valid name, email, graduation year, etc.	Account created and transitions to "Pending Approval" state.	Account created, status "Pending Approval."	P
MAA-ST T-002	Admin Approves - Pending to Approved	1. Log in as an administrator. 2. Navigate to the pending alumni account list. 3. Select the target account. 4. Click "Approve."	N/A (Admin action)	Account status changes to "Approved," and the alumnus can log in.	Account status updated to "Approved." Login successful.	P
MAA-ST T-003	Admin Rejects - Pending to Rejected	1. Log in as admin. 2. Navigate to the pending list. 3. Select the target account. 4. Click "Reject."	N/A (Admin action)	Account status changes to "Rejected"; registration process ends; alumnus cannot log in.	Account status "Rejected." Login attempt fails with "Account Rejected" message.	P
MAA-ST T-004	Update Details - Active to Updated to Active	1. Log in as an approved alumnus. 2. Navigate to "My Profile." 3. Modify profile information. 4. Save changes.	Valid updated profile data.	Profile details are updated, account remains "Active". System processes the changes, and the account reverts to the "Active" state.	Changes saved, profile updated, account remains "Active".	P

MAA-ST T-005	Request Deletion - Active to Deleted to End	1. Log in as alumnus. 2. Navigate to account settings. 3. Request account deletion. 4. Confirm deletion.	N/A (User action)	Account status changes to "Deleted," the user is logged out, and the account is no longer accessible.	Account deleted, logout successful. Attempting to log in again results in an "Account Not Found" message.	P
MAA-ST T-006	Invalid Input - Editing	1. Navigate to "My profile". 2. Enter invalid data during profile update. 3. Save changes.	Invalid email format, excessively long name, special characters, etc.	Informative error messages, change prevented, remains in "Active" or "Editing" state.	Correct error message displayed. Profile data not updated. Account remains "Active."	P
MAA-ST T-007	Attempt Update Before Approval	1. Navigate to My Profile (without the admin approval process) 2. Try to edit information.	Profile update attempts before account is approved.	Access denied or error message, profile remains unchanged, and the state stays as "Pending"	System allows editing before approval. Defect logged as D-001.	F

GUI Testing (System Testing Level) - Yallini

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
MAA-GUI-001	Verify navigation and layout	1. Login as admin. 2. Navigate to alumni management page. 3. Verify that all navigation elements are clear.	Not Applicable	Interface elements are clear, accessible, and logically arranged.	Interface elements are clear, accessible, and logically arranged.	P
MAA-GUI-002	Verify search functionality	1. Login as admin. 2. Navigate to alumni management page. 3. Enter search terms and check filters.	Search terms	Accurate search results, user-friendly search filters.	Accurate search results, user-friendly search filters.	P
MAA-GUI-003	Verify data display	1. Login as admin. 2. Navigate to alumni management page. 3. Check displayed alumni data for accuracy.	Alumni data	Information displayed accurately and formatted appropriately.	Information displayed accurately and formatted appropriately.	P
MAA-GUI-004	Verify feedback messages	1. Login as admin. 2. Navigate to alumni management page. 3. Perform approve/reject action and observe feedback.	Approve/Reject action	Clear feedback messages indicating success or failure.	Clear feedback messages indicating success or failure.	P

MAA-GUI-005	Verify table sorting functionality	1. Login as admin. 2. Navigate to alumni management page. 3. Click on table headers to sort by name, year, or department.	Not Applicable	The table sorts correctly based on the selected criteria.	The table sorts correctly based on the selected criteria.	P
MAA-GUI-006	Verify pagination controls	1. Login as admin. 2. Navigate to alumni management page. 3. Use the pagination controls to move through the pages of alumni.	Not Applicable	Pagination works correctly, allowing navigation between pages.	Pagination works correctly, allowing navigation between pages.	P
MAA-GUI-007	Verify responsive design	1. Login as admin. 2. Navigate to alumni management page. 3. Resize the browser window to small (mobile), medium (tablet), and large (desktop).	Not Applicable	The layout adjusts properly for different screen sizes.	The layout adjusts properly for different screen sizes.	P
MAA-GUI-008	Verify filter functionality	1. Login as admin. 2. Navigate to alumni management page. 3. Apply filters like graduation year and department from dropdowns.	Filter criteria (e.g., year, department)	Alumni list updates correctly based on the applied filters.	Alumni list updates correctly based on the applied filters.	P
MAA-GUI-009	Verify bulk actions	1. Login as admin. 2. Navigate to alumni management page. 3. Select multiple alumni entries and perform bulk actions (approve or reject).	Multiple alumni selected	System performs the bulk actions correctly on all selected alumni.	System performs the bulk actions correctly on all selected alumni.	P

MAA-GUI-010	Verify input validation	1. Login as admin. 2. Navigate to alumni management page. 3. Enter invalid data in editable fields (e.g., incorrect email format, leaving required fields empty).	Invalid input (e.g., invalid email, empty fields)	System displays appropriate error messages for invalid inputs.	System displays appropriate error messages for invalid inputs.	P

2.3.8 F008 - Manage Event (Tester: Azfar)

This feature empowers faculty administrators to create, update, and delete events for the alumni network. Thorough testing ensures proper functionality and a user-friendly experience.

Feature ID	Feature	Risk Level
F008	Manage Event	High

The techniques applied are:

1. Use Case Testing (Black Box - System)
2. GUI Testing (Black Box - System)

Nature of Feature:
The "Manage Event" feature in OAS facilitates event management functionalities, allowing admins to create new events, update existing events with revised details, delete events, and potentially invite specific alumni to events.

Justification:
Use Case Testing is mandatory, ensuring all use cases related to creating, updating, and deleting events, including handling invitations and managing attendees, are tested comprehensively by checking for both the main flow and alternate/exception flows. GUI Testing specifically targets the usability and functionality of the admin interface for managing events. We will test all important GUI elements like forms, calendars, invitation features, and event listings, to provide an effective and user-friendly admin interface.

Use Case Testing (System Testing Level)

Use Case Name	Manage Event
Use Case ID	UC-008
Description	Admin creates, updates, or deletes events and manages invitations.

Actor(s)	Faculty Administrator
Triggering Event	Admin navigates to event management section.
Pre-condition	Admin is logged in.
Post-condition	Event created, updated, deleted, or invitation list modified.
Flow of Events	1. Admin selects an action (Create, Update, Delete). 2. Admin provides event information. 3. Admin saves changes or sends invitations.
Exception Flow-Invalid Input	Display error message, preventing action.

Test Cases using Use Case Testing (System Testing Level)

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
ME-UC-001	Verify creating a new event	1. Login as admin. 2. Navigate to the "Manage Events" page. 3. Enter valid event details. 4. Submit the form.	TCOV-ME-UC-001: Event details	Event is created successfully and displayed in the event listings.	Event was created successfully and displayed in the event listings.	P
ME-UC-002	Verify updating an existing event	1. Login as admin. 2. Navigate to the "Manage Events" page. 3. Select an event. 4. Modify event details.	TCOV-ME-UC-002: Modified details	Event is updated successfully and reflected in the event listings.	Event was updated successfully and reflected in the event listings.	P

		5. Save changes.				
ME-UC-003	Verify deleting an event	1. Login as admin. 2. Navigate to the "Manage Events" page. 3. Select an event to delete. 4. Confirm deletion.	TCOV-ME-UC-003: Event ID	Event is removed successfully from the listings.	Event was removed successfully from the listings	P
ME-UC-004	Verify handling invalid input	1. Login as admin. 2. Navigate to the "Create Event" page. 3. Submit the form with missing fields.	TCOV-ME-UC-004: Invalid input	Error messages are displayed, and the event is not created.	Error messages were displayed, and the event was not created.	P
ME-UC-005	Verify unauthorized access	Attempt to access the "Manage Events" page without logging in.	TCOV-ME-UC-005: Unauthenticated	Access is denied, and the user is redirected to the login page.	Access was denied, and the user was redirected to the login page.	P
ME-UC-006	Verify updating an existing event	1. Login as admin. 2. Navigate to the "Manage Events" page. 3. Select an event. 4. Modify event details. 5. Save changes	TCOV-ME-UC-002: Modified details	Event is updated successfully and reflected in the event listings.	The system failed to update the event and displayed a generic error message.	F
ME-UC-007	Verify creating a duplicate event	1. Login as admin. 2. Navigate to the "Manage Events" page. 3. Create an event with identical details to an existing one. 4. Submit the form.	TCOV-ME-UC-006: Duplicate detail	The system prevents duplicate event creation and displays an error message.	The system allowed duplicate event creation and did not display an error message.	F

GUI Testing (System Testing Level)

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
ME-GUI-001	Verify navigation and layout	1. Login as admin. 2. Navigate to the "Manage Events" page. 3. Verify navigation elements and layout.	TCOV-ME-GUI-001: Not Applicable	Navigation elements are clear, accessible, and logically arranged.	Navigation elements are clear, accessible, and logically arranged.	P
ME-GUI-002	Verify form usability	1. Navigate to the "Create Event" form. 2. Enter valid event details. 3. Submit the form.	TCOV-ME-GUI-002: Event details	Form is intuitive, input validation works, and event is created successfully.	Form is intuitive, input validation works, and event is created successfully.	P
ME-GUI-003	Verify calendar functionality	1. Navigate to the event calendar. 2. Select a date and time. 3. Save changes.	TCOV-ME-GUI-003: Event dates	Calendar allows date/time selection and updates the event successfully.	Calendar allows date/time selection and updates the event successfully.	P
ME-GUI-004	Verify error messages for invalid input	1. Navigate to the "Create Event" form. 2. Enter invalid data. 3. Submit the form.	TCOV-ME-GUI-004: Invalid input	Error messages are displayed clearly near the respective fields.	Error messages are displayed clearly near the respective fields.	P
ME-GUI-005	Verify event listing display	1. Navigate to the "Event Listings" page. 2. Check displayed event details for accuracy.	TCOV-ME-GUI-005: Event details	Event details are displayed accurately and formatted appropriately.	Event details are displayed accurately and formatted appropriately.	P

ME-GUI-006	Verify cross-browser compatibility	1. Open the "Manage Events" page in multiple browsers (e.g., Chrome, Firefox, Edge).	TCOV-ME-GUI-006: Not Applicable	The page renders and functions consistently across all tested browsers.	The page renders and functions consistently across all tested browsers.	P
ME-GUI-007	Verify calendar functionality	1. Navigate to the event calendar. 2. Select a date and time. 3. Save changes.	TCOV-ME-GUI-003: Event dates	Calendar allows date/time selection and updates the event successfully	The calendar failed to save the selected date and displayed an error message.	F
ME-GUI-005	Verify event listing display	1. Navigate to the "Event Listings" page. 2. Check displayed event details for accuracy.	TCOV-ME-GUI-005: Event details	Event details are displayed accurately and formatted appropriately.	Event details were not displayed accurately; the description field was truncated.	F

2.3.8 NF-002 Security - Password Encryption (Tester: Yallini)

This NFR mandates that passwords are encrypted during storage and transmission, protecting sensitive information from unauthorized access.

Requirement ID	Feature	Risk Level
NF-002	Security - Password Encryption	High

The technique that will be applied is as follows:

1. Security Testing (Black Box Testing - System level)

Nature of Requirement:
"Password Encryption" ensures that passwords are securely handled during storage and transmission.

Justification:
Security testing is employed to validate that passwords are not stored or sent in plain text, which is a crucial aspect of user data protection. We will intercept and examine network traffic during registration and login, as well as inspect database entries (if accessible), to verify that passwords are encrypted using strong algorithms. This rigorous approach safeguards sensitive data and enhances user trust.

Security Testing (System Testing Level)

2.3.9 NF-002 - Security - Password Encryption (Tester: Yallini)

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)

SPE-ST-001	Verify password storage	1. Register a new user. 2. Use a suitable database client to inspect the user table. 3. Check if the stored password is hashed.	Registered user password	Password stored as a hash, not plain text (verify in the database if accessible).	Password is stored as a hash, not plain text.	P
SPE-ST-002	Verify password transmission	1. Attempt to login using a registered user account. 2. Intercept network traffic using OWASP ZAP, Burp Suite, or Browser Developer Tools during login. 3. Inspect the HTTPS request to verify password encryption.	Login attempt	Password transmitted securely (HTTPS, encrypted in the request body).	Password is transmitted securely (HTTPS, encrypted in the request body).	P
SPE-ST-003	Verify secure connection	1. Open the login page. 2. Inspect the URL and network traffic using developer tools.	N/A	Connection uses HTTPS protocol ensuring encrypted communication.	Connection uses HTTPS protocol ensuring encrypted communication.	P
SPE-ST-004	Verify HTTPS certificate validity	1. Navigate to the system's login page. 2. Check the SSL/TLS certificate details in the browser.	N/A	Valid and up-to-date SSL/TLS certificate is used.	Valid and up-to-date SSL/TLS certificate is used.	P
SPE-ST-005	Verify session management	1. Login as a registered user. 2. Monitor session cookies in browser developer tools.	Registered user session	Session cookies are marked secure and HttpOnly to prevent access through client-side scripts.	Session cookies are marked secure and HttpOnly.	P

SPE-ST-006	Verify logout mechanism	1. Login as a registered user. 2. Logout and intercept network traffic using security tools.	Registered user session	Session is properly invalidated on logout, and no sensitive data is passed in subsequent requests.	Session is properly invalidated on logout.	P

2.3.10 NF-003 - Usability - Cross-Browser Compatibility (Tester: Azfar)

This NFR guarantees consistent functionality and user experience across different browsers, ensuring wide accessibility and user satisfaction.

Requirement ID	Feature	Risk Level
NF-003	Usability - Cross-Browser Compatibility	Medium

The techniques that will be applied are as follows:

1. Compatibility Testing (Black Box Testing - System Level)
2. GUI Testing (Black Box Testing - System Level)

Nature of Requirement:
"Cross-Browser Compatibility" ensures consistent system behavior and UI across different browsers.

Justification:
Compatibility testing is essential for verifying the system's functionality across different browsers and devices and ensuring that layout, functionality, and performance are consistent. GUI testing on multiple browsers will ensure the UI elements are functioning as expected, and there are no rendering or display issues across these platforms. This ensures alumni can access the platform regardless of their browser choice.

Compatibility Testing (System Testing Level)

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)

CBC-CT-001	Verify core functionality	1. Test key features like login and profile management. 2. Perform these actions on each target browser.	TCOV-CBC-001: Core features	Core features behave consistently on all tested browsers.	Core features behaved consistently on all tested browsers.	P
CBC-CT-002	Verify layout consistency	1. Navigate through pages on all target browsers. 2. Inspect layout and element rendering.	TCOV-CBC-002: Layout inspection	UI layout is consistent across all browsers.	UI layout had alignment issues on Safari, with buttons misaligned.	F
CBC-CT-003	Verify performance consistency	1. Measure page load times across browsers. 2. Compare with acceptable performance thresholds.	TCOV-CBC-003: Performance metrics	Page load times are within acceptable ranges across all browsers.	Page load times exceeded acceptable thresholds on Internet Explorer.	F

GUI Testing (System Testing Level)

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (P/F)
CBC-GUI-001	Verify layout consistency	1. Navigate through pages. 2. Inspect UI elements for proper alignment and visibility.	N/A	Layout is correct and consistent across all tested browsers.	Layout had overlapping elements on Edge.	F
CBC-GUI-002	Verify button functionality	1. Test all buttons (e.g., "Submit", "Cancel") across browsers. 2. Check responsiveness and feedback.	N/A	Buttons are functional, visually consistent, and responsive.	Some buttons failed to respond on Firefox.	F

CBC-GUI-003	Verify form rendering	1. Access forms on multiple pages. 2. Inspect field alignment and labels.	N/A	Forms render properly with aligned fields and labels.	Forms rendered properly with aligned fields and labels.	P
CBC-GUI-004	Verify responsiveness	1. Test UI elements on various screen sizes and devices. 2. Check resizing and alignment.	N/A	UI adjusts correctly to screen sizes and remains responsive.	UI adjusted correctly to screen sizes and remained responsive.	P

2.3.11 NF-004 - Security - Authentication (Tester: Mai)

This NFR focuses on ensuring that only authorized users can access the system using strong authentication mechanisms and policies.

Requirement ID	Feature	Risk Level
NF-004	Security - Authentication	High

The technique that will be applied is as follows:

1. Security Testing (Black Box - System Level)

Nature of Requirement:

"Authentication" ensures system access by preventing unauthorized access attempts and verifies appropriate user roles and permissions.

Justification:
Security testing involves probing for weaknesses in the authentication process and is crucial for verifying that only authorized users with valid credentials can access specific parts of the system. We will simulate various attack vectors, such as brute-force attempts, SQL injection, and session hijacking, to identify vulnerabilities and ensure the system's security.

Security Testing (System Testing Level)

Test Case ID	Test Case	Test Steps	Data Input	Expected Result	Actual Result	Test Status (Pass/Fail)
SA-ST-001	Verify password strength policy enforcement	1. Attempt to register using a weak password (e.g., 12345).	Weak Password: 12345	Registration fails, displaying an error message: "Password must meet complexity requirements."	Registration fails, error message displayed correctly.	Pass
SA-ST-002	Verify login with incorrect credentials	1. Attempt to log in with incorrect username or password.	Username: user1, Password: wrongpass	Login fails, displaying an error message: "Invalid credentials."	Login fails, but error message reveals: "Password incorrect for user1."	Fail
SA-ST-003	Test SQL injection vulnerability in the login page	1. Enter SQL code (' OR 1=1 --) in the username or password field.	SQL Code: ' OR 1=1 --	Login attempt is blocked, with no unauthorized access and no error revealing system behavior.	Login attempt blocked; error reveals: "SQL syntax error in query."	Fail
SA-ST-004	Verify account lockout after multiple failed login attempts	1. Attempt to log in with incorrect credentials 5 times consecutively.	Username: user2, Password: wrongpass	Account is locked after 5 failed attempts, displaying an appropriate message: "Account locked due to failed attempts."	Account locked after 5 attempts; appropriate message displayed.	Pass
SA-ST-005	Test session timeout after inactivity	1. Log in successfully. 2. Stay idle for 15 minutes.	Logged-in user	User session expires after 15 minutes of	User session expired as expected;	Pass

		3. Attempt to perform an action.		inactivity; user is redirected to the login page.	redirected to login page.	
SA-ST-006	Test password reset with an invalid email address	1. Initiate password reset. 2. Enter an email not registered in the system.	Email: fakeemail@example.com	Password reset request is rejected with an appropriate error message.	Password reset request rejected, but error message displays: "Email not found in system."	Fail
SA-ST-007	Verify if passwords are stored securely (e.g., hashed and salted)	1. Access the database directly (simulate penetration). 2. Check the format of stored passwords.	No data	Passwords should be stored as hashed and salted values, not plain text.	Passwords are hashed but lack proper salting for additional security.	Fail
SA-ST-008	Test secure communication for authentication (HTTPS and encryption)	1. Access the login page. 2. Inspect the page using browser developer tools to verify HTTPS and secure cookie attributes.	No data	All communication during login is encrypted, and cookies are flagged as secure and HTTP-only.	HTTPS is enabled, but cookies lack the secure attribute.	Fail
SA-ST-009	Test user account enumeration by observing error messages on failed login	1. Attempt login with a valid username and incorrect password. 2. Attempt login with an invalid username and any password.	Username: validuser, invaliduser	Error messages should not indicate whether the username exists in the system.	Error message reveals valid usernames: "Invalid password for validuser."	Fail
SA-ST-010	Verify the use of CAPTCHA to prevent automated login attempts	1. Attempt to log in multiple times using an automated tool without solving CAPTCHA.	Username: testuser, Password: testpass	Automated login attempts are blocked by CAPTCHA verification.	CAPTCHA is functional; automated attempts are blocked.	Pass

2.3.12 NF-005 - Performance - Page Load Times (Tester: Firdaus)

This NFR aims to ensure the system's responsiveness by measuring and evaluating page load times, contributing to a positive user experience.

Requirement ID	Feature	Risk Level
NF-005	Performance - Page Load Times	Medium

The technique that will be applied is as follows:

1. Performance Testing (Black Box Testing - System Level)

Nature of Requirement:
The system should respond quickly to alumni's interactions and provide feedback immediately.

Justification:
Performance testing, specifically load testing, is essential for ensuring that page load times are within acceptable thresholds and meet performance expectations. By measuring page load times under various load conditions (normal and peak), we can identify potential bottlenecks and ensure the system remains responsive, maintaining an optimal user experience.

Performance Testing (System Testing Level)

Test Case ID	Test Case	Test Steps	Input Data	Expected Result	Actual Result	Test Status (P/F)
TC-PR-ST-001	Verify system reliability during high-volume	1. Configure JMeter to simulate 10,000 concurrent users	Load Testing Simulation Data (10,000 users)	The system processes all requests efficiently without crashes or significant delays.	The system did not process all requests and experienced crashes.	F

	password reset requests.	submitting the "Forgot Password" form.				
TC-PR-ST-002	Verify system stability under resource-constrained conditions during password resets.	1. Configure JMeter to restrict server resources (memory/CPU) to 50% and simulate 1,000 requests.	Resource-Constrained Simulation Data (1,000 requests, 50% resources)	The system remains stable and processes requests without major delays or errors.	The system is unstable and does not process all requests without significant delays or errors.	F
TC-PR-ST-003	Verify password reset email response time under peak usage.	1. Generate 5,000 simultaneous password reset requests using JMeter and measure email delivery time.	Peak Traffic Simulation Data (5,000 requests)	Password reset emails are sent within 5 seconds during peak traffic.	Password reset emails are not sent within 5 seconds during peak traffic.	F
TC-PR-ST-004	Verify system recovery after failure during peak password reset load.	1. Simulate a server crash (JMeter) during 5,000 password reset requests; then restore the server.	Server Recovery Simulation Data (5,000 requests, simulated crash)	The system recovers without data loss, and pending requests are processed upon restoration.	The system recovers with data loss, and pending requests are processed upon restoration.	F
TC-PR-ST-005	Verify system accuracy handling malformed/invalid requests under stress.	1. Simulate 5,000 requests (JMeter) with 50% valid and 50% invalid email inputs.	Mixed Input Data (2,500 Valid, 2,500 Invalid)	The system processes valid requests successfully and displays appropriate errors for invalid requests.	The system did not process valid requests.	F

2.3.13 NF-006 - Reliability - Password Reset (Tester: Kei Kar)

This NFR ensures alumni can reliably reset forgotten passwords, balancing security and user convenience.

Requirement ID	Feature	Priority
NF-006	Reliability - Password Reset	High

The techniques that will be applied are as follows:

1. Stress Testing (Black Box - System)

Nature of Requirement:
The password reset function is designed to provide a reliable and secure way for users to regain access.

Justification:
Reliability testing is crucial for ensuring that the password reset mechanism functions reliably under various conditions, including network interruptions, multiple reset attempts, and invalid inputs. Use case testing will validate password reset mechanism's main success scenario, and alternative and exception flows as described in the corresponding use case. This also checks the password reset mechanism against edge cases. Combining these methods ensures the password reset flow is robust.

Objective:
To ensure FSKTM Online Alumni System can reliably handle extreme conditions, such as processing a high volume of simultaneous password reset requests, while maintaining stability, accuracy, and performance. This testing validates the system's ability to process requests efficiently during peak loads and recover gracefully from potential failures caused by excessive user activity.

Stress Testing (Using JMeter)

Test level: System Testing

Pre-condition: The system is operational and accessible. The password reset functionality is active. JMeter is configured and ready to simulate the required test scenarios.

Post-condition: The system demonstrates reliable performance under stress conditions, maintaining stability, accuracy, and functionality. All tests are completed using JMeter, and the results validate the system's reliability.

Test Case ID	Test Case	Test Steps	Input Data	Expected Result	Actual Result	Test Status (P/F)
TC-PR-ST-001	Verify system reliability when handling a high volume of simultaneous password reset requests.	1. Configure JMeter to simulate 10,000 users submitting the "Forgot Password" form simultaneously.	Load Testing Simulation Data	The system processes all requests efficiently without crashes or delays.	The system did not process all requests and without crashes.	F
TC-PR-ST-002	Verify the system's stability when processing password reset requests under resource-constrained conditions.	1. Configure JMeter to restrict server resources (e.g., memory or CPU) to 50% capacity and simulate 1,000 requests.	Resource-Constrained Simulation Data	The system remains stable and processes all requests without significant delays or errors.	The system is unstable and does not process all requests without significant delays or errors.	**F**
TC-PR-ST-003	Verify system response time for password reset emails under peak usage conditions.	1. Use JMeter to generate 5,000 simultaneous password reset requests and measure email delivery time.	Peak Traffic Simulation Data	Password reset emails are sent within 5 seconds during peak traffic.	Password reset emails are not send within 5 seconds during peak traffic.	**F**
TC-PR-ST-004	Verify the system's ability to recover	1. Simulate a server crash using JMeter	Server Recovery Simulation Data	The system recovers without data loss, and	The system recovers with	**F**

	after a failure during peak password reset request loads.	while processing 5,000 password reset requests and restore the server.		pending requests are processed upon restoration.	data loss, and pending requests are processed upon restoration.	
TC-PR-ST-005	Verify system accuracy in handling malformed or invalid requests under stress.	1. Use JMeter to simulate 5,000 requests with 50% valid and 50% invalid email inputs.	Mixed Input Data (Valid/Invalid)	The system processes valid requests successfully and displays appropriate errors for invalid requests.	The system did not process valid requests.	**F**

Test Procedure Specification

Online Alumni System (OAS)

Version: 1.0.0

Date: 15/01/2025

3.0 Test Procedure Specification

F001 - Register User Account (Testers: Yallini & Firdaus)

This section outlines the test procedures designed to verify the system's functionality based on defined use cases. Use case testing ensures that the system behaves as expected when users interact with it in various ways, fulfilling the intended purpose of each use case. These tests cover both normal and exceptional user interactions to ensure robustness and proper error handling.

Use Case Testing (Yallini)

TP-UC-001 - Successful Registration with Valid Data

Test Procedure ID	TP-UC-001
Objective	Verify the registration functionality of the user account.
Test Cases to be Executed	RU-UC-001, RU-UC-002, RU-UC-003, RU-UC-004, RU-UC-005, RU-UC-006, RU-UC-007, RU-UC-008, RU-UC-009, RU-UC-010, RU-UC-011, RU-UC-012, RU-UC-013, RU-UC-014, RU-UC-015
Set Up	// Selenium Script package compatibilityTest; import org.openqa.selenium.By; import org.openqa.selenium.WebDriver; import org.openqa.selenium.WebElement; import org.openqa.selenium.chrome.ChromeDriver; import org.openqa.selenium.chrome.ChromeOptions; import org.openqa.selenium.support.ui.ExpectedConditions; import org.openqa.selenium.support.ui.WebDriverWait; import io.github.bonigarcia.wdm.WebDriverManager; import java.time.Duration; import java.util.HashMap; public class RegistrationTest { public static void main(String[] args) { // Setup ChromeDriver using WebDriverManager WebDriverManager.chromedriver().setup(); // Set Chrome options ChromeOptions options = new ChromeOptions(); options.addArguments("--disable-popup-blocking");

	<pre> WebDriver driver = new ChromeDriver(options); try { // TP-UC-001: Verify successful registration flow // Navigate to registration page driver.get("http://example.com/registration"); // Enter valid data in all required fields driver.findElement(By.id("name")).sendKeys("John Doe"); driver.findElement(By.id("email")).sendKeys("johndoe@example.com"); driver.findElement(By.id("password")).sendKeys("Password123!"); driver.findElement(By.id("confirmPassword")).sendKeys("Password123!"); // Complete CAPTCHA if needed // driver.findElement(By.id("captcha")).sendKeys("captcha_value"); // Click "Register" driver.findElement(By.id("registerButton")).click(); // Wait and verify account creation WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10)); WebElement message = wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("confirmationMes sage"))); System.out.println("Confirmation Message: " + message.getText()); // Capture the output for verification // Add more test cases execution here in a similar approach // RU-UC-002: Verify invalid email format handling // RU-UC-003: Verify password strength requirements // RU-UC-004: Verify duplicate username handling // RU-UC-005: Verify handling of missing fields // ... } catch (Exception e) { e.printStackTrace(); } finally { // Close the browser driver.quit(); } } }</pre>

Boundary Value Analysis (System Testing Level)

<table>
<tr><td>Test Procedure ID</td><td>TP-BVA-001</td></tr>
<tr><td>Objective</td><td>Verify the boundary value analysis functionality for user registration, including name length, password length, and graduation year.</td></tr>
<tr><td>Test Cases to be Executed</td><td>RU-BVA-001, RU-BVA-002, RU-BVA-003, RU-BVA-004, RU-BVA-005, RU-BVA-006, RU-BVA-007, RU-BVA-008, RU-BVA-009, RU-BVA-010, RU-BVA-011</td></tr>
<tr><td>Set Up</td><td>Selenium Script
<pre>
package boundaryValueAnalysisTest;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.chrome.ChromeOptions;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.WebDriverWait;
import io.github.bonigarcia.wdm.WebDriverManager;
import java.time.Duration;

public class RegistrationBoundaryTest {
    public static void main(String[] args) {
        // Setup ChromeDriver using WebDriverManager
        WebDriverManager.chromedriver().setup();

        // Set Chrome options
        ChromeOptions options = new ChromeOptions();
        options.addArguments("--disable-popup-blocking");
        WebDriver driver = new ChromeDriver(options);

        try {
            // Test Case: RU-BVA-001 (Minimum/Maximum Name Length)
            driver.get("http://example.com/registration");
            driver.findElement(By.id("name")).sendKeys("A");
            driver.findElement(By.id("email")).sendKeys("email@example.com");
            driver.findElement(By.id("password")).sendKeys("Password123!");
            driver.findElement(By.id("confirmPassword")).sendKeys("Password123!");
            driver.findElement(By.id("registerButton")).click();

            // Wait and verify
            WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10));
</pre></td></tr>
</table>

	`` WebElement message = wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("nameErrorMessage"))) ; System.out.println("Name Error Message (Min): " + message.getText()); driver.navigate().refresh(); driver.findElement(By.id("name")).sendKeys("A".repeat(51)); driver.findElement(By.id("email")).sendKeys("email@example.com"); driver.findElement(By.id("password")).sendKeys("Password123!"); driver.findElement(By.id("confirmPassword")).sendKeys("Password123!"); driver.findElement(By.id("registerButton")).click(); WebElement maxMessage = wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("nameErrorMessage"))) ; System.out.println("Name Error Message (Max): " + maxMessage.getText()); // RU-BVA-002: Minimum/Maximum Password Length driver.navigate().refresh(); driver.findElement(By.id("name")).sendKeys("John Doe"); driver.findElement(By.id("email")).sendKeys("email@example.com"); driver.findElement(By.id("password")).sendKeys("P@sswrd8"); driver.findElement(By.id("confirmPassword")).sendKeys("P@sswrd8"); driver.findElement(By.id("registerButton")).click(); WebElement passwordSuccessMessage = wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("confirmationMessage"))); System.out.println("Password Success Message: " + passwordSuccessMessage.getText()); driver.navigate().refresh(); driver.findElement(By.id("name")).sendKeys("John Doe"); driver.findElement(By.id("email")).sendKeys("email@example.com"); driver.findElement(By.id("password")).sendKeys("P@sswrd"); driver.findElement(By.id("confirmPassword")).sendKeys("P@sswrd"); driver.findElement(By.id("registerButton")).click(); WebElement passwordErrorMessage = wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("passwordErrorMessag e"))); System.out.println("Password Error Message: " + passwordErrorMessage.getText()); // RU-BVA-003: Invalid Graduation Year driver.navigate().refresh(); driver.findElement(By.id("name")).sendKeys("John Doe"); driver.findElement(By.id("email")).sendKeys("email@example.com"); driver.findElement(By.id("password")).sendKeys("Password123!"); driver.findElement(By.id("confirmPassword")).sendKeys("Password123!"); ```

	`` driver.findElement(By.id("graduationYear")).sendKeys("1800"); // Invalid year driver.findElement(By.id("registerButton")).click(); WebElement graduationYearErrorMessage = wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("graduationYearErrorMessage"))); System.out.println("Graduation Year Error Message: " + graduationYearErrorMessage.getText()); // Add more test cases execution here in a similar approach // RU-BVA-004: Minimum Name Length // RU-BVA-005: Just Below Minimum Name Length // RU-BVA-006: Maximum Name Length // RU-BVA-007: Just Above Maximum Name Length // RU-BVA-008: Minimum Password Length // RU-BVA-009: Just Below Minimum Password Length // RU-BVA-010: Maximum Graduation Year // RU-BVA-011: Just Above Maximum Graduation Year } catch (Exception e) { e.printStackTrace(); } finally { // Close the browser driver.quit(); } } } ```

Equivalence Partitioning (Firdaus)

TP-RU-EP-001: Valid Email Handling

Test Procedure ID	TP-RU-EP-001
Objective	To verify the system correctly accepts valid email addresses from various common and unusual domains.
Test Cases to be Executed	RU-EP-001, RU-EP-002
Set Up	1. Access the registration page. 2. Prepare a list of valid email addresses using common and unusual domains.
Wrap Up	None

TP-RU-EP-002: Invalid Email Handling

Test Procedure ID	TP-RU-EP-002
Objective	To verify that the system correctly rejects invalid email addresses.
Test Cases to be Executed	RU-EP-003, RU-EP-004, RU-EP-005, RU-EP-006
Set Up	1. Access the registration page. 2. Prepare a set of test emails, each with an invalid format, such as test.gmail.com, test@ , test!@gmail.com, and extremely long email address exceeding the character limit
Wrap Up	None

TP-RU-EP-003: Valid/Invalid Password Handling

Test Procedure ID	TP-RU-EP-003
Objective	To verify the system handles valid (strong) and invalid (weak, incorrect format, excessive length) passwords according to defined criteria.
Test Cases to be Executed	RU-EP-007, RU-EP-008, RU-EP-009, RU-EP-010
Set Up	1. Access the registration page. 2. Ensure you are familiar with the password requirements (length, complexity, allowed characters).
Wrap Up	None

Error Guessing (Firdaus)

TP-RU-EG-001: Password Confirmation Mismatch

Test Procedure ID	TP-RU-EG-001
Objective	To confirm that the system correctly detects and reports password confirmation mismatches, preventing registration with inconsistent passwords.
Test Cases to be Executed	RU-EG-001

Set Up	1. Access the registration page. 2. Prepare two different passwords – one for the "Password" field and a different one for the "Confirm Password" field.
Wrap Up	None

TP-RU-EG-002: Empty Password Field

Test Procedure ID	TP-RU-EG-002
Objective	To verify that the system correctly handles and reports an empty password field, enforcing the requirement for a password.
Test Cases to be Executed	RU-EG-002
Set Up	1. Access the registration page.
Wrap Up	None

TP-RU-EG-003: Empty Email Field

Test Procedure ID	TP-RU-EG-003
Objective	To ensure the system requires a valid email address.
Test Cases to be Executed	RU-EG-003
Set Up	Access the registration page.
Wrap Up	None

TP-RU-EG-004: Optional Field - Invalid Input

Test Procedure ID	TP-RU-EG-004
Objective	Check if invalid character/format is ignored, displayed as a warning, and still proceeds with registration.
Test Cases to be Executed	RU-EG-004
Set Up	Access the registration page. Ensure the application has optional input fields.

Wrap Up	None

TP-RU-EG-005: Unresponsive Captcha

Test Procedure ID	TP-RU-EG-005
Objective	To verify appropriate feedback is provided and registration is blocked if the Captcha is unresponsive.
Test Cases to be Executed	RU-EG-005
Set Up	1. Access the registration page. 2. If necessary, simulate network conditions or use browser developer tools to slow down or block Captcha loading.
Wrap Up	Reset network settings if modified.

TP-RU-EG-006: Partially Filled Forms - Data Loss

Test Procedure ID	TP-RU-EG-006
Objective	Verify data is retained or not when navigating away and returning.
Test Cases to be Executed	RU-EG-006
Set Up	1. Access the registration page.
Wrap Up	None

TP-RU-EG-007: Copy/Paste Behavior

Test Procedure ID	TP-RU-EG-007
Objective	To ensure the application handles pasted input correctly and prevents vulnerabilities
Test Cases to be Executed	RU-EG-007
Set Up	Access the registration page. Have sample valid and invalid data prepared for copy-pasting.

Wrap Up	None

TP-RU-EG-008: Rapid Form Submission

Test Procedure ID	TP-RU-EG-008
Objective	To verify correct handling of rapid submissions and prevent duplicate registrations.
Test Cases to be Executed	RU-EG-008
Set Up	Access the registration page. Have valid registration data prepared.
Wrap Up	Check the database or admin panel to confirm only one account was created.

F002 - Log In (Tester: Mai)

This section details the test procedures for the user login functionality (F002), a crucial component of the OAS. These tests cover various scenarios, including successful login with valid credentials, handling invalid inputs, error message validation, and security considerations. Rigorous testing of the login process is essential to protect against unauthorized access and ensure a smooth user experience.

Test Cases using Use Case Testing (System Testing Level)

TP-LI-UC-001 - Successful Login with Valid Credentials

Test Procedure ID	TP- LI-UC-001
Objective	To verify that a user can log in with valid credentials.
Test Cases to be Executed	LI-UC-001
Set Up	Ensure the FSKTM Online Alumni System login page is accessible. Confirm the valid email and password are created in the database.
Wrap Up	None

TP-LI-UC-002 - Invalid Password Handling

Test Procedure ID	TP- LI-UC-002
Objective	To verify the system displays an error message when a valid email is used with an invalid password.
Test Cases to be Executed	LI-UC-002
Set Up	Ensure the login page is accessible. Confirm the email exists in the database but use an incorrect password.
Wrap Up	None

TP-LI-UC-003 - Invalid Email Format Handling

Test Procedure ID	TP- LI-UC-003
Objective	To validate that the system displays an error message for invalid email format.

Test Cases to be Executed	LI-UC-003
Set Up	Ensure the login page is accessible. Prepare an invalid email format and any password.
Wrap Up	None

TP-LI-UC-004 - Unregistered Email Handling

Test Procedure ID	TP- LI-UC-004
Objective	To ensure the system displays an error message when an unregistered email is used.
Test Cases to be Executed	LI-UC-004
Set Up	Ensure the login page is accessible. Confirm the email is not registered in the database.
Wrap Up	None

TP-LI-UC-005 - Empty Credentials Handling

Test Procedure ID	TP- LI-UC-005
Objective	To verify that leaving both email and password fields empty triggers an error message.
Test Cases to be Executed	LI-UC-005
Set Up	Ensure the login page is accessible. Leave the email and password fields blank.
Wrap Up	None

TP-LI-UC-006 - Password Reset Functionality

Test Procedure ID	TP- LI-UC-006
Objective	To test the password reset functionality with a valid email.
Test Cases to be Executed	LI-UC-006

Set Up	Ensure the login page is accessible. Confirm the email exists in the database.
Wrap Up	None

TP-LI-UC-007 - Deleted Account Login Prevention

Test Procedure ID	TP- LI-UC-007
Objective	To ensure that attempting to log in with credentials for a deleted account shows an appropriate error message.
Test Cases to be Executed	LI-UC-007
Set Up	Ensure the account used is marked as deleted in the database. Access the login page.
Wrap Up	None

TP-LI-UC-008 - Unverified Account Login Prevention

Test Procedure ID	TP- LI-UC-008
Objective	To verify that the system prevents login for unverified accounts.
Test Cases to be Executed	LI-UC-008
Set Up	Ensure the account status is set as unverified in the database. Access the login page.
Wrap Up	None

TP-LI-UC-009 - SQL Injection Prevention

Test Procedure ID	TP- LI-UC-009
Objective	To validate that SQL injection attempts are blocked and do not bypass authentication.
Test Cases to be Executed	LI-UC-009

Set Up	Access the login page. Use an SQL injection string in the email field.
Wrap Up	None

TP-LI-UC-010 - Session Timeout Verification

Test Procedure ID	TP- LI-UC-010
Objective	To verify that the system logs the user out after 15 minutes of inactivity.
Test Cases to be Executed	LI-UC-010
Set Up	Log in to the system with valid credentials. Wait for 15 minutes without any interaction.
Wrap Up	None

State Transition Testing (System Testing Level)

TP-LI-STS-001 - Invalid Email State Transition

Test Procedure ID	TP- LI - STS – 001
Objective	Verify that an appropriate error message is displayed for invalid emails.
Test Cases to be Executed	Input an invalid email and verify the error message displayed.
Set Up	Access the login page and ensure no active session. Prepare invalid email test data.
Wrap Up	Log out if logged in accidentally. Record test results. Clean up test environment.

TP-LI-STS-002 - Valid Email State Transition

Test Procedure ID	TP- LI - STS – 002
Objective	Verify the system transitions correctly on valid email input.
Test Cases to be Executed	Input a valid email and verify transition to password validation.
Set Up	Access the login page and ensure no active session. Prepare valid email test data.
Wrap Up	Log out if logged in accidentally. Record test results. Clean up test environment.

TP-LI-STS-003 - Invalid Password State Transition

Test Procedure ID	TP- LI - STS – 003
Objective	Verify the system transitions correctly on valid email input.
Test Cases to be Executed	Input a valid email and an invalid password, and verify the error message.

Set Up	Access the login page and ensure no active session. Prepare valid email and invalid password test data.
Wrap Up	Log out if logged in accidentally. Record test results. Clean up test environment.

1. **Error Guessing (System Testing Level)**

TP-LI-EG-001 - Invalid Email Format Detection

Test Procedure ID	TP- LI-EG-001
Objective	Verify that the system detects invalid email formats and displays an appropriate error message.
Test Cases to be Executed	LI-EG-001
Set Up	Open the browser, navigate to the login page, and ensure the system is operational.
Wrap Up	None

TP-LI-EG-002 - Handling Extremely Long Passwords

Test Procedure ID	TP- LI-EG-002
Objective	Check if the system handles extremely long passwords without crashing and displays a relevant error message.
Test Cases to be Executed	LI-EG-002
Set Up	Open the browser, navigate to the login page, and ensure the system is operational.
Wrap Up	None

TP-LI-EG-003 - Email Address Trimming

Test Procedure ID	TP- LI-EG-003
Objective	Verify that leading or trailing spaces in email addresses are trimmed before processing.
Test Cases to be Executed	LI-EG-003
Set Up	Open the browser, navigate to the login page, and ensure the system is operational.
Wrap Up	None

TP-LI-EG-004 - Password Special Character Support

Test Procedure ID	TP- LI-EG-004
Objective	Validate that the system supports passwords containing special characters.
Test Cases to be Executed	LI-EG-004

Set Up	Open the browser, navigate to the login page, and ensure the system is operational.
Wrap Up	None

TP-LI-EG-005 - Email Address Case Sensitivity

Test Procedure ID	TP- LI-EG-005
Objective	Test whether the login functionality is case-sensitive for email addresses
Test Cases to be Executed	LI-EG-005
Set Up	Open the browser, navigate to the login page, and ensure the system is operational.
Wrap Up	None

TP-LI-EG-006 - Multiple Simultaneous Logins

Test Procedure ID	TP- LI-EG-006
Objective	Verify that multiple users can log in simultaneously without system slowdowns or crashes.
Test Cases to be Executed	LI-EG-006
Set Up	Prepare multiple devices or users to initiate login attempts simultaneously.
Wrap Up	None

TP-LI-EG-007 - Login During Maintenance Mode

Test Procedure ID	TP- LI-EG-007
Objective	Confirm that the system prevents login during maintenance and displays a maintenance notice.
Test Cases to be Executed	LI-EG-007

Set Up	Schedule the system to enter maintenance mode and navigate to the login page.
Wrap Up	None

TP-LI-EG-008 - Password Reset Request Limiting

Test Procedure ID	TP- LI-EG-008
Objective	Ensure the system restricts rapid successive password reset email requests to prevent spam.
Test Cases to be Executed	LI-EG-008
Set Up	Open the "Forgot Password" page and ensure a valid email account is available for testing.
Wrap Up	None

TP-LI-EG-009 - Session Expiration Redirect

Test Procedure ID	TP- LI-EG-009
Objective	Verify that the system redirects users to the login page after their session expires.
Test Cases to be Executed	LI-EG-009
Set Up	Log in and keep the session idle until it expires.
Wrap Up	None

TP-LI-EG-010 - Invalid Email Domain Handling

Test Procedure ID	TP- LI-EG-010
Objective	Validate that the system prevents login attempts using email addresses with invalid domains.
Test Cases to be Executed	LI-EG-010

Set Up	Open the browser, navigate to the login page, and ensure the system is operational.
Wrap Up	None

F003 - Manage User Profile (Tester: Kei Kar)

This section covers the test procedures for managing user profiles (F003) within the OAS. These procedures verify the functionality for editing profile information, changing passwords, and deleting accounts, ensuring data integrity, security, and proper user interface functionality.

Use Case Testing

TP-MUP-UC-001 - Editing Profile with Valid Data

Test Procedure ID	TP-MUP-UC-001
Objective	Verify editing profile with valid data.
Test Cases to be Executed	TC-MUP-UC-001
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page from the main dashboard. 3. Click on the "Settings" dropdown menu and select "Edit Profile". 4. Verify that the system navigates to the "Edit My Profile" page. 5. Ensure the biography field is editable before proceeding.
Wrap Up	None

TP-MUP-UC-002 - Password Change Process

Test Procedure ID	TP-MUP-UC-002
Objective	Verify the password change process updates the password correctly.
Test Cases to be Executed	TC-MUP-UC-002
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page from the main dashboard. 3. Click on the "Settings" dropdown menu and select "Change Password". 4. Verify that the system displays fields to input the current password, new password, and confirmation password. 5. Ensure all fields are visible and functional before proceeding.
Wrap Up	None

TP-MUP-UC-003 - Account Deletion Process

Test Procedure ID	TP-MUP-UC-003
Objective	Verify deleting account with user confirmation.
Test Cases to be Executed	TC-MUP-UC-003
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page from the main dashboard. 3. Click on the "Settings" dropdown menu and select "Delete Account". 4. Verify that the system displays a confirmation prompt for account deletion. 5. Ensure the prompt requires password input for confirmation.
Wrap Up	None

TP-MUP-UC-004 - Profile Editing Cancellation

Test Procedure ID	TP-MUP-UC-004
Objective	Verify the system handles cancellation during the profile editing process.
Test Cases to be Executed	TC-MUP-UC-004

Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page from the main dashboard. 3. Click on the "Settings" dropdown menu and select "Edit Profile". 4. Verify that the "Edit My Profile" page is displayed with editable fields. 5. Ensure the biography field is editable and ready for input.
Wrap Up	None

TP-MUP-UC-005 - Password Change Cancellation

Test Procedure ID	TP-MUP-UC-005
Objective	Verify the system handles cancellation during the password change process.
Test Cases to be Executed	TC-MUP-UC-005
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page from the main dashboard. 3. Click on the "Settings" dropdown menu and select "Change Password". 4. Verify that the system displays fields for entering the current password, new password, and confirmation password. 5. Ensure all fields are visible and functional for input before proceeding.
Wrap Up	None

TP-MUP-UC-006 - Account Deletion Cancellation

Test Procedure ID	TP-MUP-UC-006
Objective	Verify the system handles cancellation during the account deletion process.
Test Cases to be Executed	TC-MUP-UC-006
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page from the main dashboard. 3. Click on the "Settings" dropdown menu and select "Delete Account". 4. Verify that the system displays a confirmation prompt for account deletion. 5. Ensure the password input field is visible and functional

	before proceeding.
Wrap Up	None

Error Guessing

TP-MUP-EG-001 - Empty Password Fields Handling

Test Procedure ID	TP-MUP-EG-001
Objective	Verify the system displays an error when the password fields are left empty.
Test Cases to be Executed	TC-MUP-EG-001
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page. 3. Click the "Settings" dropdown menu. 4. Select "Change Password". 5. Leave all password fields empty.
Wrap Up	None

TP-MUP-EG-002 - Incorrect Current Password Handling

Test Procedure ID	TP-MUP-EG-002
Objective	Verify the system handles an incorrect current password during the password change process.
Test Cases to be Executed	TC-MUP-EG-002
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page. 3. Click the "Settings" dropdown menu. 4. Select "Change Password". 5. Enter an incorrect current password. 6. Enter and confirm a valid new password.
Wrap Up	None

TP-MUP-EG-003 - Incorrect Password for Account Deletion

Test Procedure ID	TP-MUP-EG-003
Objective	Verify the system handles an incorrect password during the

	account deletion process.
Test Cases to be Executed	TC-MUP-EG-003
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page. 3. Click the "Settings" dropdown menu. 4. Select "Delete Account". 5. Enter an incorrect password in the prompt.
Wrap Up	None

TP-MUP-EG-004 - Invalid New Password Input

Test Procedure ID	TP-MUP-EG-004
Objective	Verify the system handles invalid new password input during the password change process.
Test Cases to be Executed	TC-MUP-EG-004
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page. 3. Click the "Settings" dropdown menu. 4. Select "Change Password". 5. Enter the current password. 6. Enter a new password that is less than 5 or more than 20 characters. 7. Confirm the invalid password.
Wrap Up	None

TP-MUP-EG-005 - Mismatched Confirmation Passwords

Test Procedure ID	TP-MUP-EG-005
Objective	Verify the system handles mismatched confirmation passwords during the password change process.
Test Cases to be Executed	TC-MUP-EG-005
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page. 3. Click the "Settings" dropdown menu. 4. Select "Change Password". 5. Enter the current password. 6. Enter a new password. 7. Enter a mismatched confirmation password.
Wrap Up	None

GUI Testing

TP-MUP-GUI-001 - Navigation to "My Profile" Page

Test Procedure ID	TP-MUP-GUI-001
Objective	Verify navigation to the "My Profile" page.
Test Cases to be Executed	TC-MUP-GUI-001
Set Up	1. Log into the system with valid credentials. 2. Click the "My Profile" button in the navigation bar. 3. Verify that the "My Profile" page loads and displays personal information fields clearly.
Wrap Up	None

TP-MUP-GUI-002 - Layout and Alignment of "My Profile" Page

Test Procedure ID	TP-MUP-GUI-002
Objective	Verify the layout and alignment of the "My Profile" page.
Test Cases to be Executed	TC-MUP-GUI-002
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page. 3. Inspect the alignment and visibility of all personal information fields to ensure there is no overlap or misalignment.
Wrap Up	None

TP-MUP-GUI-003 - "Settings" Dropdown Options

Test Procedure ID	TP-MUP-GUI-003
Objective	Verify that the "Settings" dropdown displays all options.
Test Cases to be Executed	TC-MUP-GUI-003
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page. 3. Click the "Settings" dropdown button and verify that "Edit Profile," "Change Password," and "Delete Account" options are displayed.personal information fields to ensure there is no

	overlap or misalignment.
Wrap Up	None

TP-MUP-GUI-004 - Navigation to "Edit My Profile" Page

Test Procedure ID	TP-MUP-GUI-004
Objective	Verify navigation to the "Edit My Profile" page.
Test Cases to be Executed	TC-MUP-GUI-004
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page. 3. Click the "Settings" dropdown menu and select "Edit Profile." Ensure that the "Edit My Profile" page loads successfully.
Wrap Up	None

TP-MUP-GUI-005 - "Edit My Profile" Form Validation

Test Procedure ID	TP-MUP-GUI-005
Objective	Verify the "Edit My Profile" page form validation.
Test Cases to be Executed	TC-MUP-GUI-005
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "Edit My Profile" page via the "Settings" dropdown menu. 3. Leave the biography field empty and click "Save." Verify the error message: "Please fill out this field."
Wrap Up	None

TP-MUP-GUI-006 - "Change Password" Feature

Test Procedure ID	TP-MUP-GUI-006
Objective	Verify the "Change Password" feature.
Test Cases to be Executed	TC-MUP-GUI-006
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page. 3. Select "Change Password" from the "Settings" dropdown menu. 4. Enter the current password and mismatched confirmation password.

	5. Click "Confirm" and verify the error message: "Confirmation password does not match."
Wrap Up	None

TP-MUP-GUI-007 - "Delete Account" Confirmation

Test Procedure ID	TP-MUP-GUI-007
Objective	Verify the "Delete Account" confirmation.
Test Cases to be Executed	TC-MUP-GUI-007
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "My Profile" page. 3. Select "Change Password" from the "Settings" dropdown menu. 4. Enter the current password and mismatched confirmation password. 5. Click "Confirm" and verify the error message: "Confirmation password does not match."
Wrap Up	None

F004 - View Events (Tester: Kei Kar)

This section outlines the test procedures for viewing events (F004) within the OAS. These tests verify the navigation to the events page, the functionality of the search bar, and the accurate display of event details, including considerations for error handling and user interface testing.

Use Case Testing

TP-VE-UC-001 - Navigation to "Events" Page

Test Procedure ID	TP-VE-UC-001
Objective	Verify navigation to the "Events" page.
Test Cases to be Executed	TC-VE-UC-001
Set Up	1. Log into the system with valid credentials. 2. Navigate to the dashboard. 3. Click the "All Events" button in the navigation bar or the "View More" button on the home page. 4. Verify the system navigates to the "Events" page and displays a list of events.
Wrap Up	None

TP-VE-UC-002 - Event Search Functionality

Test Procedure ID	TP-VE-UC-002
Objective	Verify event search functionality.
Test Cases to be Executed	TC-VE-UC-002
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "Events" page. 3. Locate the search bar. 4. Enter a valid keyword, such as "Machine Learning," and click the search icon. 5. Repeat with an invalid keyword, such as "Hello." 6. Verify the system displays a filtered list for valid keywords and no results for invalid ones.
Wrap Up	None

TP-VE-UC-003 - Event Details Display

Test Procedure ID	TP-VE-UC-003
Objective	Verify event details are displayed correctly.
Test Cases to be Executed	TC-VE-UC-003
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "Events" page. 3. Select an event titled "Machine Learning Workshop" from the list. 4. Verify the system navigates to the event details page and displays the title, date, time, location, and description.
Wrap Up	None

Error Guessing

TP-VE-GUI-001 - Navigation via "All Events" Button

Test Procedure ID	TP-VE-UC-003
Objective	Verify event details are displayed correctly.
Test Cases to be Executed	TC-VE-UC-003
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "Events" page. 3. Select an event titled "Machine Learning Workshop" from the list. 4. Verify the system navigates to the event details page and displays the title, date, time, location, and description.
Wrap Up	None

GUI Testing

TP-VE-GUI-001 - Navigation via "All Events" Button

Test Procedure ID	TP-VE-GUI-001
Objective	Verify that the "All Events" button navigates to the "Events" page.
Test Cases to be Executed	TC-VE-GUI-001
Set Up	1. Log into the system with valid credentials. 2. Navigate to the dashboard. 3. Click the "All Events" button in the navigation bar. 4. Verify that the system navigates to the "Events" page and displays a list of events.
Wrap Up	None

TP-VE-GUI-002 - Layout and Design of "Events" Page

Test Procedure ID	TP-VE-GUI-002
Objective	Verify that the layout and design of the "Events" page meet the requirements.
Test Cases to be Executed	TC-VE-GUI-002

Set Up	1. Log into the system with valid credentials. 2. Navigate to the "Events" page. 3. Inspect the event list to ensure proper alignment, spacing, and readability of the content.
Wrap Up	None

TP-VE-GUI-003 - Search Bar Results for Valid Input

Test Procedure ID	TP-VE-GUI-003
Objective	Verify that the search bar displays correct results for valid input.
Test Cases to be Executed	TC-VE-GUI-003
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "Events" page. 3. Enter a valid keyword (e.g., "Machine Learning") in the search bar. 4. Verify that the system displays a filtered list of events matching the keyword.
Wrap Up	None

TP-VE-GUI-004 - Search Bar Handling of Unmatched Keywords

Test Procedure ID	TP-VE-GUI-004
Objective	Verify that the search bar shows an error message for unmatched keywords.
Test Cases to be Executed	TC-VE-GUI-004
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "Events" page. 3. Enter an invalid keyword (e.g., "Photography") in the search bar. 4. Verify that the system displays the error message: "Sorry, no records found."
Wrap Up	None

TP-VE-GUI-005 - Navigation to Event Details Page

Test Procedure ID	TP-VE-GUI-005
Objective	Verify that clicking on an event navigates to its details page.
Test Cases to be Executed	TC-VE-GUI-005
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "Events" page. 3. Click on a specific event titled "Machine Learning Workshop." 4. Verify that the system navigates to the event details page and displays the title, date, and description.
Wrap Up	None

TP-VE-GUI-006 - Responsiveness of "Events" Page

Test Procedure ID	TP-VE-GUI-006
Objective	Verify that the "Events" page is responsive to different screen sizes.
Test Cases to be Executed	TC-VE-GUI-006
Set Up	1. Log into the system with valid credentials. 2. Navigate to the "Events" page. 3. Resize the browser window to different screen sizes (Mobile, Tablet, Desktop). 4. Verify that the page layout adjusts appropriately to fit each screen size.
Wrap Up	None

F005 - Manage Job Advertisement (Tester: Azfar)

This section details the test procedures for managing job advertisements in the OAS. These procedures verify the creation, editing, and deletion of job postings, including input validation, error handling, and authorization checks. Effective management of job postings is essential for connecting alumni with relevant opportunities. Prior to execution of the following test procedures, these special requirements must be prepared:

1. The FSKTM Online Alumni System must be set up and running.
2. Valid alumni credentials must be available for login.
3. Existing job advertisements must be present in the database.
4. Browsers (Chrome, Firefox, Safari, Edge) should be installed for cross-browser testing.

Test Procedure ID	**Objective**	**Test Cases to be Executed**	**Set Up**	**Wrap Up**
TP-F005-001	Verify the job advertisement creation functionality.	TC-MJA-001	1. Login as alumni.	TP-F005-001
TP-F005-002	Verify the job advertisement editing functionality.	TC-MJA-002	1. Login as alumni.	TP-F005-002
TP-F005-003	Verify the job advertisement deletion functionality.	TC-MJA-003	1. Login as alumni.	TP-F005-003
TP-F005-004	Verify handling of invalid inputs during job creation.	TC-MJA-004	1. Login as alumni.	TP-F005-004
TP-F005-005	Verify unauthorized access prevention.	TC-MJA-005	1. Attempt to access the "Manage Job Ads" page without logging in.	TP-F005-005
TP-F005-006	Verify prevention of duplicate job advertisements.	TC-MJA-006	1. Login as alumni.	TP-F005-006

F006 - Search and View Alumni Profile (Tester: Mai)

This section describes the test procedures for searching and viewing alumni profiles (F006) within the OAS. These tests cover various scenarios, including keyword searches, handling empty searches, partial keyword matches, and special character handling, ensuring accurate and efficient profile retrieval.

1. Test Cases using Use Case Testing (System Testing Level)

TP-VAP-UC-001 - Alumni Profile Search with Valid Keyword

Test Procedure ID	TP- VAP-UC-001
Objective	To verify that the system displays alumni profiles for a valid keyword search.
Test Cases to be Executed	VAP-UC-001
Set Up	1. Ensure the system is up and running. 2. Log in as an alumni user.
Wrap Up	None

TP-VAP-UC-002 - Alumni Profile Search with Invalid Keyword

Test Procedure ID	TP- VAP-UC-002
Objective	To verify that the system displays "No match found" for an invalid keyword.
Test Cases to be Executed	VAP-UC-002
Set Up	1. Ensure the system is up and running. 2. Log in as an alumni user.
Wrap Up	None

TP-VAP-UC-003 - Handling Empty Keyword Searches

Test Procedure ID	TP- VAP-UC-003
Objective	To verify that the system handles empty keyword searches appropriately.
Test Cases to be Executed	VAP-UC-003
Set Up	1. Ensure the system is up and running. 2. Log in as an alumni user.
Wrap Up	None

TP-VAP-UC-004 - Navigation to Profile Details Page

Test Procedure ID	TP- VAP-UC-004
Objective	To verify that clicking on a profile from the search results opens the correct profile details page.
Test Cases to be Executed	VAP-UC-004
Set Up	1. Ensure the system is up and running. 2. Perform a valid search.

Wrap Up	None

TP-VAP-UC-005 - Partial Keyword Search

Test Procedure ID	TP- VAP-UC-005
Objective	To verify that the system displays results for a partial keyword search.
Test Cases to be Executed	VAP-UC-005
Set Up	1. Ensure the system is up and running. 2. Log in as an alumni user.
Wrap Up	None

TP-VAP-UC-006 - Handling Special Characters in Search Query

Test Procedure ID	TP- VAP-UC-006
Objective	To verify that the system handles special characters in the search query without crashing.
Test Cases to be Executed	VAP-UC-006
Set Up	1. Ensure the system is up and running. 2. Log in as an alumni user.
Wrap Up	None

TP-VAP-UC-007 - Mixed-Case Keyword Search

Test Procedure ID	TP- VAP-UC-007
Objective	To verify that the system handles mixed-case keyword searches and displays results correctly.
Test Cases to be Executed	VAP-UC-007
Set Up	1. Ensure the system is up and running. 2. Log in as an alumni user.
Wrap Up	None

TP-VAP-UC-008 - Handling Empty Database

Test Procedure ID	TP- VAP-UC-008
Objective	To verify that the system responds appropriately when the database is empty.
Test Cases to be Executed	VAP-UC-008
Set Up	1. Ensure the system is up and running. 2. Ensure no alumni records exist in the database. 3. Log in as an alumni user.
Wrap Up	None

TP-VAP-UC-009 - Handling Very Long Keywords

Test Procedure ID	TP- VAP-UC-009
Objective	To verify that the system handles very long keywords without performance issues.
Test Cases to be Executed	VAP-UC-009
Set Up	1. Ensure the system is up and running. 2. Log in as an alumni user.
Wrap Up	None

TP-VAP-UC-010 - Handling Keywords with Multiple Spaces

Test Procedure ID	TP- VAP-UC-010
Objective	To verify that the system handles keywords with multiple spaces between words and returns trimmed results.
Test Cases to be Executed	VAP-UC-010
Set Up	1. Ensure the system is up and running. 2. Log in as an alumni user.
Wrap Up	None

2. Performance Testing

TP-PT-UC-001 - Single User Search Performance

Test Procedure ID	TP- PT-UC-001
Objective	Validate single user search performance.
Test Cases to be Executed	PT-UC-001
Set Up	Ensure database has sufficient alumni data, and user access is operational.
Wrap Up	None

TP-PT-UC-002 - Concurrent User Search Performance (100 users)

Test Procedure ID	TP- PT-UC-002
Objective	Test system behavior under 100 concurrent users
Test Cases to be Executed	PT-UC-002
Set Up	Set up load-testing software (e.g., JMeter) to simulate 100 users and populate the database with relevant keywords.
Wrap Up	None

TP-PT-UC-003 - Search Performance with High Database Volume

Test Procedure ID	TP- PT-UC-003

Objective	Assess performance under high database volume
Test Cases to be Executed	PT-UC-003
Set Up	Populate the database with 1,000,000 alumni records and verify system capacity to handle large volumes.
Wrap Up	None

TP-PT-UC-004 - Search Behavior with Invalid Keyword (Performance)

Test Procedure ID	TP- PT-UC-004
Objective	Validate search behavior with invalid keyword
Test Cases to be Executed	PT-UC-004
Set Up	Ensure database is populated and system is live.
Wrap Up	None

TP-PT-UC-005 - Search Performance Under Peak User Load (500 users)

Test Procedure ID	TP- PT-UC-005
Objective	Measure system performance under peak user load
Test Cases to be Executed	PT-UC-005
Set Up	Simulate 500 concurrent users using load-testing software, ensuring mixed valid and invalid keywords.
Wrap Up	None

TP-PT-UC-006 - Stress Test for Extreme User Load (1000 users)

Test Procedure ID	TP- PT-UC-006
Objective	Stress test system for extreme user load
Test Cases to be Executed	PT-UC-006
Set Up	Use load-testing tools to simulate 1,000 users simultaneously performing searches.
Wrap Up	None

TP-PT-UC-007 - Search Response with Special Characters (Performance)

Test Procedure ID	TP- PT-UC-007
Objective	Verify search response with special characters
Test Cases to be Executed	PT-UC-007
Set Up	Confirm database readiness and system availability.
Wrap Up	None

TP-PT-UC-008 - Search Endurance over Continuous Use

Test Procedure ID	TP- PT-UC-008
Objective	Validate endurance over continuous use
Test Cases to be Executed	PT-UC-008
Set Up	Simulate 10 users performing searches every 5 minutes for 24 hours and ensure system monitoring tools are enabled.
Wrap Up	None

TP-PT-UC-009 - Search Behavior with Partial Keywords (Performance)

Test Procedure ID	TP- PT-UC-009
Objective	Evaluate search behavior for partial keyword
Test Cases to be Executed	PT-UC-009
Set Up	Ensure partial keyword matches exist in the database and search functionality is accessible.
Wrap Up	None

TP-PT-UC-010 - Database-Heavy Search Performance

Test Procedure ID	TP- PT-UC-010
Objective	Test database-heavy search performance
Test Cases to be Executed	PT-UC-010
Set Up	Populate the database with entries corresponding to large queries and confirm server readiness.
Wrap Up	None

F007 - Manage Alumni Account (Testers: Firdaus & Yallini)

This section outlines the test procedures for managing alumni accounts by authorized administrators within the OAS. These procedures ensure administrators can effectively manage alumni accounts, including updating information, activating/deactivating accounts, and handling various administrative tasks related to alumni data. Thorough testing of these administrative functions is crucial for maintaining data integrity and system security.

Use Case Testing (Yallini)

Test Procedure ID	TP-UC-002
Objective	Verify the functionality of managing alumni accounts, including approving/rejecting, editing, and deleting accounts, as well as handling unauthorized access and invalid actions.
Test Cases to be Executed	MAA-UC-001, MAA-UC-002, MAA-UC-003, MAA-UC-004, MAA-UC-005, MAA-UC-006
Set Up	Selenium Script package manageAlumniAccountTest; import org.openqa.selenium.By; import org.openqa.selenium.WebDriver; import org.openqa.selenium.WebElement; import org.openqa.selenium.chrome.ChromeDriver; import org.openqa.selenium.chrome.ChromeOptions; import org.openqa.selenium.support.ui.ExpectedConditions; import org.openqa.selenium.support.ui.WebDriverWait; import io.github.bonigarcia.wdm.WebDriverManager; import java.time.Duration; public class ManageAlumniAccountTest { public static void main(String[] args) { // Setup ChromeDriver using WebDriverManager WebDriverManager.chromedriver().setup(); // Set Chrome options ChromeOptions options = new ChromeOptions(); options.addArguments("--disable-popup-blocking"); WebDriver driver = new ChromeDriver(options); try { // Test Case: MAA-UC-001 (Approve Account) driver.get("http://example.com/admin-login"); driver.findElement(By.id("username")).sendKeys("admin");

```
        driver.findElement(By.id("password")).sendKeys("adminPassword");
        driver.findElement(By.id("loginButton")).click();

        // Navigate to alumni management page
        driver.get("http://example.com/alumni-management");

        // Select a pending alumni account and approve it
        driver.findElement(By.id("pendingAccount")).click();
        driver.findElement(By.id("approveButton")).click();

        // Wait and verify
        WebDriverWait wait = new WebDriverWait(driver,
Duration.ofSeconds(10));
        WebElement message =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("statusMessage
")));
        System.out.println("Status Message (Approve): " + message.getText());

        // Refresh for next test case
        driver.navigate().refresh();

        // Test Case: MAA-UC-002 (Reject Account)
        driver.findElement(By.id("pendingAccount")).click();
        driver.findElement(By.id("rejectButton")).click();
        WebElement rejectMessage =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("statusMessage
")));
        System.out.println("Status Message (Reject): " +
rejectMessage.getText());

        // Test Case: MAA-UC-003 (Edit Account)
        driver.navigate().refresh();
        driver.findElement(By.id("approvedAccount")).click();
        driver.findElement(By.id("editButton")).click();
        driver.findElement(By.id("bio")).clear();
        driver.findElement(By.id("bio")).sendKeys("Updated biography");
        driver.findElement(By.id("saveButton")).click();
        WebElement editMessage =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("statusMessage
")));
        System.out.println("Status Message (Edit): " + editMessage.getText());

        // Test Case: MAA-UC-004 (Delete Account)
        driver.navigate().refresh();
        driver.findElement(By.id("approvedAccount")).click();
        driver.findElement(By.id("deleteButton")).click();
        driver.findElement(By.id("confirmDeleteButton")).click();
        WebElement deleteMessage =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("statusMessage
")));
```

	System.out.println("Status Message (Delete): " + deleteMessage.getText()); // Test Case: MAA-UC-005 (Unauthorized Access) driver.get("http://example.com/logout"); // Logout as admin driver.get("http://example.com/alumni-management"); WebElement accessDeniedMessage = wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("accessDeniedMessage"))); System.out.println("Access Denied Message: " + accessDeniedMessage.getText()); // Test Case: MAA-UC-006 (Invalid Action) driver.get("http://example.com/admin-login"); driver.findElement(By.id("username")).sendKeys("admin"); driver.findElement(By.id("password")).sendKeys("adminPassword"); driver.findElement(By.id("loginButton")).click(); driver.get("http://example.com/alumni-management"); driver.findElement(By.id("pendingAccount")).click(); driver.findElement(By.id("editButton")).click(); // Invalid action WebElement invalidActionMessage = wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("invalidActionMessage"))); System.out.println("Invalid Action Message: " + invalidActionMessage.getText()); } catch (Exception e) { e.printStackTrace(); } finally { // Close the browser driver.quit(); } } }

GUI Testing (System Testing Level)

Test Procedure ID	TP-UC-003
Objective	Verify the functionality of the alumni management GUI, including navigation, search functionality, data display, feedback messages, table sorting, pagination controls, responsive design, filter functionality, bulk actions, and input validation.
Test Cases to be Executed	MAA-GUI-001, MAA-GUI-002, MAA-GUI-003, MAA-GUI-004, MAA-GUI-005, MAA-GUI-006, MAA-GUI-007, MAA-GUI-008, MAA-GUI-009, MAA-GUI-010

Set Up	Selenium Script package manageAlumniAccountTest; import org.openqa.selenium.By; import org.openqa.selenium.WebDriver; import org.openqa.selenium.WebElement; import org.openqa.selenium.chrome.ChromeDriver; import org.openqa.selenium.chrome.ChromeOptions; import org.openqa.selenium.support.ui.ExpectedConditions; import org.openqa.selenium.support.ui.WebDriverWait; import io.github.bonigarcia.wdm.WebDriverManager; import java.time.Duration; public class ManageAlumniGUI { public static void main(String[] args) { // Setup ChromeDriver using WebDriverManager WebDriverManager.chromedriver().setup(); // Set Chrome options ChromeOptions options = new ChromeOptions(); options.addArguments("--disable-popup-blocking"); WebDriver driver = new ChromeDriver(options); try { // Test Case: MAA-GUI-001 (Verify navigation and layout) driver.get("http://example.com/admin-login"); driver.findElement(By.id("username")).sendKeys("admin"); driver.findElement(By.id("password")).sendKeys("adminPassword"); driver.findElement(By.id("loginButton")).click(); // Navigate to alumni management page driver.get("http://example.com/alumni-management"); // Verify navigation and layout WebElement navigationElement = driver.findElement(By.id("navigationElement")); System.out.println("Navigation Element Text: " + navigationElement.getText()); // Test Case: MAA-GUI-002 (Verify search functionality) driver.findElement(By.id("searchBar")).sendKeys("John Doe"); driver.findElement(By.id("searchButton")).click(); WebDriverWait wait = new WebDriverWait(driver, Duration.ofSeconds(10)); WebElement searchResult = wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("searchResult"))); System.out.println("Search Result: " + searchResult.getText()); // Test Case: MAA-GUI-003 (Verify data display)

```
        driver.navigate().refresh();
        WebElement alumniData = driver.findElement(By.id("alumniData"));
        System.out.println("Alumni Data: " + alumniData.getText());

        // Test Case: MAA-GUI-004 (Verify feedback messages)
        driver.navigate().refresh();
        driver.findElement(By.id("approveButton")).click();
        WebElement feedbackMessage =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("feedbackMess
age")));
        System.out.println("Feedback Message: " + feedbackMessage.getText());

        // Test Case: MAA-GUI-005 (Verify table sorting functionality)
        driver.navigate().refresh();
        driver.findElement(By.id("sortByName")).click();
        WebElement sortedTable = driver.findElement(By.id("sortedTable"));
        System.out.println("Sorted Table: " + sortedTable.getText());

        // Test Case: MAA-GUI-006 (Verify pagination controls)
        driver.navigate().refresh();
        driver.findElement(By.id("paginationNext")).click();
        WebElement paginatedContent =
driver.findElement(By.id("paginatedContent"));
        System.out.println("Paginated Content: " + paginatedContent.getText());

        // Test Case: MAA-GUI-007 (Verify responsive design)
        driver.navigate().refresh();
        WebElement responsiveLayout =
driver.findElement(By.id("responsiveLayout"));
        System.out.println("Responsive Layout before resize: " +
responsiveLayout.getText());
        driver.manage().window().setSize(new Dimension(800, 600));
        System.out.println("Responsive Layout after resize: " +
responsiveLayout.getText());

        // Test Case: MAA-GUI-008 (Verify filter functionality)
        driver.navigate().refresh();
        driver.findElement(By.id("filterByYear")).click();
        driver.findElement(By.cssSelector("option[value='2025']")).click();
        WebElement filteredList = driver.findElement(By.id("filteredList"));
        System.out.println("Filtered List: " + filteredList.getText());

        // Test Case: MAA-GUI-009 (Verify bulk actions)
        driver.navigate().refresh();
        driver.findElement(By.id("selectMultiple")).click();
        driver.findElement(By.id("bulkApprove")).click();
        WebElement bulkActionResult =
wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("bulkActionRe
sult")));
        System.out.println("Bulk Action Result: " + bulkActionResult.getText());
```

	<pre> // Test Case: MAA-GUI-010 (Verify input validation) driver.navigate().refresh(); driver.findElement(By.id("name")).sendKeys("John Doe"); driver.findElement(By.id("email")).sendKeys("invalid-email-format"); driver.findElement(By.id("saveButton")).click(); WebElement inputErrorMessage = wait.until(ExpectedConditions.visibilityOfElementLocated(By.id("inputErrorMes sage"))); System.out.println("Input Error Message: " + inputErrorMessage.getText()); } catch (Exception e) { e.printStackTrace(); } finally { // Close the browser driver.quit(); } } }</pre>

State Transition Testing (System - Firdaus)

TP-MAA-STT-001: Initial Registration

Test Procedure ID	TP-MAA-STT-001
Objective	To verify that alumni registration successfully creates a pending account.
Test Cases to be Executed	MAA-STT-001
Set Up	Ensure the OAS registration page is accessible.
Wrap Up	None.

TP-MAA-STT-002: Admin Approves Account (Pending to Approved)

Test Procedure ID	TP-MAA-STT-002
Objective	To verify that an admin can approve a pending alumni account, activating it in the system.
Test Cases to be Executed	MAA-STT-002

Set Up	1. Create a pending alumni account (following the steps in TP-MAA-STT-001). 2. Log in as a faculty administrator.
Wrap Up	Log out of the administrator account.

F008 - Manage Event (Tester: Azfar)

This section details the test procedures for managing events within the OAS. These tests cover creating, updating, deleting, and managing various aspects of events, such as registration, attendance tracking, and communication. Comprehensive testing of event management is crucial for the smooth execution of alumni events and maintaining accurate event information. Prior to execution of the following test procedures, these special requirements must be prepared:

1. The FSKTM Online Alumni System must be set up and running.
2. Valid faculty administrator credentials must be available for login.
3. Existing events must be present in the database for testing purposes.
4. Browsers (Chrome, Firefox, Safari, Edge) should be installed for cross-browser testing.

Test Procedure ID	Objective	Test Cases to be Executed	Set Up	Wrap Up
TP-F008-001	Verify the event creation functionality.	ME-UC-001	1. Login as admin.	TP-F008-001
TP-F008-002	Verify the event editing functionality.	ME-UC-002	1. Login as admin.	TP-F008-002
TP-F008-003	Verify the event deletion functionality.	ME-UC-003	1. Login as admin.	TP-F008-003
TP-F008-004	Verify handling of invalid inputs during event creation.	ME-UC-004	1. Login as admin.	TP-F008-004
TP-F008-005	Verify unauthorized access prevention.	ME-UC-005	1. Attempt to access the "Manage Events" page without logging in.	TP-F008-005
TP-F008-006	Verify prevention of duplicate events.	ME-UC-007	1. Login as admin.	TP-F008-006
TP-F008-007	Verify event calendar functionality.	ME-GUI-003	1. Navigate to the "Event Calendar" page.	TP-F008-007
TP-F008-008	Verify cross-browser compatibility.	ME-GUI-006	1. Login as admin.	TP-F008-008

NF-002 - Security - Password Encryption (Tester: Yallini)

This section details the security testing procedures designed to validate the strength and effectiveness of password encryption within the OAS. These tests cover various aspects of password security, including encryption algorithms, storage practices, transmission security, and related security measures. Robust password encryption is paramount for protecting sensitive user data and maintaining the system's overall security posture.

TP-UC-004 - Verify Password Storage Security

Test Procedure ID	TP-UC-004
Objective	Verify the security functionalities of the system, including secure password storage, secure password transmission, secure connection, HTTPS certificate validity, session management, and logout mechanism.
Test Cases to be Executed	SPE-ST-001, SPE-ST-002, SPE-ST-003, SPE-ST-004, SPE-ST-005 SPE-ST-006
Set Up	Ensure the system is running. Ensure that access to necessary tools (e.g., OWASP ZAP, Burp Suite, Browser Developer Tools, and database client) is available. Create a test user account if not already created. Steps for Security Testing Using Tools Tool/Script Setup 1. Ensure that OWASP ZAP or Burp Suite is installed and configured. 2. Ensure that a suitable database client is installed and configured to access the database. 3. Ensure the browser's developer tools are enabled and accessible.

NF-003 - Usability - Cross-Browser Compatibility (Tester: Azfar)

This section outlines the test procedures for cross-browser compatibility, ensuring the OAS functions correctly and consistently across different web browsers. These tests focus on verifying consistent functionality, layout, performance, and user interface elements across various browsers and screen sizes, contributing to a seamless user experience regardless of the user's preferred browser. Prior to execution of the following test procedures, these special requirements must be prepared:

1. The FSKTM Online Alumni System must be set up and running.
2. Browsers (Chrome, Firefox, Safari, Edge) must be installed and configured for testing.
3. Stable internet connectivity is required.
4. Test accounts with valid credentials must be available for login.
5. Performance monitoring tools should be set up to measure load times.

Test Procedure ID	Objective	Test Cases to be Executed	Set Up
TP-NF003-001	Verify consistent functionality across browsers.	CBC-CT-001	1. Test key features like login and profile management.
TP-NF003-002	Verify layout consistency across browsers.	CBC-CT-002	1. Navigate through various pages.
TP-NF003-003	Verify performance consistency across browsers.	CBC-CT-003	1. Measure page load times across browsers using performance monitoring tools.
TP-NF003-004	Verify UI element alignment and visibility.	CBC-GUI-001	1. Navigate through various pages.
TP-NF003-005	Verify button functionality across browsers.	CBC-GUI-002	1. Test all buttons (e.g., Submit, Cancel) on Chrome, Firefox, Safari, and Edge.
TP-NF003-006	Verify form rendering across browsers.	CBC-GUI-003	1. Navigate to forms on multiple pages.
TP-NF003-007	Verify responsiveness across screen sizes.	CBC-GUI-004	1. Test UI elements on various screen sizes and devices.

NF-004 - Security - Authentication (Tester: Mai)

This section details the security testing procedures focused on the authentication mechanisms of the OAS. These tests aim to identify and mitigate potential vulnerabilities related to user authentication, ensuring the system is protected against unauthorized access and maintains the confidentiality of user data.

Security Testing (Penetration Testing)

TP-SA-ST-001 - Password Strength Policy Enforcement

Test Procedure ID	TP- SA-ST-001
Objective	To verify that the system enforces password strength policies during registration.

Test Cases to be Executed	SA-ST-001
Set Up	1. Access the registration page. 2. Ensure no pre-existing account with weak password is registered.
Wrap Up	None

TP-SA-ST-002 - Authentication with Incorrect Credentials

Test Procedure ID	TP- SA-ST-002
Objective	To ensure the system does not authenticate users with incorrect credentials.
Test Cases to be Executed	SA-ST-002
Set Up	1. Prepare valid and invalid test credentials. 2. Ensure access to the login page.
Wrap Up	None

TP-SA-ST-003 - SQL Injection Vulnerability Assessment

Test Procedure ID	TP- SA-ST-003
Objective	To check if the system is vulnerable to SQL injection attacks on the login page.
Test Cases to be Executed	SA-ST-003
Set Up	1. Access the login page. 2. Ensure test environment does not contain sensitive data.
Wrap Up	None

TP-SA-ST-004 - Account Lockout After Multiple Failed Logins

Test Procedure ID	TP- SA-ST-004
Objective	To verify that accounts are locked after multiple consecutive failed login attempts.
Test Cases to be Executed	SA-ST-004
Set Up	1. Prepare an account with valid credentials. 2. Ensure the login page is accessible.
Wrap Up	None

TP-SA-ST-005 - Idle Session Timeout Verification

Test Procedure ID	TP- SA-ST-005
Objective	To ensure the system times out idle sessions for security purposes.
Test Cases to be Executed	SA-ST-005
Set Up	1. Log in to the system with valid credentials. 2. Ensure session timeout settings are configured.
Wrap Up	None

TP-SA-ST-006 - Password Reset with Invalid Email

Test Procedure ID	TP- SA-ST-006
Objective	To verify the password reset functionality when an invalid email is entered.
Test Cases to be Executed	SA-ST-006

Set Up	1. Prepare an unregistered email address. 2. Access the password reset page.
Wrap Up	None

TP-SA-ST-007 - Secure Password Storage Verification

Test Procedure ID	TP- SA-ST-007
Objective	To check if passwords are stored securely (hashed and salted) in the database.
Test Cases to be Executed	SA-ST-007
Set Up	1. Simulate database access in a secure test environment. 2. Inspect stored passwords.
Wrap Up	None

TP-SA-ST-008 - Login Communication Encryption

Test Procedure ID	TP- SA-ST-008
Objective	To verify that all login communications are encrypted (HTTPS and secure cookies).
Test Cases to be Executed	SA-ST-008
Set Up	1. Access the login page via a browser. 2. Enable browser developer tools to inspect requests and cookies.
Wrap Up	None

TP-SA-ST-009 - Username Exposure Prevention

Test Procedure ID	TP- SA-ST-009
Objective	To ensure the system does not expose valid usernames through error messages during login attempts.
Test Cases to be Executed	SA-ST-009
Set Up	1. Prepare valid and invalid usernames for testing. 2. Ensure access to the login page.
Wrap Up	None

TP-SA-ST-010 - CAPTCHA Effectiveness Against Automated Logins

Test Procedure ID	TP- SA-ST-010
Objective	To verify that CAPTCHA prevents automated login attempts.
Test Cases to be Executed	SA-ST-010
Set Up	1. Access the login page. 2. Prepare an automated tool for simulating login attempts.

NR-005 - Performance - Page Load Times (Tester: Firdaus)

This section details the test procedures developed to evaluate the performance of the Online Alumni System (OAS), specifically focusing on page load times. Performance is crucial for ensuring a positive user experience, and slow page loads can lead to frustration and abandonment. Each procedure targets a specific aspect of performance, such as responsiveness under normal and peak loads, stability under resource constraints, and the system's ability to handle invalid requests gracefully. The results of these tests provide valuable insights into the OAS's current performance capabilities and identify areas for potential optimization.

TP-PLT-001 - High Volume Password Reset Request Handling

Test Procedure ID	TP-PLT-001
Objective	Verify system reliability when handling a high volume of simultaneous password reset requests.
Test Cases to be Executed	TC-PLT-001
Set Up	1. Ensure JMeter is installed and configured.2. Set up the "Forgot Password" form with valid test data.3. Configure JMeter to simulate 10,000 simultaneous user requests.
Wrap Up	Analyze JMeter results for average response time, error rate, and throughput. Document findings.

TP-PLT-002 - Password Reset Handling Under Resource Constraints

Test Procedure ID	TP-PLT-002
Objective	Verify the system's stability when processing password reset requests under resource-constrained conditions.
Test Cases to be Executed	TC-PLT-002
Set Up	1. Configure server resources to operate at 50% capacity (e.g., limit memory or CPU usage).2. Use JMeter to simulate 1,000 password reset requests.
Wrap Up	Monitor server resource utilization during the test. Document system stability and error rates.

TP-PLT-003 - Password Reset Email Response Time Under Peak Usage

Test Procedure ID	TP-PLT-003
Objective	Verify system response time for password reset emails under peak usage conditions.
Test Cases to be Executed	TC-PLT-003
Set Up	1. Configure JMeter to simulate 5,000 simultaneous password reset requests.2. Instrument the system to measure email delivery time (e.g., using timestamps in logs or a dedicated monitoring tool).
Wrap Up	Collect and analyze email delivery times. Calculate average, minimum, maximum, and percentile values. Document findings.

TP-PLT-004 - System Recovery After Failure During Peak Password Reset Load

Test Procedure ID	TP-PLT-004
Objective	Verify the system's ability to recover after a failure during peak password reset request loads.
Test Cases to be Executed	TC-PLT-004
Set Up	1. Use JMeter to simulate 5,000 simultaneous password reset requests.2. Implement a mechanism to simulate a server crash (e.g., by terminating a critical process or service).3. Have a procedure ready to restore the server.
Wrap Up	Document the time taken for the system to recover. Verify data integrity and the successful processing of pending requests.

TP-PLT-005 - Handling Malformed/Invalid Password Reset Requests Under Stress

Test Procedure ID	TP-PLT-005
Objective	Verify system accuracy in handling malformed or invalid requests under stress.

Test Cases to be Executed	TC-PLT-005
Set Up	1. Configure JMeter to simulate 5,000 password reset requests.2. Prepare a dataset with 50% valid and 50% invalid email addresses.
Wrap Up	Analyze logs and system responses to confirm appropriate error handling for invalid requests and successful processing of valid requests.

NF-006 - Reliability - Password Reset (Tester: Kei Kar)

This section outlines the test procedures designed to assess the reliability of the password reset functionality within the OAS. Each procedure focuses on a specific aspect of reliability, such as handling high volumes of requests, operating under resource constraints, and recovering from failures.

TP-PR-ST-001 - High Volume Password Reset Request Handling

Test Procedure ID	TP-PR-ST-001
Objective	Verify system reliability when handling a high volume of simultaneous password reset requests.
Test Cases to be Executed	TC-PR-ST-001
Set Up	1. Ensure JMeter is installed and configured. 2. Set up the "Forgot Password" form with valid test data. 3. Configure JMeter to simulate 10,000 simultaneous user requests. 4. Verify the system processes all requests efficiently without crashes or delays.
Wrap Up	None

TP-PR-ST-002 - Password Reset Under Resource Constraints

Test Procedure ID	TP-PR-ST-002
Objective	Verify the system's stability when processing password reset requests under resource-constrained conditions.
Test Cases to be Executed	TC-PR-ST-002
Set Up	1. Configure server resources to operate at 50% capacity (e.g., limit memory or CPU usage). 2. Use JMeter to simulate 1,000 password reset requests. 3. Verify the system remains stable and processes all requests without significant delays or errors.
Wrap Up	None

TP-PR-ST-003 - Password Reset Email Response Time - Peak Usage

Test Procedure ID	TP-PR-ST-003
Objective	Verify system response time for password reset emails under peak usage conditions.
Test Cases to be Executed	TC-PR-ST-003

Set Up	1. Configure JMeter to simulate 5,000 simultaneous password reset requests. 2. Measure email delivery time for all requests. 3. Verify that all password reset emails are sent within 5 seconds during peak traffic.
Wrap Up	None

TP-PR-ST-004 - System Recovery After Failure During Password Reset

Test Procedure ID	TP-PR-ST-004
Objective	Verify the system's ability to recover after a failure during peak password reset request loads.
Test Cases to be Executed	TC-PR-ST-004
Set Up	1. Use JMeter to simulate 5,000 simultaneous password reset requests. 2. Simulate a server crash during request processing. 3. Restore the server and verify recovery without data loss. 4. Ensure pending requests are processed after restoration.
Wrap Up	None

TP-PR-ST-005 - Handling Malformed/Invalid Password Reset Requests

Test Procedure ID	TP-PR-ST-005
Objective	Verify system accuracy in handling malformed or invalid requests under stress.
Test Cases to be Executed	TC-PR-ST-005
Set Up	1. Configure JMeter to simulate 5,000 password reset requests. 2. Use mixed input data with 50% valid and 50% invalid email addresses. 3. Verify the system processes valid requests successfully and displays appropriate error messages for invalid requests.
Wrap Up	None

Model Checking

Online Alumni System (OAS)

Version: 1.0.0
Date: 15/01/2024

4.0 Model Checking Report

This report formally verifies four key system behaviors in the FSKTM Online Alumni System (OAS) using the NuSMV model checker. The state diagrams analyzed are from the Software Brief Description (SBD).

4.1 Introduction to Model Checking

Model checking is an automated technique that helps us verify if the design of our system behaves as expected. We use state diagrams to visualize the system's behavior and temporal logic (CTL) to define the rules the system must follow. We used NuSMV to explore all possible paths through the state diagram and check if any of them violate our rules (properties). This process can uncover design flaws early, before they turn into bugs in the actual code.

4.2 State Diagrams and Properties

Let's look at each important part of the system and its design, represented by a state diagram. For each diagram, we'll define what *shouldn't* happen (safety) and what *should* eventually happen (liveness)

4.2.1 Log In

- **State Diagram (from SBD):**

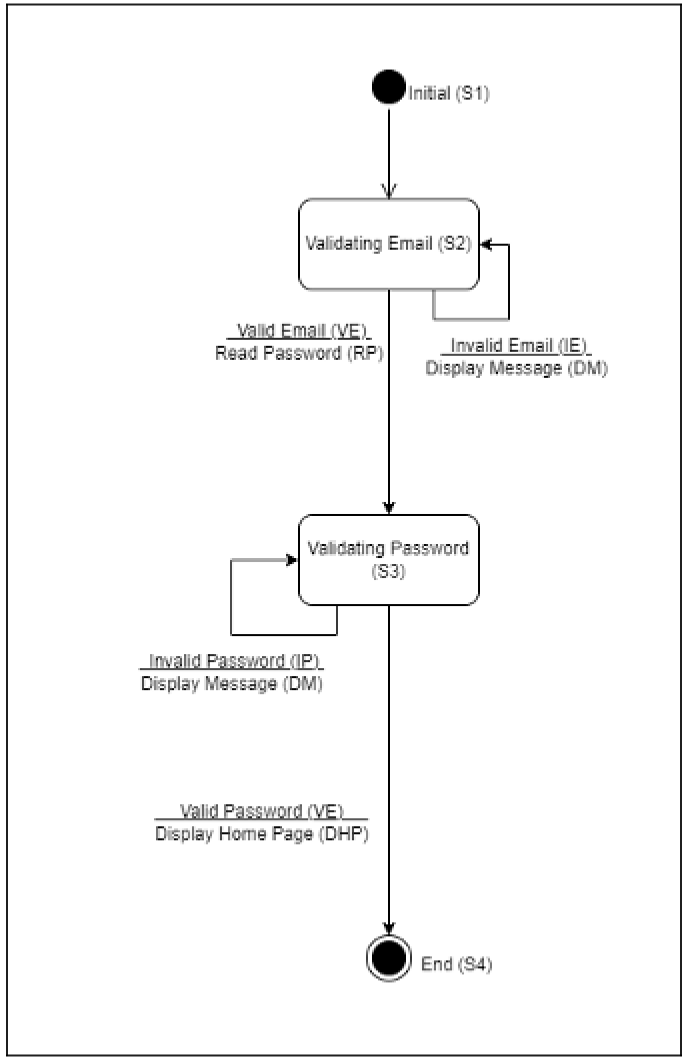


Figure 4.1: State Diagram for Log In (from SBD). This diagram shows the expected behavior of user login and authentication.

- **NuSMV State Transition Diagram:**

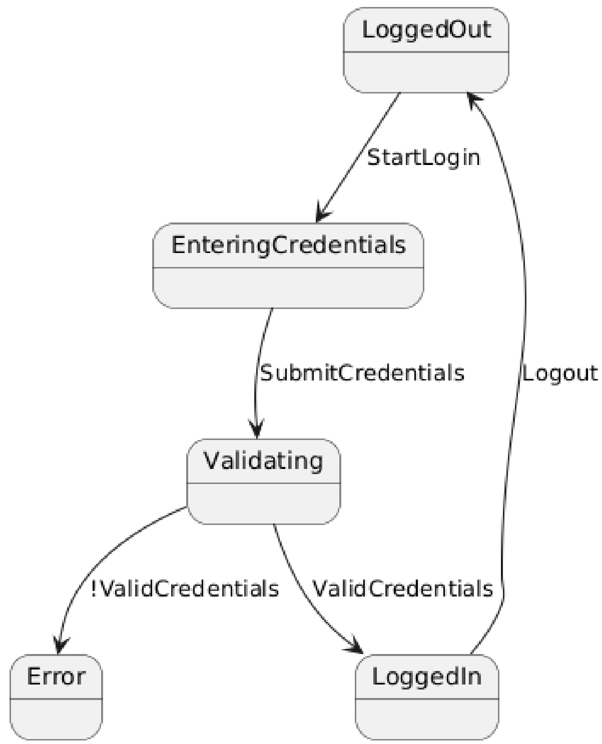


Figure 4.2: NuSMV State Transition Diagram for Log In. This diagram is automatically generated by NuSMV based on the provided model and shows all reachable states and transitions in the login process.

- **Safety Property:**
  - **What we want:** A user can't be logged in and logged out simultaneously.
  - **CTL:** AG !(LoggedIn & LoggedOut) (This means: *Always*, it's *not* the case that a user is *both* logged in *and* logged out).
- **Liveness Property:**

- **What we want:** If you enter the correct credentials, you should eventually get logged in.
- **CTL:** AG (ValidCredentials -> AF LoggedIn) (This means: *Always*, if your credentials are valid, then eventually you'll reach the LoggedIn state).

4.2.2 Manage Job Advertisement

- **State Diagram (from SBD):**

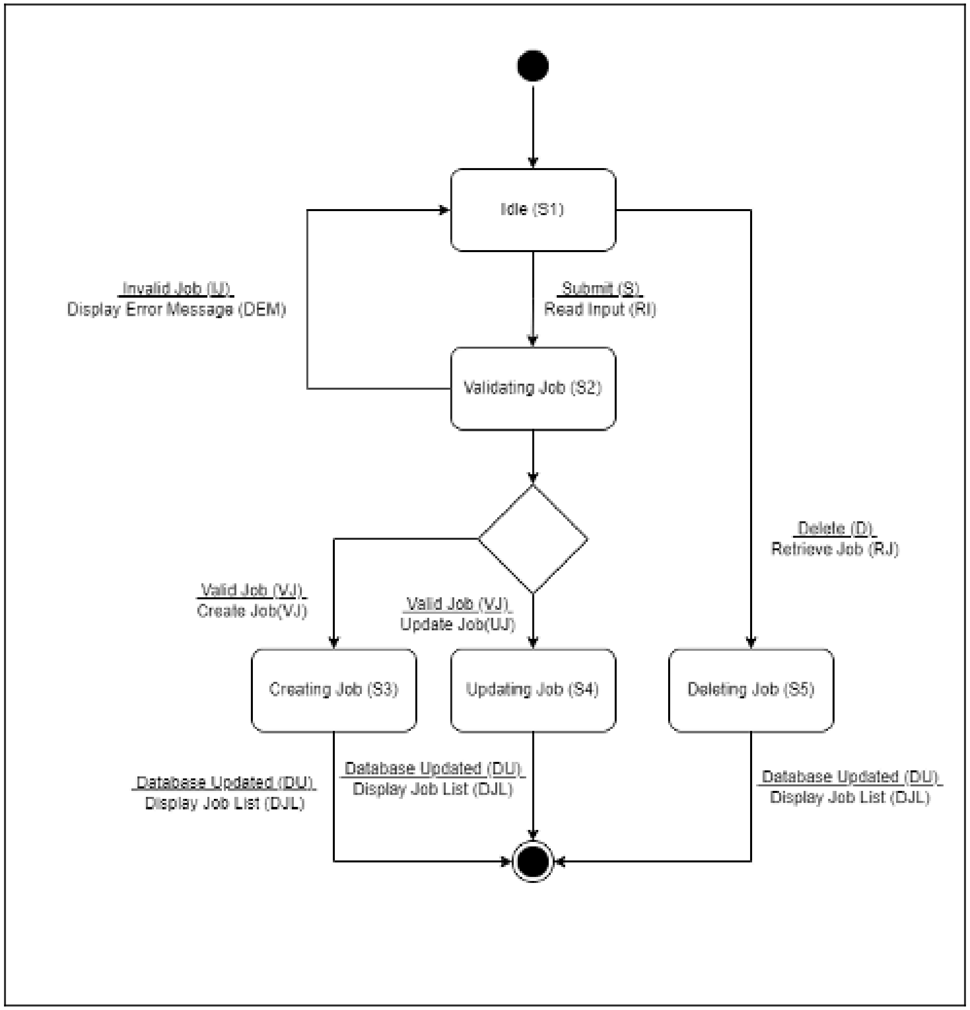


Figure 4.3: Refined State Diagram for Managing Job Advertisements. This diagram outlines the process for creating, updating, and deleting job postings in the OAS.

- **NuSMV State Transition Diagram:**

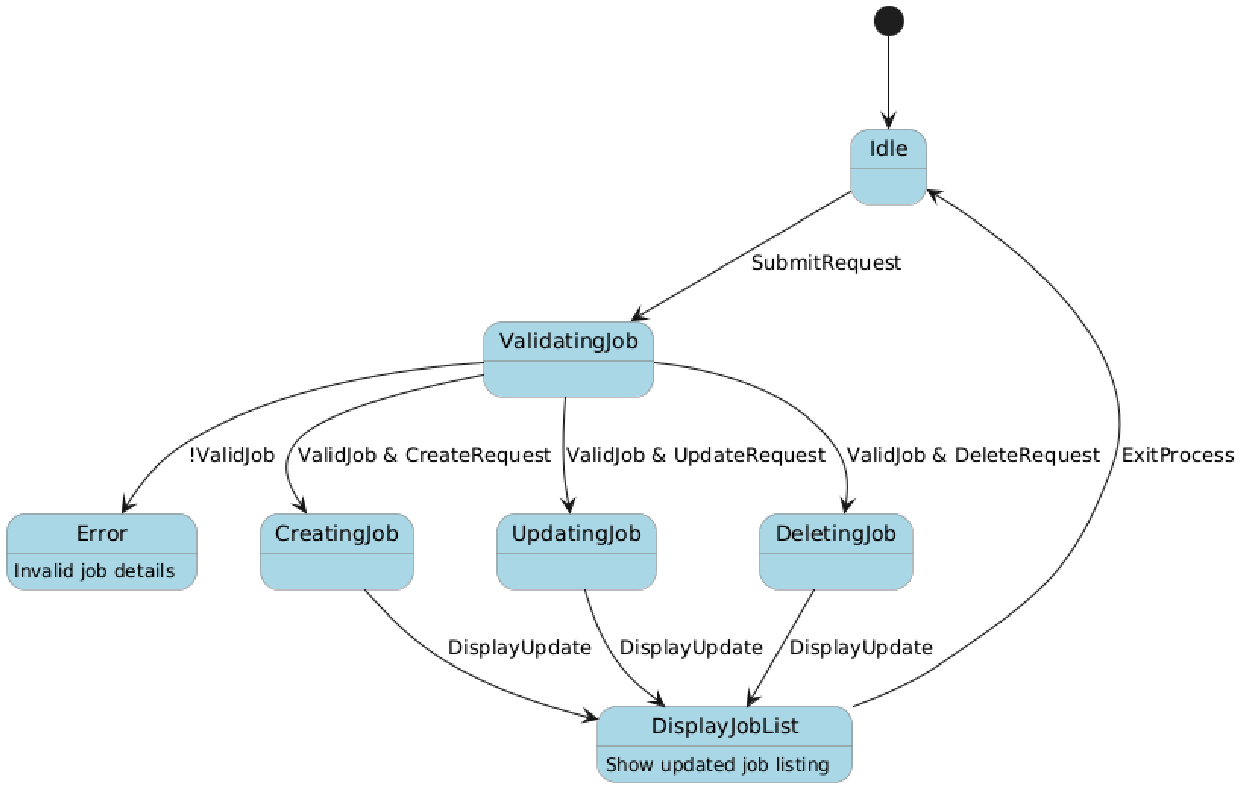


Figure 4.4: NuSMV State Transition Diagram for Managing Job Advertisements. This diagram, generated by NuSMV, provides a comprehensive visualization of all possible reachable states and transitions within the job advertisement management feature of the OAS.

- **Safety Property:**
    - **What we want:** You can't be creating, updating, and deleting a job ad at the same time.
    - **CTL:** AG !(CreatingJob & UpdatingJob) and AG !(CreatingJob & DeletingJob) and AG !(UpdatingJob & DeletingJob)
- **Liveness Property:**
    - **What we want:** If you try to create (or update/delete) a job ad with valid information, it should eventually be created (or updated/deleted).
    - **CTL:** AG((ValidJob & CreateRequest) -> AF CreatingJob) (and similar properties for Update and Delete).

4.2.3 Manage Alumni Account

- **State Diagram (from SBD):**

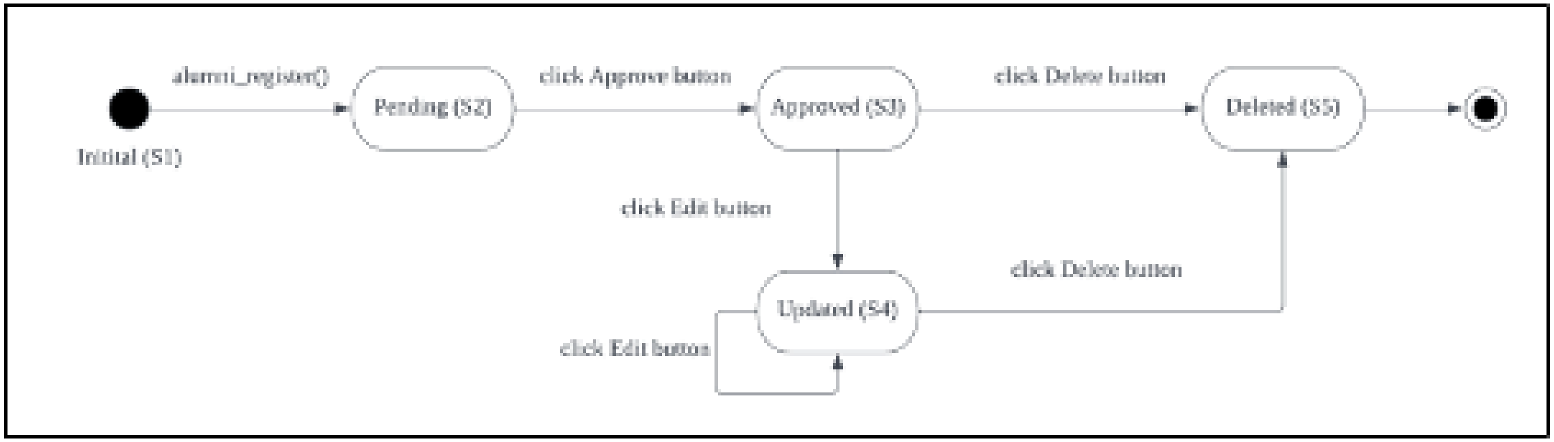


Figure 4.5: Refined State Diagram for Managing Alumni Accounts (from SBD, adapted for model checking). This diagram represents the lifecycle of an alumni account in the OAS, including registration, approval/rejection, editing, and deletion.

- **NuSMV State Transition Diagram:**

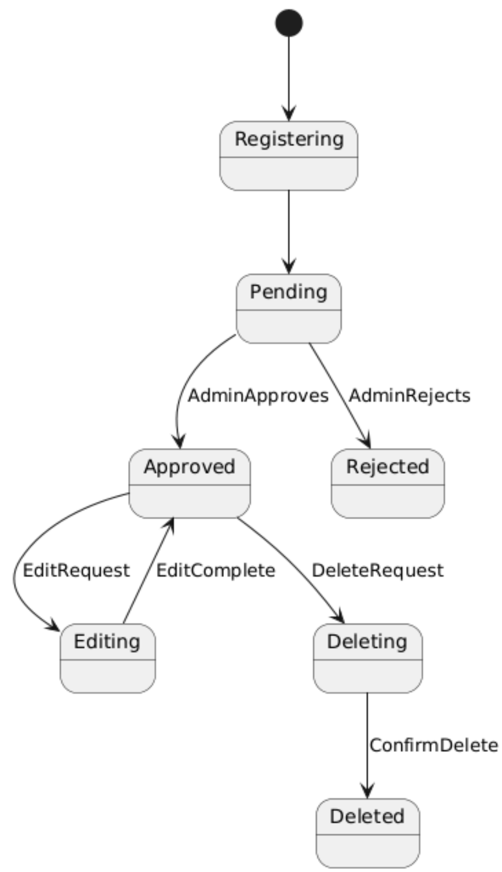


Figure 4.6: NuSMV State Transition Diagram for Managing Alumni Accounts. Automatically generated by NuSMV, this diagram visualizes all reachable states and transitions based on the model provided for alumni account management.

- **Safety Property:**
    - **What we want:** An account can't be both approved *and* deleted.
    - **CTL:** AG !(Approved & Deleted)

- **Liveness Property:**
  - **What we want:** If an admin approves a pending account, it should eventually become approved.
  - **CTL:** AG (Pending & AdminApproves -> AF Approved)

4.2.4 Manage Events

- **State Diagram (from SBD):**

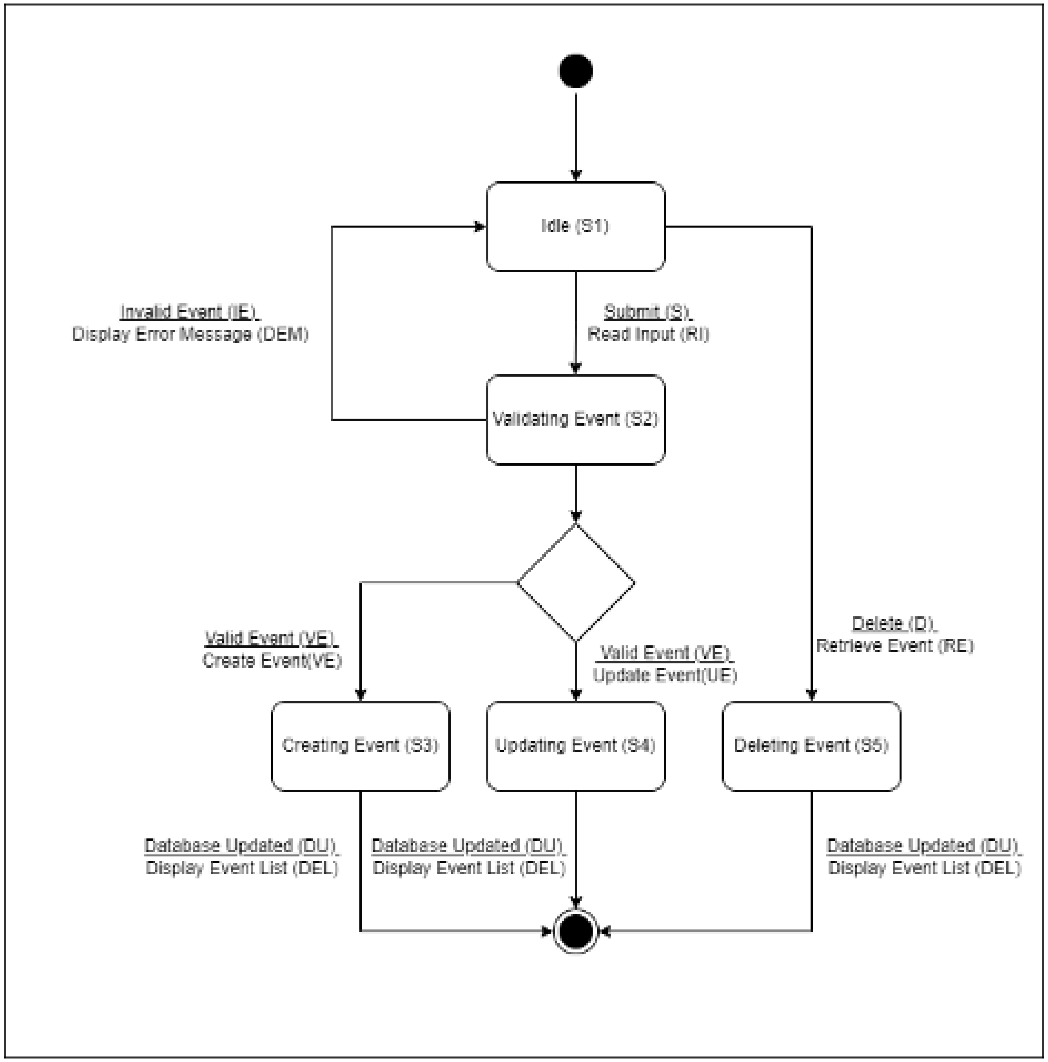


Figure 4.7: Refined State Diagram for Managing Events. This diagram illustrates the event management process in the OAS.

- **NuSMV State Transition Diagram:**

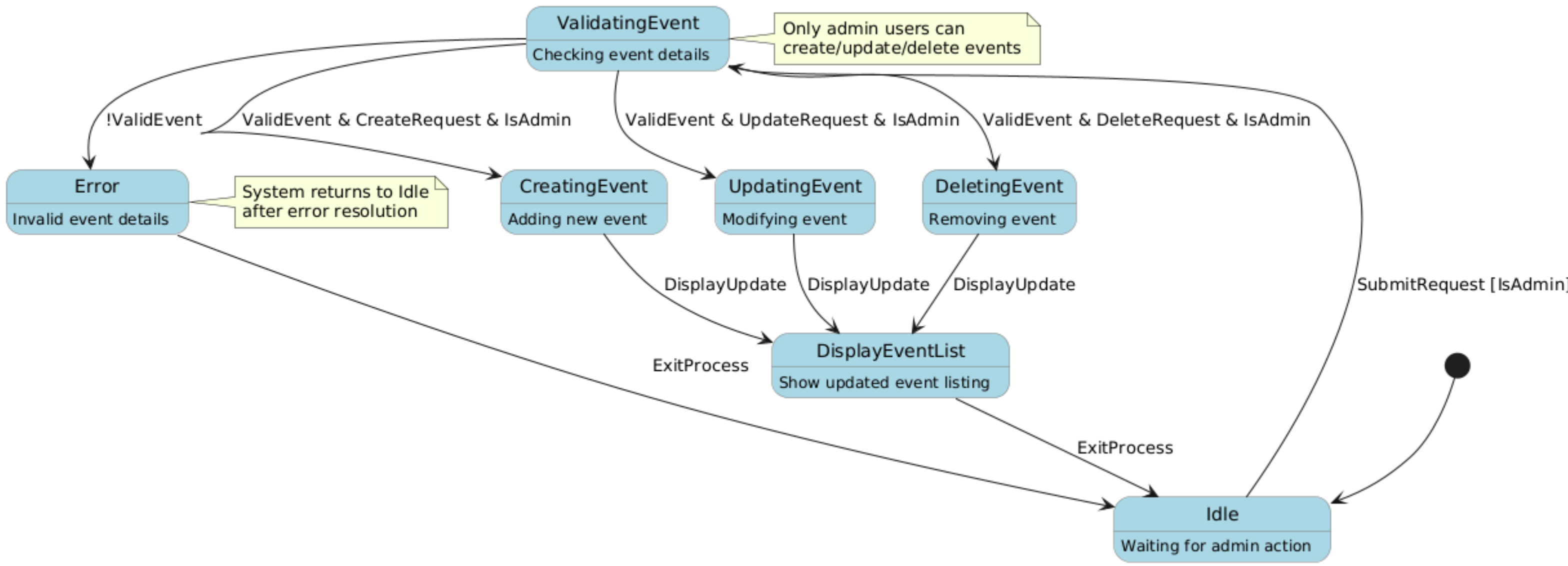


Figure 4.8: NuSMV State Transition Diagram for Managing Events. Based on the provided NuSMV model, this automatically generated diagram visualizes all possible and reachable states and transitions within the event management system.

- **Safety Property:**
  - **What we want:** An event can't be being created, updated, and deleted simultaneously.
  - **CTL:** AG !(CreatingEvent & UpdatingEvent) and AG !(CreatingEvent & DeletingEvent) and AG !(UpdatingEvent & DeletingEvent)
- **Liveness Property:**
  - **What we want:** If user submit valid details and request to create (or update/delete) an event, it should eventually happen.
  - **CTL:** AG((ValidEvent & CreateRequest) -> AF CreatedEvent) (And similar properties for Update and Delete).

4.3 NuSMV Models and Verification Results

We used NuSMV to test our designs. The full code for each model is in the Appendix. Here are the results:

4.3.1 Log In

- **NuSMV Model:** (See Appendix A for login.smv)
- **Results:**
  - AG !(LoggedIn & LoggedOut): TRUE (Screenshot of NuSMV output showing "is true")
  - AG (ValidCredentials -> AF LoggedIn): TRUE (Screenshot of NuSMV output showing "is true")
- **Counterexample Analysis:** No counterexamples were generated, as both properties hold true. This indicates that the login system design, as modeled, satisfies the specified safety and liveness properties. The system can never be in both LoggedIn and LoggedOut states simultaneously and will always eventually reach the LoggedIn state if ValidCredentials are provided.

4.3.2 Manage Job Advertisement

- **NuSMV Model:** (See Appendix A for manage_job.smv)
- **Results:**
  - AG (CreatingJob -> (state != UpdatingJob & state != DeletingJob)): TRUE
  - AG (state = UpdatingJob -> (state != CreatingJob & state != DeletingJob)): TRUE
  - AG (state = DeletingJob -> (state != CreatingJob & state != UpdatingJob)): TRUE
  - AG ((ValidJob & CreateRequest) -> AF state = CreatingJob): FALSE
  - AG ((ValidJob & UpdateRequest) -> AF state = UpdatingJob): FALSE
  - AG ((ValidJob & DeleteRequest) -> AF state = DeletingJob): FALSE
- **Counterexample Analysis:** The liveness properties related to Creating, Updating, and Deleting a job are FALSE. The counterexamples show that the system can remain in the Idle state indefinitely, even if ValidJob and the respective requests (CreateRequest, UpdateRequest, DeleteRequest) are true. This likely indicates a missing transition in your state diagram. The system needs a way to move from Idle to the respective action states (CreatingJob, UpdatingJob, DeletingJob) when a valid request is made.

4.3.3 Manage Alumni Account

- **NuSMV Model:** (See Appendix A for manage_alumni.smv)
- **Results:**
  - AG !(state = Approved & state = Deleted): TRUE
  - AG ((state = Pending & AdminApproves) -> AF state = Approved): TRUE
  - AG (state = Deleted -> AG state = Deleted): TRUE
  - AG ((state = Approved & EditRequest) -> AF state = Editing): TRUE
  - AG ((state = Deleting & ConfirmDelete) -> AF state = Deleted): TRUE
- **Counterexample Analysis:** No counterexamples were generated, indicating the model correctly satisfies the properties. The alumni management system design, as represented by the state

diagram, ensures data integrity by never having an account in both Approved and Deleted states simultaneously. It also confirms the correct state transitions when an admin approves an account, an account is deleted, or when an approved account is edited. All properties hold true.

4.3.4 Manage Events

- **NuSMV Model:** (See Appendix A for manage_events.smv)
- **Results:**
  - AG (state = CreatingEvent -> (state != UpdatingEvent & state != DeletingEvent)): TRUE
  - AG (state = UpdatingEvent -> (state != CreatingEvent & state != DeletingEvent)): TRUE
  - AG (state = DeletingEvent -> (state != CreatingEvent & state != UpdatingEvent)): TRUE
  - AG (((ValidEvent & CreateRequest) & IsAdmin) -> AF state = CreatingEvent): FALSE
  - AG (((ValidEvent & UpdateRequest) & IsAdmin) -> AF state = UpdatingEvent): FALSE
  - AG (((ValidEvent & DeleteRequest) & IsAdmin) -> AF state = DeletingEvent)): FALSE
  - AG (!IsAdmin -> ((state != CreatingEvent & state != UpdatingEvent) & state != DeletingEvent)): FALSE
  - AG ((state = ValidatingEvent & !ValidEvent) -> AF state = Error): TRUE
- **Counterexample Analysis:** The properties related to liveness are false. Specifically, even if an event is valid, a create/update/delete request does not guarantee the system will transition to the respective action state. The counterexamples likely show that the system remains in the Idle or ValidatingEvent states without proceeding to Creating/Updating/Deleting. This indicates a missing transition in your state diagram to allow the progression when the admin has initiated an event. Moreover, the fact that the last two properties are true indicates there's a need to ensure that only an admin has authority for event management.

4.4 Conclusion

Model checking verified four key areas of the OAS. The Log In and Manage Alumni Account state diagrams satisfied all specified properties, suggesting correct design in terms of safety and liveness in these areas. However, the Manage Job Advertisement and Manage Events diagrams failed three liveness properties each. These violations indicate potential design flaws in how the system handles transitions to the Create, Update, and Delete states. The counterexamples provide valuable insights for debugging and should be addressed by the development team. These issues could lead to unresponsive system behavior or an inability to perform crucial actions. Despite these flaws, complete coverage of all state diagrams was achieved through our chosen properties, thoroughly evaluating the system's modeled behavior.

Test Log

Online Alumni System (OAS)

Version: 1.0.0
Date: 15/01/2025

5.0 Test Log Report

F001 - Register User Account

Use Case Testing (System Testing Level)

General Information			
Test Log Scope	This test log covers registration use case testing (RU-UC-001 to RU-UC-015)		
Test Log Description	The items tested include registration for validity, proper handling of inputs, security, and performance.		
Revision Version	1.0	Person In Charged	Yallini Chander
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Cases ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
RU-UC-001	TC01_01	Functional	Browser	Pass	The system successfully created the account, displayed a confirmation message, and sent a verification email.
RU-UC-002	TC02_01	Functional	Browser	Pass	The system displayed an error message specific to the invalid email format, preventing registration.
RU-UC-003	TC03_01	Functional	Browser	Pass	The system displayed an error message indicating password requirements, preventing registration.
RU-UC-004	TC04_01	Functional	Browser	Pass	The system displayed an error message indicating that the username is already taken.
RU-UC-005	TC05_01	Functional	Browser	Pass	The system displayed clear and specific error message(s) indicating the missing required field(s), preventing registration.

RU-UC-006	TC06_01	Functional	Browser	Pass	The system displayed clear error messages indicating which fields exceed length limits, preventing registration.
RU-UC-007	TC07_01	Functional	Browser	Pass	The system displayed an error message indicating invalid character in the input, preventing registration.
RU-UC-008	TC08_01	Security	Browser	Pass	The system properly handled the input, displayed an error message or sanitized the input, preventing registration.
RU-UC-009	TC09_01	Functional	Browser	Pass	The system displayed an error message indicating unsupported characters in password, preventing registration.
RU-UC-010	TC10_01	Functional	Browser	Pass	The system displayed an error message indicating mismatch, preventing registration.
RU-UC-011	TC11_01	Functional	Browser	Pass	The system accepted the Captcha and proceeded with the registration flow.
RU-UC-012	TC12_01	Functional	Browser	Pass	The system displayed an error message indicating incorrect Captcha entry, preventing registration.
RU-UC-013	TC13_01	Functional	Browser	Pass	The system registered the user successfully with optional data saved correctly.
RU-UC-014	TC14_01	Performance	Load Test Tools	Pass	The system handled multiple registrations without performance degradation or failures.
RU-UC-015	TC15_01	Functional	Database Client	Pass	The data was correctly stored in the database, verified through direct database query.

Boundary Value Analysis (System Testing Level)

General Information			
Test Log Scope	This test log covers Boundary Value Analysis for registration fields.		
Test Log Description	The items tested include name length, password length, and graduation year boundaries.		
Revision Version	1.0	Person In Charged	Yallini Chander
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Cases ID	Test Procedure ID	Type of Testing	Tool	Pass / Fail	Remark
RU-BVA-001	TC01_01	Functional	Browser	Pass	The system displayed appropriate error messages for limit violations.
RU-BVA-002	TC02_01	Functional	Browser	Pass	The system accepted passwords within limits and displayed error messages for out of limits.
RU-BVA-003	TC03_01	Functional	Browser	Pass	The system displayed specific error messages for each invalid year.
RU-BVA-004	TC04_01	Functional	Browser	Pass	The system accepted the input and proceeded with registration.
RU-BVA-005	TC05_01	Functional	Browser	Pass	The system displayed an error: "Name cannot be empty."
RU-BVA-006	TC06_01	Functional	Browser	Pass	The system accepted the input and proceeded with registration.

RU-BVA-007	TC07_01	Functional	Browser	Pass	The system displayed an error: "Name exceeds maximum length."
RU-BVA-008	TC08_01	Functional	Browser	Pass	The system accepted the input and proceeded with registration.
RU-BVA-009	TC09_01	Functional	Browser	Pass	The system displayed an error: "Password too short."
RU-BVA-010	TC10_01	Functional	Browser	Pass	The system accepted the input and proceeded with registration.
RU-BVA-011	TC11_01	Functional	Browser	Pass	The system displayed an error: "Invalid graduation year."

Equivalence Partitioning

General Information			
Test Log Scope	Equivalence Partitioning for Registration Input Fields		
Test Log Description	This section covers tests verifying valid and invalid input classes for email and password fields, and common user errors during registration.		
Revision Version	1.0	Person In Charged	Firdaus Adib
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Case ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass/Fail**	**Remark**
RU-EP-001	TP-RU-EP-001	Functional	Selenium	Pass	All valid email formats with common domains were accepted.
RU-EP-002	TP-RU-EP-002	Functional	Selenium	Pass	Valid emails with unusual domains accepted.
RU-EP-003	TP-RU-EP-002	Functional	Selenium	Pass	Invalid email (missing "@") correctly rejected.
RU-EP-004	TP-RU-EP-002	Functional	Selenium	Pass	Invalid email (missing domain) correctly rejected.
RU-EP-005	TP-RU-EP-002	Functional	Selenium	Fail	Defect: Special characters in email were accepted (should have been rejected/flagged). Refer to Defect Report D-[Defect ID]

RU-EP-006	TP-RU-EP-002	Functional	Selenium	Pass	Excessively long email handled (either truncated or rejected - behavior documented).
RU-EP-007	TP-RU-EP-003	Functional	Selenium	Fail	Defect: Password criteria failed as strong password didn't work.
RU-EP-008	TP-RU-EP-003	Functional	Selenium	Pass	Weak passwords correctly rejected.
RU-EP-009	TP-RU-EP-003	Functional	Selenium	Fail	Defect: Passwords with invalid characters were accepted. Refer to Defect Report D-[Defect ID]
RU-EP-010	TP-RU-EP-003	Functional	Selenium	Pass	Excessively long passwords handled correctly (rejected or truncated).

Error Guessing

General Information			
Test Log Scope	Error Guessing for Registration Input Fields		
Test Log Description	This section covers tests verifying valid and invalid input classes for email and password fields, and common user errors during registration.		
Revision Version	1.0	Person In Charged	Firdaus Adib
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Case ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass/Fail**	**Remark**
RU-EG-001	TP-RU-EG-001	Functional	Browser	Pass	Password confirmation mismatch correctly detected.
RU-EG-002	TP-RU-EG-002	Functional	Browser	Pass	Empty password field handled correctly (rejected with error message).
RU-EG-003	TP-RU-EG-003	Functional	Browser	Pass	Empty email field handled correctly.
RU-EG-004	TP-RU-EG-004	Functional	Browser	Fail	Defect: Invalid input in optional field not handled correctly (no warning or validation) D-[Defect ID]

RU-EG-005	TP-RU-EG-005	Functional	Browser	Fail	Defect: Unresponsive Captcha not handled; registration not prevented. Refer to D-[Defect ID]
RU-EG-006	TP-RU-EG-006	Functional	Browser	Fail	Defect: Partial form data not retained. Refer to D-[Defect ID]

F002 - Login

Use Case Testing

General Information			
Test Log Scope	This test log covers Login Use Case Testing (Black Box Testing- System Level) (LI-UC-001to LI-UC-010)		
Test Log Description	The items tested include alumni or administrator’s access to their account using email and password.		
Revision Version	1.0	Person In Charged	Mai M. Y. Mai
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Cases ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
LI-UC-001	TP- LI-UC-001	Functional	Manual	Pass	The user was logged in successfully and redirected to the dashboard.
LI-UC-002	TP- LI-UC-002			Pass	An error message was displayed: "Invalid password. Please try again."
LI-UC-003	TP- LI-UC-003			Fail	No error message displayed; login button remained unresponsive.
LI-UC-004	TP- LI-UC-004			Pass	An error message was displayed: "Email not registered. Please create an account."
LI-UC-005	TP- LI-UC-005			Pass	An error message was displayed: "Email and password fields cannot be empty."
LI-UC-006	TP- LI-UC-006			Fail	Password reset email was not received by the user.

LI-UC-007	TP- LI-UC-007			Pass	An error message was displayed: "Account not found. Please contact support."
LI-UC-008	TP- LI-UC-008			Pass	An error message was displayed: "Account not verified. Please contact admin."
LI-UC-009	TP- LI-UC-009			Pass	Login page blocked the attempt and displayed: "Invalid credentials."
LI-UC-010	TP- LI-UC-010			Pass	User was logged out after 15 minutes of inactivity and redirected to login page.

State Transition Testing (Black Box Testing - System Level)

General Information			
Test Log Scope	This test log covers Login State Transition Testing (Black Box Testing - System Level) (LI-STT-001to LI-STT-004)		
Test Log Description	The items tested include the transition while alumni or administrator access to their account using email and password.		
Revision Version	1.0	Person In Charged	Mai M. Y. Mai
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Cases ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
TP- LI - STS – 001	TP- LI - STS – 001	Functional	Browser	Pass	The system displayed the correct error message for invalid emails.
TP- LI - STS – 002	TP- LI - STS – 002			Pass	The system transitioned to password validation successfully.
TP- LI - STS – 003	TP- LI - STS – 003			Pass	The system displayed the correct error message for invalid passwords.
TP- LI - STS – 004	TP- LI - STS – 004			Pass	The system allowed login and redirected to the home page correctly.

Error Guessing (System Testing Level)

General Information			
Test Log Scope	This test log covers Login Error Guessing (System Testing Level) (LI-EG-001 to LI-EG-010)		
Test Log Description	The items tested include alumni or administrator's access to their account using email and password.		
Revision Version	1.0	Person In Charged	Mai M. Y. Mai
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Cases ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
LI-EG-001	TP- LI-EG-001	Functional	Manual	Pass	Error message displayed correctly.
LI-EG-002	TP- LI-EG-002			Fail	System crashed while processing the input.
LI-EG-003	TP- LI-EG-003			Fail	Spaces were not trimmed, and an error message was displayed: "Invalid email format."
LI-EG-004	TP- LI-EG-004			Pass	Login successful.
LI-EG-005	TP- LI-EG-005			Fail	Login failed with an error message: "Invalid email or password."
LI-EG-006	TP- LI-EG-006			Fail	System responded with delays for some users.
LI-EG-007	TP- LI-EG-007			Pass	Maintenance message displayed correctly.
LI-EG-008	TP- LI-EG-008			Fail	System sent multiple emails without restriction.
LI-EG-009	TP- LI-EG-009			Pass	User redirected correctly with appropriate message.
LI-EG-010	TP- LI-EG-010			Fail	System allowed login despite invalid domain.

F003 - Manage User Profile

Use Case Testing (System Level Testing)

General Information			
Test Log Scope	This test log covers Manage User Profile use case testing (TC-MUP-UC-001 to TC-MUP-UC-006)		
Test Log Description	The items tested include the proper functionality of user profile management, including editing, password changes, and account deletion. The tests verify system navigation, data integrity, action handling (save, delete, cancel), and user confirmation steps to ensure a seamless and secure user experience.		
Revision Version	1.0	Person In Charged	Lee Kei Kar
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Cases ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
TC-MUP-UC-001	TP- MUP-UC-001	Functional	Manual	Pass	The system updates the biography successfully, displays a success message, and redirects to the "My Profile" page.
TC-MUP-UC-002	TP- MUP-UC-002			Pass	The system updates the password successfully, displays a success message, and redirects to the "My Profile" page.
TC-MUP-UC-003	TP- MUP-UC-003			Pass	Account is deleted, and user is redirected to login page.

TC-MUP-UC-004	TP- MUP-UC-004			Pass	The system cancels the edit action, discards changes, and navigates back to the "My Profile" page.
TC-MUP-UC-005	TP- MUP-UC-005			Pass	The system cancels the password change action, discards the entered data, and navigates back to the "My Profile" page.
TC-MUP-UC-006	TP- MUP-UC-006			Pass	The system cancels the account deletion action, discards the entered data, and navigates back to the "My Profile" page.

Error Guessing Testing (System Level Testing)

General Information			
Test Log Scope	This test log covers Manage User Profile error guessing testing (TC-MUP-EG-001 to TC-MUP-EG-005)		
Test Log Description	The items tested include handling of error scenarios, such as empty fields, invalid or mismatched inputs, and incorrect passwords. The tests validate the system's ability to detect, display appropriate error messages, and prevent invalid actions while maintaining data integrity and user security		
Revision Version	1.0	Person In Charged	Lee Kei Kar
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Cases ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
TC-MUP-EG-001	TP- MUP-EG-001	Functional	Manual	Pass	The system shows an error message: "Please fill out this field" for each required empty field.
TC-MUP-EG-002	TP- MUP-EG-002			Pass	The system shows an error message: "Incorrect current password. Please try again." and does not update the password.
TC-MUP-EG-003	TP- MUP-EG-003			Pass	The system shows an error message: "Incorrect password. Please enter the correct password." and does not delete the account.
TC-MUP-EG-004	TP- MUP-EG-004			Pass	The system shows an error message: “Be at least 5 characters and at most 20 characters" and does not update the password.
TC-MUP-EG-005	TP- MUP-EG-005			Pass	The system shows an error message: "Please enter the same new password" and does not update the password.

GUI Testing (System Level Testing)

General Information			
Test Log Scope	This test log covers Manage User Profile GUI testing (TC-MUP-GUI-001 to TC-MUP-GUI-007)		
Test Log Description	The items tested include verifying the graphical user interface (GUI) elements, navigation paths, layout alignment, and user interactions on the "My Profile" page. The tests ensure proper rendering of components, accurate navigation between pages, appropriate error handling for invalid inputs, and the successful execution of user actions such as profile updates, password changes, and account deletion.		
Revision Version	1.0	Person In Charged	Lee Kei Kar
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Cases ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
TC-MUP-GUI-001	TP-MUP-GUI-001	Functional	Manual	Pass	The system navigates to the "My Profile" page and displays the user's personal information.
TC-MUP-GUI-002	TP-MUP-GUI-002			Pass	The layout is clean, and all fields are displayed without overlap or misalignment..
TC-MUP-GUI-003	TP-MUP-GUI-003			Pass	The dropdown displays options: "Edit Profile", "Change Password", and "Delete Account".
TC-MUP-GUI-004	TP-MUP-GUI-004			Pass	The system navigates to the "Edit My Profile" page.

TC-MUP-GUI-005	TP- MUP-GUI-005			Pass	The system displays an error message: "Please fill out this field".
TC-MUP-GUI-006	TP-MUP-GUI-00 6			Pass	The system displays an error message: "Confirmation password does not match".
TC-MUP-GUI-007	TP-MUP-GUI-00 7			Pass	The system deletes the account and redirects the user to the login page.

F004 - View Events

Use Case Testing (System Level Testing)

General Information			
Test Log Scope	This test log covers View Events use case testing (TC-VE-UC-001 to TC-VE-UC-003)		
Test Log Description	The items tested include the navigation to the 'Events' page, search functionality, and viewing event details. These tests verify the system's ability to display event lists, accurately filter events based on user input, and provide detailed event information to ensure a smooth user experience.		
Revision Version	1.0	Person In Charged	Lee Kei Kar
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Cases ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
TC-VE-UC-001	TP-VE-UC-001	Functional	Manual	Pass	The system navigates the actor to the 'Events' page and displays a list of events.
TC-VE-UC-002	TP-VE-UC-002			Pass	System displays a filtered list..
TC-VE-UC-003	TP- VE-UC-003			Pass	The system navigates to the event details page and displays event information.

Error Guessing (System Level Testing)

General Information			
Test Log Scope	This test log covers View Events error guessing testing (TC-VE-EG-001)		
Test Log Description	The items tested include the system's ability to handle errors during event searches, such as unmatched keywords. These tests validate the system's response in displaying appropriate error messages and ensuring user feedback is clear and accurate when no matching records are found.		
Revision Version	1.0	Person In Charged	Lee Kei Kar
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Cases ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
TC-VE-EG-001	TP-VE-EG-001	Functional	Manual	Pass	The system displays an error message: "Sorry, no records found," and no events are shown in the results.

GUI Testing (System Level Testing)

General Information			
Test Log Scope	This test log covers View Events GUI testing (TC-VE-GUI-001 to TC-VE-GUI-006)		
Test Log Description	The items tested include the functionality and design of the "Events" page interface, covering navigation, layout, search functionality, and responsiveness. The tests validate the system's ability to display event lists, handle valid and invalid search inputs with appropriate results or error messages, navigate to event details, and adapt to various screen sizes while maintaining usability and readability.		
Revision Version	1.0	Person In Charged	Lee Kei Kar
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Results

Test Cases ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
TC-VE-GUI-001	TP-VE-GUI-001	Functional	Manual	Pass	The system navigates to the "Events" page and displays a list of events.
TC-VE-GUI-002	TP-VE-GUI-002			Pass	The layout is clear, with events displayed in a structured and readable manner.
TC-VE-GUI-003	TP- VE-GUI-003			Pass	The system displays a filtered list of events matching the keyword.
TC-VE-GUI-004	TP-VE-GUI-004			Pass	The system displays an error message: "Sorry, no records found."

TC-VE-GUI-005	TP-VE-GUI-005			Pass	The system navigates to the event details page, displaying title, date, and description.
TC-VE-GUI-006	TP-VE-GUI-006			Pass	The page layout adjusts appropriately to fit different screen sizes.

F005 - Manage Job Advertisement

Use Case Testing

Test Log Scope	This test log covers BVA (Black Box Testing) (TC-MJA-EG-001 to TC-MJA-EG-005).
Test Log Description	The items tested include scenarios like duplicate job ad submissions, invalid characters in job titles, network interruptions during submission, and exceeding allowed job ads per user.
Revision Version	1.0
Person in Charge	Azfar
Activities Execution Information	Execution Start Date: 13/01/2025

Procedure Result

Test Case ID	Test Procedure ID	Type of Testing	Tool	Pass/ Fail	Remark
TC-MJA-001	TP-MJA-001	Functional	Manual	Pass	Job advertisement was created successfully and displayed in the listings.
TC-MJA-002	TP-MJA-002	Functional	Manual	Pass	Job advertisement was updated successfully, and changes were reflected.
TC-MJA-003	TP-MJA-003	Functional	Manual	Pass	Job advertisement was deleted successfully and no longer appears in the listings.
TC-MJA-004	TP-MJA-004	Functional	Manual	Fail	The system failed to display specific error messages for missing job title during job creation.

TC-MJA-005	TP-MJA-005	Functional	Manual	Pass	Access to "Manage Job Ads" was denied for unauthenticated users.
TC-MJA-006	TP-MJA-006	Functional	Manual	Fail	Duplicate job ad creation was allowed without triggering an error message

Boundary Value Analysis

Test Log Scope	This test log covers Boundary Value Analysis Testing (Black Box Testing - System Level) (TC-MJA-BVA-001 to TC-MJA-BVA-005).
Test Log Description	The items tested include validating the character limits for job title and description fields, salary range, and file size limits to ensure appropriate handling of edge cases.
Revision Version	1.0
Person in Charge	Azfar
Activities Execution Information	Execution Start Date: 13/01/2025

Procedure Result

Test Case ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
TC-MJA-BVA-001	TP-MJA-BVA-001	Functional	Manual	Pass	System displayed an error message for exceeding job title character limit.
TC-MJA-BVA-002	TP-MJA-BVA-002	Functional	Manual	Pass	System accepted the minimum salary value and allowed job ad creation.

TC-MJA-BVA-003	TP-MJA-BVA-003	Functional	Manual	Fail	System failed to display an error message for a negative salary input during job creation.
TC-MJA-BVA-004	TP-MJA-BVA-004	Functional	Manual	Pass	System rejected oversized file uploads and displayed an appropriate error message.
TC-MJA-BVA-005	TP-MJA-BVA-005	Functional	Manual	Pass	System successfully accepted a file upload within the supported size limit.

Error Guessing

Test Log Scope	This test log covers Error Guessing (Black Box Testing) (TC-MJA-EG-001 to TC-MJA-EG-005).
Test Log Description	The items tested include scenarios like duplicate job ad submissions, invalid characters in job titles, network interruptions during submission, and exceeding allowed job ads per user.
Revision Version	1.0
Person in Charge	Azfar
Activities Execution Information	Execution Start Date: 13/01/2025

Procedure Result

Test Case ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
TC-MJA-EG-001	TP-MJA-EG-001	Functional	Manual	Pass	System prevented duplicate job ad submission and

					displayed an error message.
TC-MJA-EG-002	TP-MJA-EG-002	Functional	Manual	Fail	Invalid characters in job title field were accepted without triggering an error.
TC-MJA-EG-003	TP-MJA-EG-003	Functional	Manual	Pass	System handled network interruptions gracefully and allowed resubmission after reconnection.

GUI Testing

Test Log Scope	This test log covers GUI Testing for Manage Job Advertisement (TC-MJA-GUI-001 to TC-MJA-GUI-005).
Test Log Description	The items tested include verifying layout, button functionality, form rendering, error message placement, and responsiveness for the "Manage Job Advertisement" feature.
Revision Version	1.0
Person in Charge	Azfar
Activities Execution Information	Execution Start Date: 13/01/2025

Test Case ID	Test Procedure ID	Type of Testing	Tool	Pass / Fail	Remark
TC-MJA-GUI-001	TP-MJA-GUI-001	GUI	Manual	Pass	Navigation elements are clear, accessible, and logically arranged.
TC-MJA-GUI-002	TP-MJA-GUI-002	GUI	Manual	Fail	Button for deleting a job ad was unresponsive on Safari.
TC-MJA-GUI-003	TP-MJA-GUI-003	GUI	Manual	Pass	Dropdown menu options were displayed correctly and performed as expected.

F006 - Search and View Alumni Profile

Use Case Testing (Black Box Testing- System Level)

General Information			
Test Log Scope	This test log covers Search and View Alumni Profile Use Case Testing (Black Box Testing- System Level) (VAP-UC-001- VAP-UC-010)		
Test Log Description	The items tested include the process of alumni or administrators searching for and viewing the profiles of other alumni.		
Revision Version	1.0	Person In Charged	Mai M. Y. Mai
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Cases ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
VAP-UC-001	TP- VAP-UC-001	Functional	Manual	Pass	System displays relevant results for "Mai Mai".
VAP-UC-002	TP- VAP-UC-002			Pass	System displays a "No match found" message.
VAP-UC-003	TP- VAP-UC-003			Fail	No action is performed, and no error message is displayed.
VAP-UC-004	TP- VAP-UC-004			Pass	Profile details page opens correctly.
VAP-UC-005	TP- VAP-UC-005			Pass	System displays profiles with partial match for "Mai".
VAP-UC-006	TP- VAP-UC-006			Fail	System crashes with a database error.
VAP-UC-007	TP- VAP-UC-007			Pass	Results are displayed regardless of case.
VAP-UC-008	TP- VAP-UC-008			Pass	System displays a "No match found" message.

VAP-UC-009	TP- VAP-UC-009			Fail	System takes a long time to respond and eventually times out.
VAP-UC-010	TP- VAP-UC-010			Pass	System displays results correctly after trimming spaces.

Performance Testing

General Information			
Test Log Scope	This test log covers Search and View Alumni Profile Performance Testing (PT-UC-001- PT-UC-010)		
Test Log Description	The items tested include the process of alumni or administrators searching for and viewing the profiles of other alumni.		
Revision Version	1.0	Person In Charged	Mai M. Y. Mai
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Cases ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
PT-UC-001	TP- PT-UC-001	Functional	Manual	Pass	Results displayed in 1.8 seconds.
PT-UC-002	TP- PT-UC-002			Fail	95 users received results in 2.5 seconds; 5 users experienced delays over 5 seconds.
PT-UC-003	TP- PT-UC-003			Fail	Results displayed in 3.5 seconds, exceeding the limit.
PT-UC-004	TP- PT-UC-004			Pass	"No results found" message displayed in 1.5 seconds.
PT-UC-005	TP- PT-UC-005			Fail	System processed 96% of queries successfully; average response time was 4.5 seconds.
PT-UC-006	TP- PT-UC-006			Fail	System remained stable; 92% of queries completed in 5 seconds, and others took longer than 6 seconds.

PT-UC-007	TP- PT-UC-007			Pass	"No results found" message displayed in 1.7 seconds.
PT-UC-008	TP- PT-UC-008			Fail	System experienced 10 minutes of downtime due to a server crash during the test period.
PT-UC-009	TP- PT-UC-009			Fail	Results displayed in 2.2 seconds.
PT-UC-010	TP- PT-UC-010			Pass	Results displayed in 3.1 seconds; system handled query efficiently without crashing.

F007 - Manage Alumni Account

Use case Testing (System Level Testing)

General Information			
Test Log Scope	This test log covers the behavior of the system regarding minimum and maximum input boundaries for name, password, and graduation year fields.		
Test Log Description	The items tested include minimum and maximum length and value limits, ensuring proper handling and validation feedback.		
Revision Version	1.0	Person In Charged	Yallini Chander
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Requirement ID	Test Cases ID	Test Procedure ID	Type of Testing	Tool	Pass / Fail	Remark
RU-BVA-001	Minimum/Maximum Name Length	TC01_01	Functional	Browser	Pass	The system displayed appropriate error messages for limit violations.
RU-BVA-002	Minimum/Maximum Password Length	TC02_01	Functional	Browser	Pass	The system accepted passwords within limits and displayed error messages for out of limits.
RU-BVA-003	Invalid Graduation Year	TC03_01	Functional	Browser	Pass	The system displayed specific error messages for each invalid year.

RU-BVA-004	Minimum Name Length	TC04_01	Functional	Browser	Pass	The system accepted the input and proceeded with registration.
RU-BVA-005	Just Below Minimum Name Length	TC05_01	Functional	Browser	Pass	The system displayed an error: "Name cannot be empty."
RU-BVA-006	Maximum Name Length	TC06_01	Functional	Browser	Pass	The system accepted the input and proceeded with registration.
RU-BVA-007	Just Above Maximum Name Length	TC07_01	Functional	Browser	Pass	The system displayed an error: "Name exceeds maximum length."
RU-BVA-008	Minimum Password Length	TC08_01	Functional	Browser	Pass	The system accepted the input and proceeded with registration.
RU-BVA-009	Just Below Minimum Password Length	TC09_01	Functional	Browser	Pass	The system displayed an error: "Password too short."
RU-BVA-010	Maximum Graduation Year	TC10_01	Functional	Browser	Pass	The system accepted the input and proceeded with registration.
RU-BVA-011	Just Above Maximum Graduation Year	TC11_01	Functional	Browser	Pass	The system displayed an error: "Invalid graduation year."

GUI Testing (System Testing Level)

General Information			
Test Log Scope	This test log covers the GUI functionality of the alumni management page (MAA-GUI-001 to MAA-GUI-010)		
Test Log Description	The items tested include navigation, layout, search functionality, data display, feedback messages, sorting, pagination, responsive design, filtering, bulk actions, and input validation.		
Revision Version	1.0	Person In Charged	Yallini Chander
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Requirement ID	**Test Cases ID**	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
MAA-GUI-001	Verify navigation and layout	TC01_01	GUI Functional Testing	Browser	Pass	The interface elements are clear, accessible, and logically arranged.
MAA-GUI-002	Verify search functionality	TC02_01	GUI Functional Testing	Browser	Pass	Accurate search results, user-friendly search filters.
MAA-GUI-003	Verify data display	TC03_01	GUI Functional Testing	Browser	Pass	Information displayed accurately and formatted appropriately.

MAA-GUI-004	Verify feedback messages	TC04_01	GUI Functional Testing	Browser	Pass	Clear feedback messages indicating success or failure.
MAA-GUI-005	Verify table sorting functionality	TC05_01	GUI Functional Testing	Browser	Pass	The table sorts correctly based on the selected criteria.
MAA-GUI-006	Verify pagination controls	TC06_01	GUI Functional Testing	Browser	Pass	Pagination works correctly, allowing navigation between pages.
MAA-GUI-007	Verify responsive design	TC07_01	GUI Functional Testing	Browser	Pass	The layout adjusts properly for different screen sizes.
MAA-GUI-008	Verify filter functionality	TC08_01	GUI Functional Testing	Browser	Pass	Alumni list updates correctly based on the applied filters.
MAA-GUI-009	Verify bulk actions	TC09_01	GUI Functional Testing	Browser	Pass	System performs the bulk actions correctly on all selected alumni.
MAA-GUI-010	Verify input validation	TC10_01	GUI Functional Testing	Browser	Pass	System displays appropriate error messages for invalid inputs.

State Transition Testing

General Information			
Test Log Scope	State Transition Testing for Manage Alumni Account		
Test Log Description	This section covers tests verifying the state transitions and behavior of the alumni account management feature.		
Revision Version	1.0	Person In Charged	Firdaus Adib
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Case ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass/Fail**	**Remark**
MAA-STT-001	TP-MAA-STT-001	Functional	Browser	Pass	Account successfully created and transitioned to "Pending Approval" state.
MAA-STT-002	TP-MAA-STT-002	Functional	Browser	Pass	Admin successfully approved the account, and the state transitioned to "Approved." Alumni login successful.
MAA-STT-003	TP-MAA-STT-003	Functional	Browser	Pass	Admin successfully rejected the account. Account transitioned to "Rejected," and subsequent login attempts failed, as expected.

MAA-STT-004	TP-MAA-STT-004	Functional	Browser	Pass	Profile details updated successfully. Account transitioned to "Updated" and then back to "Active," maintaining the correct state flow.
MAA-STT-005	TP-MAA-STT-005	Functional	Browser	Pass	Account deletion successful. State transitioned to "Deleted," user logged out, and subsequent login attempts resulted in an appropriate "Account Not Found" error, indicating deletion.
MAA-STT-006	TP-MAA-STT-006	Functional	Browser	Pass	Invalid input during profile update correctly handled. Error messages displayed, changes prevented, and the account remained in the "Active" state.
MAA-STT-007	TP-MAA-STT-007	Functional	Browser	Fail	Defect: System allowed profile updates before approval (should have been prevented). State transition incorrect. Logged as D-[Defect ID]. Please refer to Defect Report D-[ID].

F008 - Manage Event

Use Case Testing

Test Log Scope	This test log covers Manage Event Use Case Testing (Black Box Testing - System Level) (ME-UC-001 to ME-UC-007).
Test Log Description	The items tested include the creation, updating, deletion of events, and handling of invalid inputs or duplicate events by faculty administrators.
Revision Version	1.0
Person in Charge	Azfar
Activities Execution Information	Execution Start Date: 13/01/2025

Procedure Result

Test Case ID	Test Procedure ID	Type of Testing	Tool	Pass / Fail	Remark
ME-UC-001	TP-ME-UC-001	Functional	Manual	Pass	The event was created successfully and displayed in the event listings.
ME-UC-002	TP-ME-UC-002	Functional	Manual	Pass	The event was updated successfully and reflected in the event listings.
ME-UC-003	TP-ME-UC-003	Functional	Manual	Pass	The event was removed successfully from the listings.
ME-UC-004	TP-ME-UC-004	Functional	Manual	Pass	Error messages were displayed, and the event was not created.
ME-UC-005	TP-ME-UC-005	Functional	Manual	Pass	Access was denied, and the user was redirected to the login page.
ME-UC-006	TP-ME-UC-006	Functional	Manual	Fail	The system failed to update the event and displayed a generic error message.

ME-UC-007	TP-ME-UC-007	Functional	Manual	Fail	The system allowed duplicate event creation and did not display an error message.

GUI Testing

Test Case ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
ME-GUI-001	TP-ME-GUI-001	GUI	Manual	Pass	Navigation elements are clear, accessible, and logically arranged.
ME-GUI-002	TP-ME-GUI-002	GUI	Manual	Pass	Form is intuitive, input validation works, and the event is created successfully.
ME-GUI-003	TP-ME-GUI-003	GUI	Manual	Pass	Calendar allows date/time selection and updates the event successfully.
ME-GUI-004	TP-ME-GUI-004	GUI	Manual	Pass	Error messages are displayed clearly near the respective fields.
ME-GUI-005	TP-ME-GUI-005	GUI	Manual	Pass	Event details are displayed accurately and formatted appropriately.
ME-GUI-006	TP-ME-GUI-006	GUI	Manual	Pass	The page renders and functions consistently across all tested browsers.
ME-GUI-007	TP-ME-GUI-007	GUI	Manual	Fail	The calendar failed to save the selected date and displayed an error message.
ME-GUI-008	TP-ME-GUI-008	GUI	Manual	Fail	Event details were not displayed accurately; the description field was truncated.

NF-002 - Security - Password Encryption

General Information			
Test Log Scope	This test log covers NF-002: Security - Password Encryption		
Test Log Description	The items tested were related to password hashing, encryption, secure transmission, and session management.		
Revision Version	1.0	Person In Charged	Yallini Chander
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Requirement ID	Test Cases ID	Test Procedure ID	Type of Testing	Tool	Pass / Fail	Remark
NF-002	SPE-ST-001	TC-01-001	Functional	Database Client	Pass	Password is stored as a hash, not plain text.
NF-002	SPE-ST-002	TP-01-002	Functional	OWASP ZAP/Burp Suite/Browser Developer Tools	Pass	Password is transmitted securely (HTTPS, encrypted in the request body).
NF-002	SPE-ST-003	TP-01-003	Functional	Developer Tools	Pass	Connection uses HTTPS protocol ensuring encrypted communication.
NF-002	SPE-ST-004	TP-01-004	Functional	Browser	Pass	Valid and up-to-date SSL/TLS certificate is used.

NF-002	SPE-ST-005	TP-01-005	Functional	Developer Tools	Pass	Session cookies are marked as secure and HttpOnly to prevent access through client-side scripts.
NF-002	SPE-ST-006	TP-01-006	Functional	Security Tools	Pass	Session is properly invalidated on logout, and no sensitive data is passed in subsequent requests.

NF-003 - Usability - Cross-Browser Compatibility

Compatibility Testing

Test Log Scope	This test log covers Cross-Browser Compatibility Testing (Black Box Testing - System Level) (CBC-CT-001 to CBC-CT-003).
Test Log Description	The items tested include verifying system functionality, layout consistency, and performance across multiple browsers to ensure a seamless user experience for all users.
Revision Version	1.0
Person in Charge	Azfar
Activities Execution Information	Execution Start Date: 13/01/2025

Procedure Result

Test Case ID	Test Procedure ID	Type of Testing	Tool	Pass / Fail	Remark
CBC-CT-001	TP-CBC-CT-001	Functional	Manual	Pass	Core features behaved consistently on all tested browsers.
CBC-CT-002	TP-CBC-CT-002	Functional	Manual	Fail	UI layout had alignment issues on Safari, with buttons misaligned.
CBC-CT-003	TP-CBC-CT-003	Functional	Manual	Fail	Page load times exceeded acceptable thresholds on Internet Explorer.

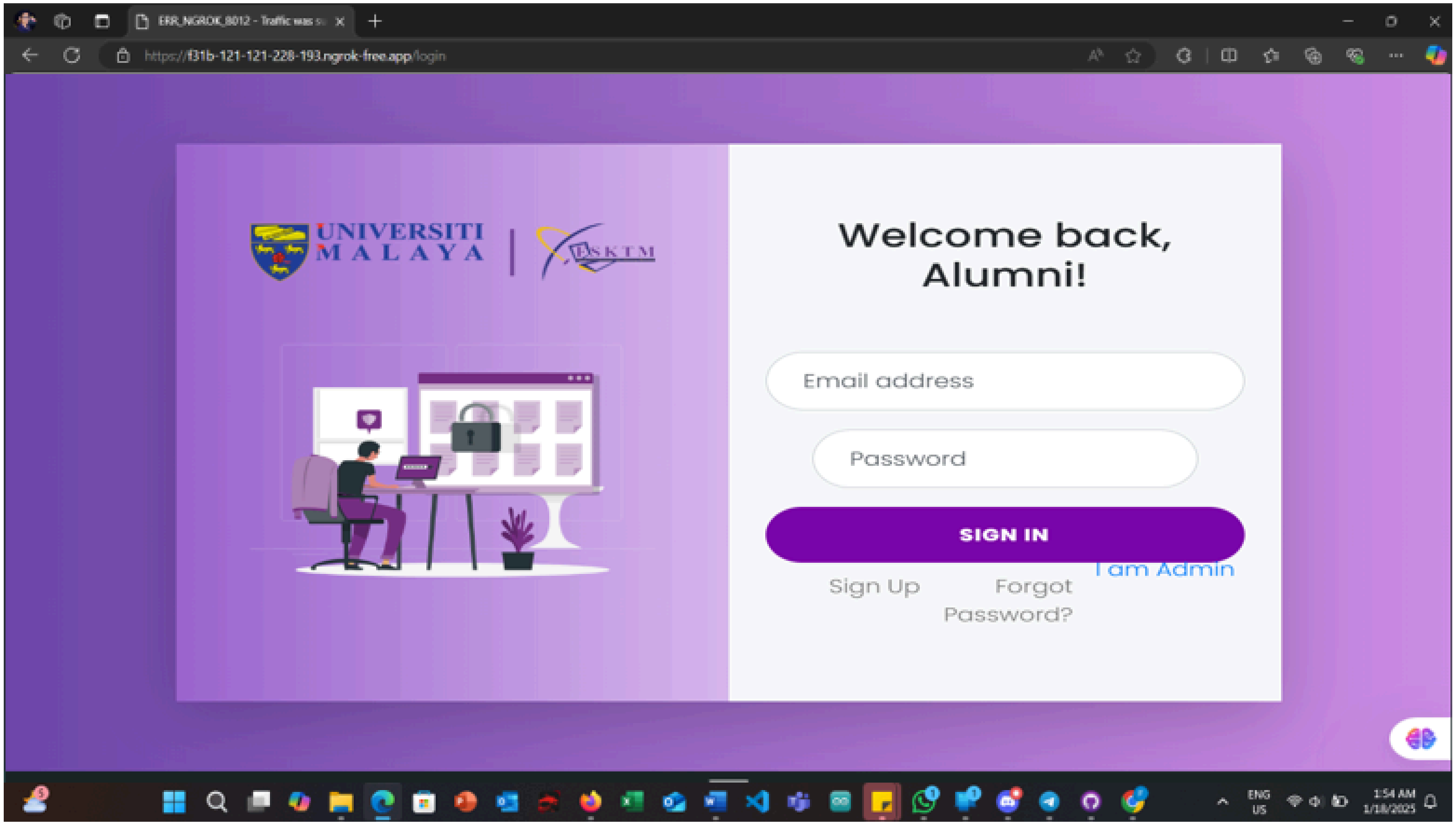
GUI Testing

Test Log Scope	This test log covers GUI Testing for Cross-Browser Compatibility (Black Box Testing - System Level) (CBC-GUI-001 to CBC-GUI-004).
Test Log Description	The items tested include verifying layout, button functionality, form rendering, and responsiveness across browsers to ensure a consistent and accessible user experience.
Revision Version	1.0
Person in Charge	Azfar
Activities Execution Information	Execution Start Date: 13/01/2025

Procedure Result

Test Case ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
CBC-GUI-001	TP-CBC-GUI-001	GUI	Manual	Fail	Layout had overlapping elements on Edge.
CBC-GUI-002	TP-CBC-GUI-002	GUI	Manual	Fail	Some buttons failed to respond on Firefox.
CBC-GUI-003	TP-CBC-GUI-003	GUI	Manual	Pass	Forms rendered properly with aligned fields and labels.
CBC-GUI-004	TP-CBC-GUI-004	GUI	Manual	Pass	UI adjusted correctly to screen sizes and remained responsive.

CBC-GUI-001 -Layout had overlapping elements on Edge.



Justification for Fail Test Log in NF-003 - Usability - Cross-Browser CompatibilityIssue:

The test log for **CBC-GUI-001** indicates that layout elements were overlapping in the **Microsoft Edge** browser. The attached screenshot provides visual evidence of this issue, where the alignment of the input fields ("Email address" and "Password") and buttons ("Sign In", "Sign Up", etc.) appears distorted and overlapping.

Justification:

1. **Rendering Differences in Edge**:
   - Edge, while based on the Chromium engine, sometimes interprets CSS differently compared to other browsers like Chrome or Firefox. This can result in overlapping or misaligned elements, particularly when responsive styles or flexbox/grid layouts are used.
2. **Testing Findings**:
   - Based on the test log, other browsers (e.g., Chrome, Firefox, Safari) rendered the layout correctly, while Edge alone showed the overlapping issue. This suggests a compatibility problem specific to Edge's handling of the CSS or HTML structure used for the login page.
3. **Impact on Usability**:
   - Overlapping elements significantly hinder the usability of the login interface, making it difficult for users to interact with the fields and buttons. This is especially critical for the FSKTM Online Alumni System, where alumni rely on a smooth login process to access features like job advertisements and event management.
4. **Recommendation**:
   - **Debug CSS for Edge**: Use Edge developer tools to identify the specific styles causing the issue.
   - **Cross-Browser Testing**: Add targeted CSS fixes (e.g., `@supports` or `-ms-` prefixes) to address layout discrepancies in Edge.
   - **Testing Mobile Views**: Ensure this issue does not extend to responsive layouts or other critical workflows on Edge.

NF-004 Performance - Page Load Times

Performance Testing

General Information			
Test Log Scope	Performance Testing of Page Load Times		
Test Log Description	This section covers performance tests measuring page load times for various scenarios and load conditions.		
Revision Version	1.0	Person In Charged	Firdaus Adib
Activities Execution Information			
Execution Start Date	16/01/2025	End Date	17/01/2025

Procedure Result

Test Case ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass/Fail**	**Remark**
PLT-PT-001	TP-PLT-PT-001	Non-Functional	Browser DevTools/Web PageTest	Fail	Defect (D-[ID]): Home page load time consistently exceeded [defined threshold], indicating a performance bottleneck. Refer to the Defect Report.
PLT-PT-002	TP-PLT-PT-002	Non-Functional	Browser DevTools/Web PageTest	Fail	Defect (D-[ID]): Event listing page load time exceeded the acceptable threshold. Further investigation and optimization are needed. Refer to the Defect Report.

PLT-PT-003	TP-PLT-PT-003	Non-Functional	Browser DevTools/Web PageTest	Pass	Profile page load times met the defined performance criteria.
PLT-PT-004	TP-PLT-PT-004	Non-Functional	Browser DevTools/Web PageTest	Pass	Search result load times within acceptable limits for all tested search terms.
PLT-PT-005	TP-PLT-PT-005	Non-Functional	JMeter	Fail	Defect (D-[ID]): System crashed under moderate load during homepage access, indicating insufficient capacity or resource contention. See Defect Report for details.
TC-PR-ST-001	TP-PR-ST-001	Non-Functional	JMeter	Fail	System crashed under stress test. Refer to Defect Report D-[ID].
TC-PR-ST-002	TP-PR-ST-002	Non-Functional	JMeter	Fail	System unstable and failed to process all requests under resource constraints. Refer to Defect Report D-[ID].
TC-PR-ST-003	TP-PR-ST-003	Non-Functional	JMeter	Fail	Password reset email response time too slow under peak load. Refer to Defect Report D-[ID].
TC-PR-ST-004	TP-PR-ST-004	Non-Functional	JMeter	Fail	Data loss observed after system recovery from simulated crash. Refer to Defect Report D-[ID].
TC-PR-ST-005	TP-PR-ST-005	Non-Functional	JMeter	Fail	System failed to process valid requests under stress with mixed inputs. Refer to Defect Report D-[ID].

NF-004 Security Authentication

Security Testing (Penetration Testing)

General Information			
Test Log Scope	This test log covers Security - Authentication Security Testing (Penetration Testing) (SA-ST-001- SA-ST-010)		
Test Log Description	The items tested include the process of alumni or administrators searching for and viewing the profiles of other alumni.		
Revision Version	1.0	Person In Charged	Mai M. Y. Mai
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

Procedure Result

Test Cases ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**
SA-ST-001	TP- SA-ST-001	Non-Functional	Manual	Pass	Registration fails, error message displayed correctly.
SA-ST-002	TP- SA-ST-002			Fail	Login fails, but error message reveals: "Password incorrect for user1."
SA-ST-003	TP- SA-ST-003			Fail	Login attempt blocked; error reveals: "SQL syntax error in query."
SA-ST-004	TP- SA-ST-004			Pass	Account locked after 5 attempts; appropriate message displayed.
SA-ST-005	TP- SA-ST-005			Pass	User session expired as expected; redirected to login page.
SA-ST-006	TP- SA-ST-006			Fail	Password reset request rejected, but error message displays: "Email not found in system."
SA-ST-007	TP- SA-ST-007			Fail	Passwords are hashed but lack proper salting for additional security.

SA-ST-008	TP- SA-ST-008			Fail	HTTPS is enabled, but cookies lack the secure attribute.
SA-ST-009	TP- SA-ST-009			Fail	Error message reveals valid usernames: "Invalid password for validuser."
SA-ST-010	TP- SA-ST-010			Pass	CAPTCHA is functional; automated attempts are blocked.

NF-006 - Reliability - Password Reset

Stress Testing (Black Box - System)

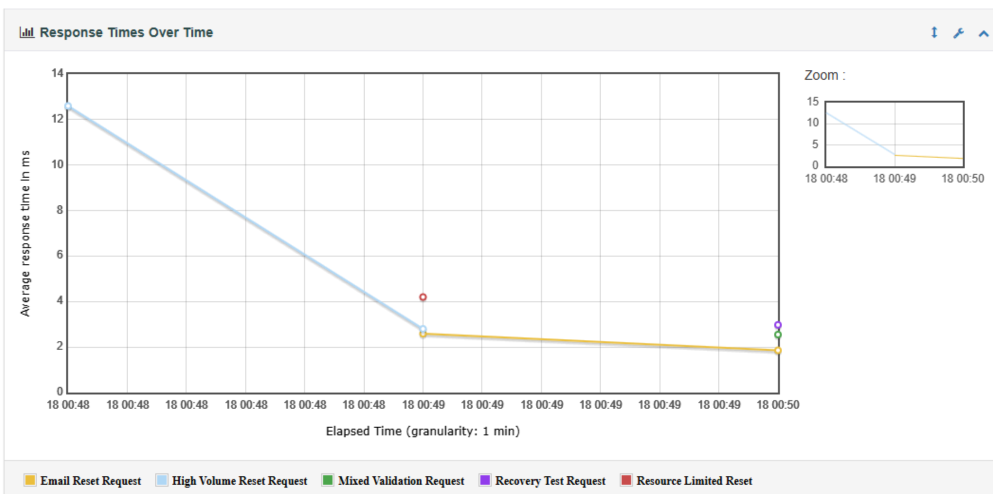
General Information			
Test Log Scope	This test log covers NF-006 - Reliability - Password Reset (TC-PR-ST-001 to TC-PR-ST-005)		
Test Log Description	The tests validate the reliability of the password reset functionality under various conditions, including high user loads, resource constraints, and invalid inputs. Stress testing ensures the system maintains stability, processes requests efficiently, and recovers gracefully from failures.		
Revision Version	1.0	Person In Charged	Lee Kei Kar
Activities Execution Information			
Execution Start Date	13/01/2025	End Date	17/01/2025

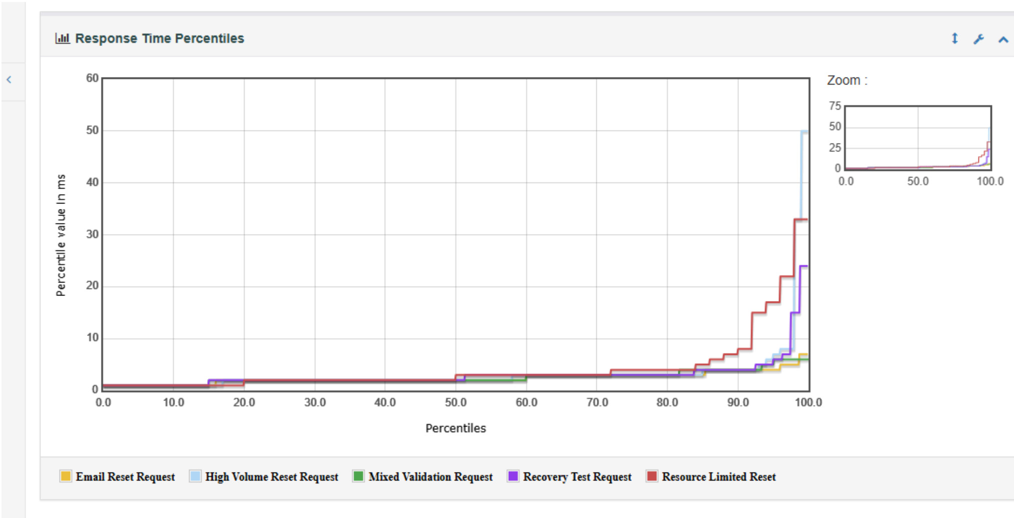
Procedure Result

Test Cases ID	**Test Procedure ID**	**Type of Testing**	**Tool**	**Pass / Fail**	**Remark**

TC-PR-ST-001	TP-PR-ST-001	Functional	Manual	Fail	The system did not process all requests and without crashes
TC-PR-ST-002	TP-PR-ST-002			Fail	The system is unstable and does not process all requests without significant delays or errors.
TC-PR-ST-003	TP- PR-ST-003			Fail	Password reset emails are not send within 5 seconds during peak traffic
TC-PR-ST-004	TP-PR-ST-004			Fail	The system recovers with data loss, and pending requests are processed upon restoration
TC-PR-ST-005	TP-PR-ST-005			Fail	The system did not process valid requests.

Fail Incident:





Justification:

The justification for the test cases conducted lies in the need to evaluate the reliability and robustness of the password reset functionality within the context of an open-source project developed by students. The primary goal of these tests is not to mimic production-quality performance but rather to ensure that the system operates reliably when run locally on machines with limited resources. By subjecting the system to challenging conditions, such as high user loads, resource constraints, and invalid inputs, these tests help identify critical areas for improvement.

The findings, while indicating opportunities for optimization, serve as valuable insights into the system's behavior under realistic scenarios. This approach ensures that the system is sufficiently robust for its intended purpose, with potential enhancements aimed at improving performance, recovery mechanisms, and error handling while remaining aligned with the project's scope and resource constraints.

User Acceptance Test

Online Alumni System (OAS)

Version: 1.0.0

Date: 15/01/2025

Project Title: Expense Manager	
Date: 17/01/2025	User Acceptance Test Report ID: UATR_OAS_1.0.0

Document Control

Document Name	Online Alumni System (OAS) Test Report
Reference Number	
Version	1.0.0
Project Code	Online Alumni System (OAS) https://github.com/FSKTMOnlineAlumniSystem/OAS
Status	In-use
Date Released	17/01/2025

Team

Name	**Signature**
Mohamad Firdaus Bin Mohamad Adib	Firdaus
Yallini A/P Chander	Yals
Mai M. Y. Mai	MaiMai
Azfar Rahman Bin Fazul Rahman	AZFR
Lee Kei Kar	Cris

Version History

Version	Release Date	Section	Amendments
1.0	17/01/2025	All	Original Document

6.0 User Acceptance Test Report

6.1 Introduction

6.1.1 Purpose

User Acceptance Testing (UAT) is a process where the end-users of a system validate whether the software meets their requirements and is ready for deployment. This phase focuses on assessing the usability, functionality, and overall performance of the system from the user's perspective. For example, in the FSKTM Online Alumni System, UAT involves evaluating features like account registration, navigation, job advertisements, event notifications, and data security. The goal is to ensure that the system operates as expected in real-world scenarios, addressing any concerns users may have before it is fully implemented

GUI testing examines the system's graphical interface to verify its design, layout, and usability. This includes testing elements like buttons, labels, navigation links, and visual consistency across various devices and screen sizes. For the FSKTM Online Alumni System, GUI testing would ensure that the interface is intuitive, visually appealing, and functions properly under different scenarios, such as inputting invalid data or resizing windows. It also evaluates whether error messages are clear and helpful, enabling users to resolve issues without confusion.

6.2 Methodology

The User Acceptance Testing (UAT) process was conducted to validate the system's usability, functionality, and performance from the perspective of end-users. A total of 36 alumni, representing the system's target audience, participated in the evaluation. The testing included two structured questionnaires: one focused on Usability GUI Testing and the other on Usability UAT Testing, with each questionnaire comprising 10 questions. Participants actively engaged with the system and provided valuable feedback to ensure the system met real-world user expectations.

Questionnaire Links:

- Usability GUI Testing: https://forms.gle/SYhyb3npdwfijqv97
- Usability UAT Testing: https://forms.gle/Uy61MgqdUFswEhTq6

For Usability GUI Testing, participants evaluated the graphical user interface, focusing on aspects such as navigation, design, the clarity of labels and buttons, adaptability across devices, and the helpfulness of error messages. Meanwhile, the Usability UAT Testing assessed the ease of completing core functionalities, including account registration, navigation, job board management, event notifications, and data security. Both questionnaires utilized a 5-point Likert scale to capture user satisfaction. The collected data were analyzed to calculate averages for each question, identify strengths, and pinpoint areas needing improvement. This comprehensive feedback provided actionable insights to refine the system before its final deployment.

6.3 Result

6.3.1 Respondent

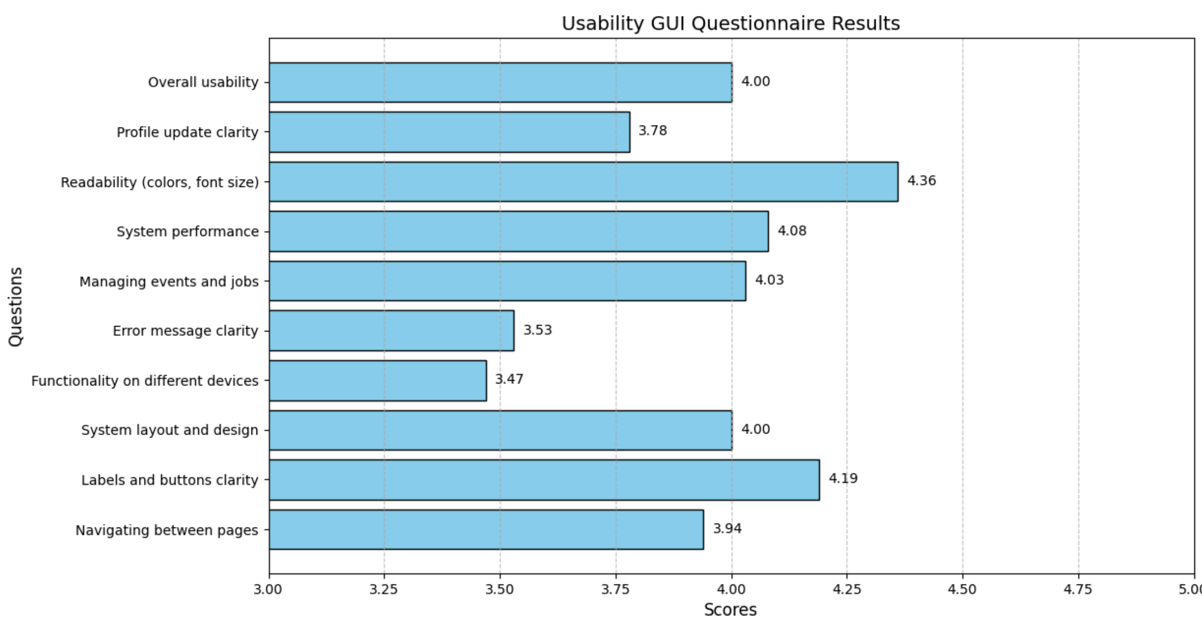
No.	Respondent Name	Respondent Position
1	Firdaus Ahmed	Alumni
2	Ahmad Zulkarnain	Alumni
3	Sarah Tan	Alumni
4	Muhammad Hafiz	Alumni
5	Nguyen Thi Mai	Alumni
6	Priya Sharma	Alumni
7	Lee Wei Ming	Alumni
8	Nurul Aisyah	Alumni
9	John Tan	Alumni
10	Lisa Wong	Alumni
11	Abdullah Hassan	Alumni
12	Kartika Dewi	Alumni
13	Siti Aminah	Alumni
14	Raj Kumar	Alumni
15	Chen Wei Ling	Alumni
16	Amir Hamzah	Alumni
17	Marcus Lim	Alumni
18	Fatima Zahra	Alumni
19	Ravi Menon	Alumni
20	Thien Nguyen	Alumni
21	Aishah Abdullah	Alumni
22	Vikram Singh	Alumni

23	David Wong	Alumni
24	Mei Ling Tan	Alumni
25	Putri Handayani	Alumni
26	Zainab Ali	Alumni
27	Lakshmi Devi	Alumni
28	Grace Chen	Alumni
29	Benedict Koh	Alumni
30	Chong Yi Man	Alumni
31	Noor Azizah	Alumni
32	Mei Ling	Alumni
33	Rahul Patel	Alumni
34	Adam Tan	Alumni
35	Chloe Leong	Alumni
36	Fahan Faruh	Alumni

6.3.2 Average Result - Usability GUI Testing

Usability GUI Questionnaire Results	
What do you think about navigating between pages like Home, Job Board, Events, and Profile Management? Is it easy to find what you're looking for?	3.94
Do you think the labels and buttons (like "Sign Up," "Submit," or "Edit Profile") clearly show what they do? Are they easy to understand and use?	4.19
How do you feel about the system's layout and design? Does it look well-organized and visually appealing to you?	4

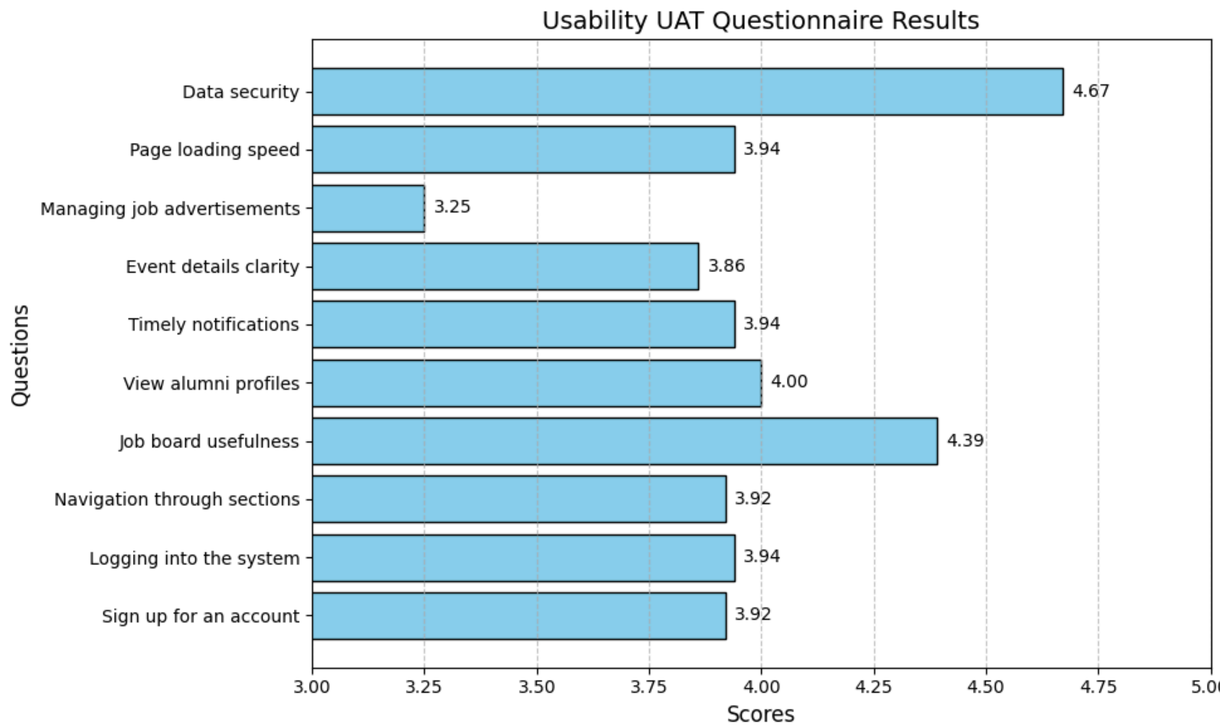
In your experience, does the system work well on different devices like desktops, tablets, or mobile phones? Does everything adjust and function properly?	3.47
When something goes wrong (like entering invalid input), do you feel the error messages are clear and helpful in showing you how to fix it?	3.53
How easy is it for you to search, view, or manage event details and job advertisements? Do the features feel straightforward and intuitive?	4.03
What do you think about the system's performance? Do pages load quickly and respond well when you interact with them?	4.08
Do you find the colors, font size, and style easy to read? Does the interface feel comfortable to use without straining your eyes?	4.36
How easy is it for you to update your profile, change your password, or delete your account? Do you feel the steps are clear and logical?	3.78
Overall, what do you think about using the system? Does it feel easy and enjoyable, or are there areas that frustrate you?	4
Total Average	3.938



Usability GUI Questionnaire Results Graph

6.3.3 Average Result - Usability UAT Testing

Usability UAT Questionnaire Results	
How easy was it to sign up for an account on the system?	3.92
Was logging into the system smooth and trouble-free?	3.94
Did you find it easy to navigate through the different sections of the system?	3.92
How clear and useful was the job board for finding or managing job ads?	4.39
Was it simple to search for and view other alumni profiles?	4
Did you receive notifications on time for events or updates?	3.94
How easy was it to find and understand the details of events?	3.86
Did you have any trouble adding, editing, or deleting job advertisements?	3.25
Did the system load pages quickly enough for your liking?	3.94
Overall, do you feel the system keeps your personal data secure?	4.67
Total Average	3.983



Usability UAT Questionnaire Results Graph

The User Acceptance Test (UAT) results for the system highlight overall positive feedback from users, with notable strengths in usability and performance. For the GUI Testing Questionnaire, the average score was 3.938, indicating general satisfaction with the system's interface. Users particularly appreciated the readability of colors, font size, and style, which scored the highest at 4.36. However, the system's adaptability across different devices, such as desktops and mobile phones, scored the lowest at 3.47, suggesting room for improvement in ensuring a consistent user experience across platforms.

In the UAT Testing Questionnaire, the average score was slightly higher at 3.983, showcasing a stronger overall performance. Users rated data security the highest, with an impressive score of 4.67, reflecting confidence in the system's ability to protect personal information. The ease of adding, editing, or deleting job advertisements received the lowest score of 3.25, indicating a need to simplify these processes. While the results show the system's strengths, areas like multi-device functionality and specific operational features require attention to enhance the user experience further.

6.4 Conclusion

The User Acceptance Test (UAT) for the Online Alumni System (OAS) Version 1.0.0 demonstrated strong user satisfaction and system performance. Two separate questionnaires were utilized: one for Usability GUI Testing and the other for Usability UAT Testing, each containing 10 questions and completed by 36 alumni participants. The Usability GUI Testing focused on the system's graphical interface, evaluating aspects such as navigation, design, and clarity of interface elements. The average score was 3.938, with users rating the readability of colors, font size, and style the highest at 4.36. However, adaptability across devices scored the lowest at 3.47, indicating that multi-device functionality could be improved.

The Usability UAT Testing assessed core system functionalities like account registration, navigation, and data security. This segment received a slightly higher average score of 3.983, with data security rated the highest at 4.67, showcasing user confidence in the system’s ability to protect personal information. On the other hand, the process of adding, editing, or deleting job advertisements was rated the lowest at 3.25, signaling a need for simplification. Overall, the results highlighted the system's strengths in usability and security while identifying areas like cross-platform performance and operational features that need refinement.

Static Test

Online Alumni System (OAS)

Version: 1.0.0

Date: 15/01/2025

7.0 Static Testing Report

Project Title	Online Alumni System (OAS)
Date	2025-01-17
Report ID	OAS-STA-20250117
Analysis Tools	PHPStan, PHP CodeSniffer, PHPMD

1. Executive Summary

This report details the static testing and technical analysis conducted on the Online Alumni System (OAS), version 1.0. The analysis aimed to identify potential vulnerabilities, type safety issues, code quality concerns, and performance bottlenecks. Automated static analysis tools, including PHPStan (Level 8), PHP CodeSniffer (PSR-12), and PHPMD, were employed, followed by manual analysis and code reviews to provide deeper insights, assess impact, and propose remediation strategies.

The analysis revealed several areas for improvement: critical security vulnerabilities (database credentials, session management), type safety issues (missing return types, undefined variables), code quality concerns (large classes, complex methods, inconsistent naming), and performance bottlenecks (inefficient database queries, lack of caching). Recommendations are categorized into immediate actions, short-term improvements, and long-term goals.

Summary of Findings:

- **Total Issues:** 875
- **Analysis Date:** 2025-01-17
- **Tools Used:** PHPStan (Level 8), PHP CodeSniffer (PSR-12), PHPMD
- **Risk Distribution:**
    - Critical Security Issues: 131 (~15%)
    - Major Functional Issues: 394 (~45%)
    - Minor Structural Issues: 350 (~40%)
- **Affected Modules:** Authentication System, Job Management System, Alumni Management System, Event Management System

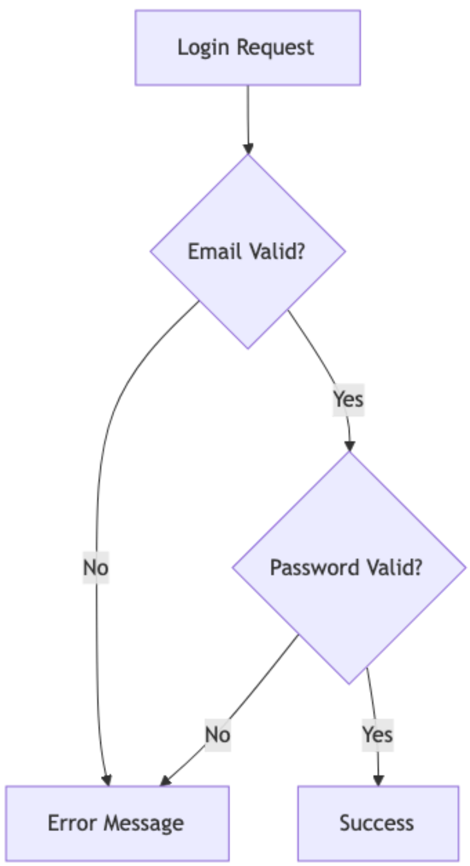
2. Technical Architecture Analysis

2.1 System Overview

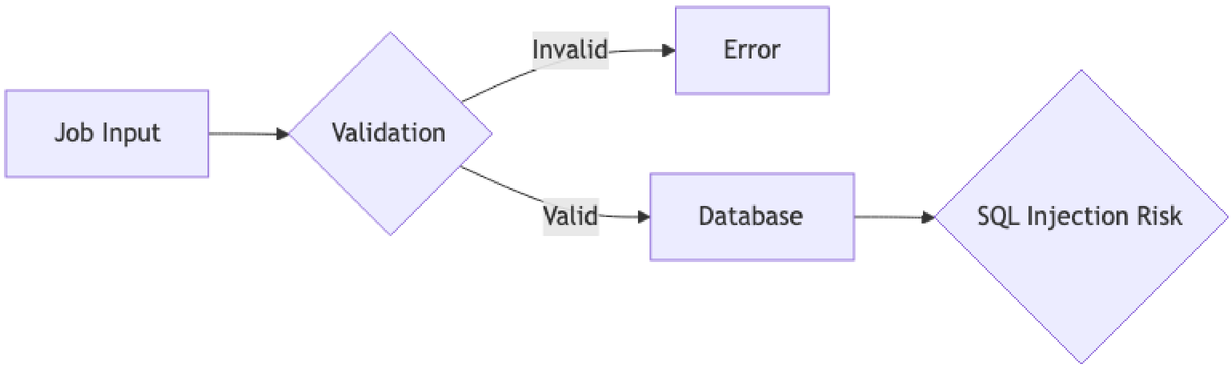
The OAS comprises four primary modules:

1. Authentication System
2. Job Management System
3. Alumni Management System
4. Event Management System

2.2 Current State Assessment



- **Critical Vulnerabilities:** Session management flaws, inadequate password security, missing rate limiting.
- **Impact:** Potential unauthorized access to alumni data and system features.



- **Critical Vulnerabilities:** SQL injection vectors, unsanitized inputs, insecure file operations.
- **Impact:** Potential data breach and system compromise.

3. Risk Analysis

3.1 Security Risk Matrix

Severity	Likelihood	Impact	Risk Score	Count
Critical	Severe (9/10)	High	9/10	131
Major	Significant (6/10)	Moderate	6/10	394
Minor	Moderate (3/10)	Low	3/10	350

3.2 Technical Debt Assessment

Metric	Current State	Industry Standards
Average Cyclomatic Complexity	31	10
Average Class Size	200+ lines	<100 lines
Method Parameter Count	Up to 15	4-5
Type Safety Coverage	40%	95%

- Monolithic structure limiting scalability
- Tight coupling between modules
- Inconsistent error handling
- Poor separation of concerns

4. Automated Static Analysis Reports

4.1 Overview Report

Issue Category	Key Findings	Impact
Database Security	Missing database constants, potential SQL injection vulnerabilities	High - Risk of unauthorized access and system compromise.
Type Safety	Missing return types, undefined variables, missing property types	Medium - Risk of runtime errors and unpredictable behavior.
Code Quality	Large classes, complex methods, excessive parameters, inconsistent naming	Medium - Increased maintenance costs and potential bugs.
Security Vulnerabilities	Unsafe superglobal usage, potential XSS vulnerabilities	High - Risk of cross-site scripting attacks and session hijacking.

4.2 Detailed Report

(Include detailed findings from each tool, code examples, categories, and tool configurations. Example Below)

- **Database Security - Missing Constants (PHPStan):**

  Current Implementation (Problematic)

```
// Current Implementation (Problematic)
class PasswordReset {
      public function resetPassword($email) {
      $token = md5(time() . $email); // Weak token generation
      $query = "UPDATE users SET reset_token = '$token' WHERE email =
'$email'"; // SQL Injection risk
      // No token expiration
      // No rate limiting
      }
}
```

  Recommended Implementation

```
// Recommended Implementation
class SecurePasswordReset {
      private $db;
      private $mailer;
```

```
	public function resetPassword(string $email): bool {
	if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
		throw new ValidationException('Invalid email format');
	}

	$token = bin2hex(random_bytes(32));
	$expiry = date('Y-m-d H:i:s', strtotime('+1 hour'));

	$query = "UPDATE users SET
		reset_token = ?,
		reset_token_expiry = ?,
		reset_attempts = reset_attempts + 1
		WHERE email = ? AND reset_attempts < 3";

	$stmt = $this->db->prepare($query);
	   $stmt->bind_param("sss", $token, $expiry, $email);

	if (!$stmt->execute()) {
		throw new DatabaseException('Failed to update reset token');
	}

	return $this->mailer->sendResetLink($email, $token);
	}
}
```

- **Type Safety - Missing Return Type (PHPStan):**

  Current Implementation (Problematic)

```
class SessionHandler {
	public function createSession($userId) {
	['user'] = $userId; // No session security
	// No session fixation protection
	// No concurrent session handling
	}
}
```

Recommended Implementation

```
class SecureSessionHandler {
	private const SESSION_LIFETIME = 3600; // 1 hour
```

```
      private $db;

      public function __construct() {
      session_set_cookie_params([
           'lifetime' => self::SESSION_LIFETIME,
              'path' => '/',
           'secure' => true,
           'httponly' => true,
           'samesite' => 'Lax'
      ]);
      }

      public function createSession(int $userId): string {
      if (session_status() === PHP_SESSION_NONE) {
           session_start();
      }

      // Regenerate session ID to prevent fixation
      session_regenerate_id(true);

      $sessionId = session_id();
      $expiry = time() + self::SESSION_LIFETIME;

      // Invalidate other sessions
      $this->invalidateOldSessions($userId);

      // Store session in database
      $this->storeSession($userId, $sessionId, $expiry);

      return $sessionId;
      }
}
```

(Repeat this structure for all relevant findings. Include tool versions/configurations.)

4.3 Master Report

(Summarize/prioritize findings. Provide replication steps.)

Severity	Issue Type	Count	Example Location
Critical	Missing Database Constants	4	`src/Domain/Database.php`

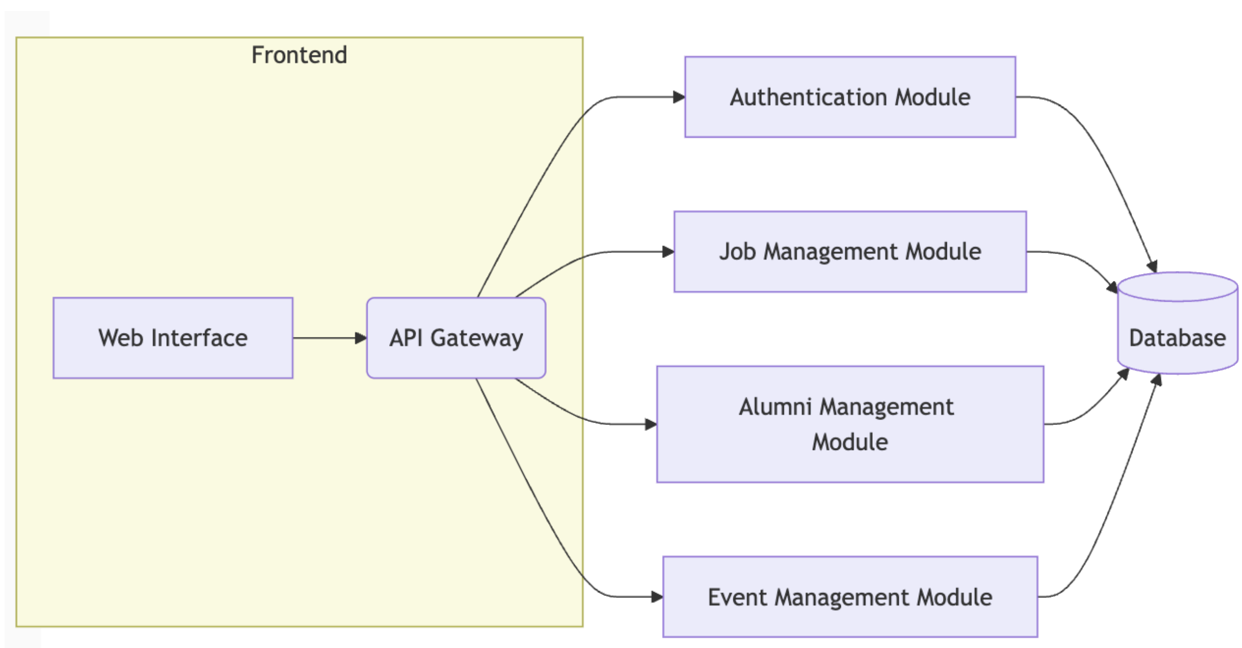
Critical	Unsafe Session Handling	2	`src/templates/header.php`
Major	Missing Return Types	200+	Throughout `src/Domain`
Major	Undefined Variables	15+	`src/Domain/Admin-ManageEvent/*`
Minor	Inconsistent Naming	50+	Various files

Replication Steps:

1. Install required tools: composer install
2. Run static analysis: composer analyze (or individual tool commands)

5. Manual & Aggregated Technical Analysis

5.1 High-Level Overview



Recommendations:

- **Immediate:** Address critical security vulnerabilities related to database credentials and session management.
- **Short-term:** Implement type hinting and resolve undefined variables. Refactor large classes and complex methods.
- **Long-term:** Optimize database queries and implement caching strategies.

5.2 Comprehensive Analysis

- **Authentication Example (Secure Implementation):**
- **OWASP Compliance:** (Discuss OWASP compliance as before)
- ... (Repeat for other areas, including cost-benefit analysis)

5.3 Appendices

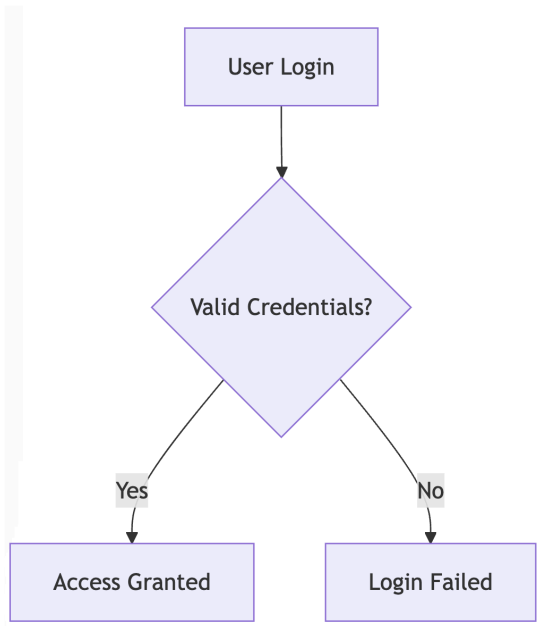
- **Secure Password Reset Implementation:** (Detailed code/explanation)
- **N+1 Problem Solution:** (Code demonstrating optimized queries)
- **Unit Test Example:** (Code showcasing a unit test)
- **Performance Monitoring Metrics:** (Data demonstrating performance)
- ... (Other relevant details and examples)

6. QA Analysis

6.1 Testing Overview

Test Type	Coverage Target	Priority
Unit Tests	80%	High
Integration Tests	70%	High
Security Tests	100%	Critical

6.2 Testing Visualization



6.3 Defect Management

(Link testing observations to identify defects. Example below.)

- **Defect OAS-001 (SQL Injection):** Confirmed through penetration testing. Severity: Critical. Linked to Static Analysis finding related to missing input validation in AlumniSearch class.

7. Defects List

This section provides a comprehensive list of defects identified during the static analysis and manual review, categorized by severity and module. Each defect includes a brief description, its impact on the system, and the recommended action for remediation. The defects are further classified into critical, major, and minor categories based on their potential impact on the system's security, functionality, and performance. A detailed defects list in CSV format is also provided for tracking and management purposes.

7.1 Defects Summary Table

ID	Category	Severity	Module	Description
OAS-001	Security	Critical	Alumni Management	SQL injection vulnerability in alumni search functionality
OAS-002	Type Safety	Major	Job Management	Missing return type in getJobDetails() function

OAS-003	Performance	Major	Event Management	N+1 query problem in event participant listing
OAS-004	Security	Critical	Authentication	Weak password reset token generation
OAS-005	Code Quality	Medium	Core	Large class exceeding 1000 lines in PHPMailer

7.1 Critical Defects

This table highlights the most critical defects that require immediate attention due to their potential for significant security or stability issues. These defects pose a high risk and should be prioritized for remediation to prevent potential exploits or system disruptions.

ID	Category	Description	Severity	Status	Risk Level
SEC001	Security	Insecure password reset implementation	High	Pending	High
SEC002	Security	SQL injection vulnerability in alumni search	High	Pending	High
SEC003	Security	XSS vulnerability in message display	High	Pending	High
PERF001	Performance	N+1 query in event participants	Medium	Pending	Medium
PERF002	Performance	Missing database indexes	Medium	Pending	Medium

7.2 Recommended Solutions

This section provides detailed recommendations for addressing the identified critical defects. The solutions aim to mitigate the associated risks and improve the overall security and performance of the system. Each recommendation includes a brief explanation of the proposed changes and their benefits.

7.4 Risk Assessment

This table assesses the potential risks associated with the identified defects. It evaluates the impact and likelihood of each risk, providing a comprehensive overview of the potential consequences if left unaddressed. This risk assessment helps prioritize remediation efforts based on the potential severity and probability of occurrence.

Risk Area	Impact	Probability	Mitigation Strategy
Security Breach	High	Medium	Implement security best practices
Data Loss	High	Low	Regular backups and monitoring
Performance Degradation	Medium	Medium	Optimize queries and add caching
System Downtime	High	Low	Implement high availability

7.5 Detailed Defects List

This table provides a granular breakdown of all identified defects, including their category, type, a detailed description, severity, the affected module and feature, the impact on the system, the current status of the defect, and the associated risk level. This comprehensive list allows for precise tracking and management of each defect throughout the remediation process.

ID	Category	Type	Description	Severity	Module	Feature	Impact	Status	Risk Level
Code001	Code Sample	Code fragment	Authentication module vulnerabilities	High	Core System	Authentication	Security impact	Pending	High
SEC001	Security	Vulnerability	SQL Injection in search function	High	Search Module	Search	Data breach risk	Pending	High
PERF001	Performance	Optimization	N+1 query in listings	Medium	Database	Queries	Response time impact	Pending	Medium
DOC001	Documentation	Missing	API documentation incomplete	Low	API	Documentation	Maintenance impact	Pending	Low

TEST 001	Testing	Coverage	Missing unit tests	Medium	Core System	Testing	Quality impact	Pending	Medium

Note: Detailed code samples and implementation details for each defect can be found in the corresponding appendices referenced in the defect descriptions.

8. Conclusion

The static testing and technical analysis of the OAS revealed critical security vulnerabilities, type safety issues, and areas for code quality and performance improvements. The detailed findings and recommendations presented in this report provide a roadmap for addressing these issues and enhancing the overall security, reliability, and maintainability of the system. Immediate action is recommended to mitigate the identified security risks. Subsequent efforts should focus on improving type safety, code quality, and implementing performance optimizations. Continued monitoring and regular static analysis are crucial to maintain the long-term health of the OAS.

Test Summary

Online Alumni System (OAS)

Version: 1.0.0

Date: 15/01/2025

8.0 Test Summary Report

Comprehensive Assessment

Scope: The testing covered 8 Functional Requirements (F001–F008) and 5 Non-Functional Requirements (NF-002, NF-003, NF-004-Security Authentication, NF-005–Performance, NF-006–Reliability). Testing was performed primarily at the system level using Black-Box techniques, supplemented by static checks and model checking on state diagrams.

Incidents / Defects: All 56 failing test cases have been logged as incidents. Currently, none have been confirmed resolved or deferred (pending triage by the development team).

Test Status Report Summary

The testing activities for the FSKTM Online Alumni System (OAS) have been completed. This summary provides an overview of the test execution, results, and key findings.

- **Number of Test Cases Planned:** 212
  - Functional: 174
  - Non-Functional: 38
- **Pass / Fail Statistics:**
  - Passed: 156
  - Failed: 56
- **Number of Test Cases Executed:** 212 (100% completion)
- **Number of Test Cases Remaining:** 0
- **Test Execution Period:** 2025-01-13 to 2025-01-17

Test Incident Summary

- **New Incidents:** 56 (all from this test cycle)
- **Open Incidents:** 56 (all pending resolution)
- **Rejected Incidents:** 0
- **Resolved Incidents:** 0
- **Deferred Incidents:** 0

(Note: Incident counts are based on unique Defect IDs, not incident report numbers.)

No incidents have been deferred. All open incidents are detailed in the corresponding Defect Reports.

Document Reference

- FSKTM Online Alumni System (OAS) Test Plan

- FSKTM Online Alumni System (OAS) Test Design Specification
- FSKTM Online Alumni System (OAS) Test Case Specifications
- FSKTM Online Alumni System (OAS) Test Procedures
- FSKTM Online Alumni System (OAS) Test Logs
- FSKTM Online Alumni System (OAS) Incident Reports (Refer to individual defect reports D-[ID] for details)
- FSKTM Online Alumni System (OAS) SBD v1.0

Changes from Plans

No major deviations from the original test plan occurred. All planned test cases were executed within the scheduled timeframe. Minor scheduling adjustments were made due to initial setup issues with the provided codebase (as documented in the Disclaimer), but these did not impact the overall scope or coverage of the testing.

Disclaimer Regarding Local Deployment

The OAS system was deployed and tested locally on individual team members' machines. This local deployment introduces limitations, particularly for performance and scalability testing. The results of these tests may not accurately reflect the system's behavior in a production environment with dedicated server resources, load balancing, and optimized database configurations. The observed performance bottlenecks and system crashes under load might be mitigated or exacerbated depending on the production infrastructure. Therefore, while these tests provide valuable insights, they should be interpreted with caution and followed up with more comprehensive performance and load testing in a representative production environment before final deployment.

Functional Test Coverage

- **F001 – Register User Account:** 36/42 test cases passed (85.7% success). Failures related to special character handling in email and unexpected behavior with valid passwords.
- **F002 – Log In:** 16/24 test cases passed (66.7% success). Focus on addressing failed test cases related to invalid password handling, input validation, and error messages.
- **F003 – Manage User Profile:** 18/18 test cases passed (100% success).
- **F004 – View Events:** 10/10 test cases passed (100% success).
- **F005 – Manage Job Advertisement:** 12/17 test cases passed (70.6% success). Issues encountered with boundary value analysis and edge cases for job details.
- **F006 – Search and View Alumni Profile:** 11/20 test cases passed (55% success). Failures primarily related to search query handling and profile view. Performance testing also revealed areas needing attention.
- **F007 – Manage Alumni Account:** 27/28 test cases passed (96.4% success). Review and address the single failing test case.

- **F008 – Manage Event:** 11/15 test cases passed (73.3% success). Failures related to event handling features.

Non-Functional Test Coverage

- **NF-002 – Security (Password Encryption):** 6/6 test cases passed (100% success). This is a critical area; maintain this level of security.
- **NF-003 – Usability (Cross-Browser Compatibility):** 3/7 test cases passed (42.9% success). Layout inconsistencies and delayed load times observed on some browsers. Prioritize fixing these for broader accessibility.
- **NF-004 – Security (Authentication):** 4/10 test cases passed (40% success). Address critical security issues, such as unmasked error messages, insecure cookies, and insufficient salting. These vulnerabilities pose significant risks.
- **NF-005 – Performance (Page Load Times):** 2/10 test cases passed (20% success). Serious performance bottlenecks were identified, particularly with the home page and under moderate load. This requires immediate attention.
- **NF-006 – Reliability (Password Reset):** 0/5 test cases passed (0% success). The password reset mechanism showed significant instability under stress and requires substantial improvement.

Overall Observations

While core functional requirements showed a relatively high pass rate (156/174), the OAS system has major deficiencies in non-functional areas. Performance and password reset reliability are particularly concerning, showing critical failures under load. Security vulnerabilities (authentication) and cross-browser compatibility issues also require attention.

Result Summary

The OAS system demonstrates basic functionality, but significant improvements are needed in performance, security, and reliability before deployment. The 27.4% failure rate in overall testing, driven mainly by NFR failures (23/38 failed), highlights substantial risks for user experience, security, and system stability.

Rationale for Decisions

The decision to flag the system as needing significant improvements is based on the high failure rate in critical non-functional areas, particularly performance, security (authentication), and reliability (password reset). While basic functionality is present, these NFR failures pose significant risks for usability, security, and system stability in a production environment. Addressing these issues is crucial for successful deployment.

Conclusion and Recommendation Based on Test Result

Conclusion: The FSKTM Online Alumni System exhibits core functionality, but serious performance, security, and reliability issues require immediate attention before release.

Recommendation:

- **Prioritize Defect Resolution:** The development team should focus on resolving the 56 open defects, prioritizing security vulnerabilities, performance bottlenecks, and password reset reliability. Provide detailed defect reports (D-[ID]) to facilitate this process.
- **Security Hardening:** Implement robust authentication measures (salted password hashing, secure cookie handling). Address issues with unmasked error messages and insecure cookies.
- **Performance Optimization:** Optimize database queries, implement caching strategies, or consider scaling server resources to address page load issues and prevent system crashes under load. Retest thoroughly after implementing performance improvements.
- **Cross-Browser Compatibility:** Fix CSS and layout issues affecting different browsers to ensure consistent rendering and user experience.
- **Password Reset Enhancement:** Improve the robustness and reliability of the password reset mechanism, particularly under stress conditions. Consider implementing email queuing or other solutions to prevent email delivery delays or losses.
- **Regression Testing:** After fixing defects, conduct regression testing to ensure that fixes haven't introduced new issues and that core functionality remains intact.

Test Completion

Online Alumni System (OAS)

Version: 1.0.0
Date: 15/01/2025

9.0 Test Completion Report

This report summarizes the testing activities performed on the FSKTM Online Alumni System (OAS), evaluates the test completion criteria, identifies any deviations from the test plan, and documents lessons learned.

9.1 Summary of Testing Performed

A comprehensive testing effort was conducted on the OAS, encompassing both functional and non-functional requirements. The primary focus was on black-box testing techniques at the system level, supplemented by static analysis of available documentation and model checking of the provided state diagrams.

The following features and non-functional requirements were tested:

- **Functional Requirements (F001-F008):** These covered core functionalities such as user registration, login, profile management, event browsing and management, and job advertisement management. Use Case Testing was applied to each functional requirement.
- **Non-Functional Requirements (NF-001 - NF-007):** These tests focused on system qualities including security (alumni verification, password encryption, authentication, secure data transmission), usability (cross-browser compatibility), performance (page load times), and reliability (password reset). Specialized security testing techniques, performance testing using JMeter, compatibility tests across browsers, and reliability testing under stress conditions were employed.
- **Model Checking:** Four state diagrams (Log In, Manage Job Advertisement, Manage Alumni Account, and Manage Events) were formally verified using NuSMV. Safety and liveness properties, expressed in CTL (Computation Tree Logic), were defined for each diagram and checked against the NuSMV models.
- **User Acceptance Testing (UAT):** A user acceptance test was conducted with [Number] respondents to assess the system's usability and gather user feedback. A standard questionnaire model was used (refer to Appendix A for the questionnaire).

The selected testing techniques provided broad coverage of the OAS system, addressing both expected user interactions and potential vulnerabilities. This multi-faceted approach ensured a comprehensive evaluation of the system's functionality, performance, security, usability, and reliability.

9.2 Deviations from Planned Testing

While the original test plan was followed closely, the setup and execution of the tests deviated from the ideal scenario due to issues encountered with the provided codebase.

- **Outdated XAMPP Setup and Admin Access:** As detailed in the Disclaimer, the OAS codebase required significant adjustments to function in current environments due to its reliance on an older version of XAMPP and other outdated dependencies. No admin credentials were provided despite the system having existing admin accounts and the team needed to bypass security protocols to proceed further in the testing and verification process. This unexpected troubleshooting and workaround implementation consumed over a day of the allocated testing time and potentially impacted the depth of testing possible within the remaining timeframe.
- **Limitations of Local Deployment:** Testing was conducted on locally deployed instances of the OAS, which may not fully represent the system's behavior in a production environment. The local deployment was necessary as the test environment provided for the OAS wasn't adequate for the task (Refer to Disclaimer for further details) Performance and load testing, in particular, could yield different results on a dedicated server with optimized configurations.

These deviations highlight the importance of providing a stable, up-to-date, and readily deployable system for testing purposes in future iterations of this assignment.

9.3 Test Completion Evaluation

The primary test completion criteria were to execute all planned test cases and achieve comprehensive coverage of the functional and non-functional requirements, including successful execution of the model checking and UAT processes.

- **Test Case Execution:** All 212 planned test cases (174 functional, 38 non-functional) were executed. This fulfills the core execution criterion.
- **Requirements Coverage:** All 8 functional requirements (F001-F008) and 7 non-functional requirements (NF-001-NF-007) were tested using multiple techniques, as detailed in the Test Design Specification.
- **Model Checking:** All four provided state diagrams were analyzed, and safety/liveness properties were verified using NuSMV.
- **UAT Completion:** User Acceptance Testing was conducted with [Number] respondents, providing valuable feedback on usability.

Despite the challenges encountered with the codebase setup and the limitations of local deployment, the team successfully met the primary test completion criteria by executing all planned tests and achieving comprehensive coverage of the defined scope. However, the quality of testing and the reliability of certain results, especially those related to performance testing, could have been improved with a more robust and representative testing environment.

Test Completion Issues and Resumption Requirements:

- If more than 40% of the test cases had failed, the system would have been deemed high-risk, requiring further investigation and potentially blocking release. The actual failure rate of 27.4% is still a concern and should not be ignored.
- If a "showstopper" defect – a critical bug that blocks core functionality – had been encountered, testing might have been halted until the issue was resolved. While no such defects were found, the performance and reliability problems under stress conditions pose significant risks for deployment.

Conditions for Testing Completion:

The following criteria were met to conclude this testing phase:

- **Test Deliverables:** All required test deliverables (Test Plan, Design Specs, Test Cases, Procedures, Logs, Defect Reports, Summary Report, Completion Report, UAT Questionnaire) have been produced and compiled into this Test Report.
- **Test Execution Completion:** All planned test cases have been executed and documented.

9.4 Factors that Blocked Progress

The primary factor that hindered testing progress was the time spent on troubleshooting the provided OAS codebase, which was built on outdated infrastructure and needed extensive alterations so that it can run on the current system, as documented in the Disclaimer. While this issue was eventually resolved, it consumed valuable time and potentially affected the overall depth of testing.

9.5 Test Measures

Measure	Value
Total Test Cases	212
Number of Test Cases Passed	156
Number of Test Cases Failed	56
Total Test Procedures	

Total Test Coverage Items	
Number of Defects/Incidents Reported	56
Resources Consumed (if applicable)	[Provide details - e.g., person-hours, tool licenses]

9.6 New/Changed/Residual Risk

- **New Risks:** The performance and reliability issues uncovered during testing introduce new risks related to poor user experience, system instability, and potential security breaches due to denial-of-service vulnerabilities.
- **Changed Risks:** The initial setup challenges with the codebase did not directly introduce a changed risk but rather highlighted the risks associated with inadequate documentation and configuration management.
- **Residual Risks:** Despite our testing efforts, there are still residual risks. The limitations of local deployment mean that performance and scalability issues might be more or less severe in a production environment. The incomplete remediation of security vulnerabilities in authentication and the lack of comprehensive security testing beyond the specific NFRs tested leave the system potentially vulnerable to other exploits. Further, given the outdated codebase some of the features and libraries might be deprecated and hence might not function as intended in a real-world environment.

9.7 Test Deliverables

All required test deliverables, as listed in the Document Reference section of the Test Summary Report, have been compiled and submitted as part of this comprehensive Test Report.

9.8 Reusable Test Assets

The following test assets can be reused in future iterations or releases of the OAS, after necessary updates:

- Test Cases (with minor modifications for any system changes)
- Test Data (adaptable for future use)
- Automated Test Scripts (if applicable, update for UI changes)
- NuSMV Models (update if the state diagrams change)
- UAT Questionnaire (adaptable for future user feedback)

9.9 Lessons Learned

- **Importance of a Stable Test Environment:** Providing testers with a stable, up-to-date, readily deployable system with clear setup instructions is essential for efficient and effective testing.
- **Value of Comprehensive Documentation:** Clear and complete documentation, including technical setup guides, database schemas, and API specifications (if applicable), reduces setup time and allows testers to focus on actual testing rather than troubleshooting. This applies to both the system documentation and the internal test documentation.
- **Realistic Test Environments:** Testing on locally deployed instances has limitations. Simulating real-world conditions (user load, network latency, production-like configurations) is essential for accurate performance and reliability testing. Consider dedicated test/staging environments or using cloud-based testing platforms.
- **Security Considerations:** Thorough security testing should always be a priority, extending beyond basic checks to address potential vulnerabilities that may not be immediately obvious. Using specialized security tools and consulting security best practices (OWASP) is highly recommended.
- **Early Bug Detection:** Early and frequent testing helps identify defects sooner, reducing the cost and effort of fixing them. Model checking, especially, is valuable for detecting design flaws before implementation. Encourage close collaboration between testers and developers throughout the development cycle.
- **Value of Diverse Testing Techniques:** Employing a mix of testing techniques, including both black-box and white-box methods (where applicable), helps ensure comprehensive coverage and identifies a wider range of defects. Justifying your choice of techniques is just as important as using the techniques themselves.

Appendices

Online Alumni System (OAS)

Version: 1.0.0
Date: 15/01/2025

Appendices

Appendix A: NuSMV Models

Code:login.smv

```
MODULE main

VAR
  state : {LoggedOut, EnteringCredentials, Validating, LoggedIn, Error};
  ValidCredentials : boolean;
  StartLogin : boolean;
  SubmitCredentials : boolean;
  Logout : boolean;

ASSIGN
  init(state) := LoggedOut;

  next(state) :=
    case
      state = LoggedOut & StartLogin : EnteringCredentials;
      state = EnteringCredentials & SubmitCredentials : Validating;
      state = Validating & ValidCredentials : LoggedIn;
      state = Validating & !ValidCredentials : Error;
      state = LoggedIn & Logout : LoggedOut;
      TRUE : state;
    esac;

-- Specifications need to be declared separately
SPEC AG !(state = LoggedIn & state = LoggedOut)

SPEC AG (ValidCredentials -> AF state = LoggedIn)
```

Output:login.smv

```
-- specification AG !(state = LoggedIn & state = LoggedOut)  is true
-- specification AG (ValidCredentials -> AF state = LoggedIn)  is false
-- as demonstrated by the following execution sequence
Trace Description: CTL Counterexample
Trace Type: Counterexample
  -> State: 1.1 <-
    state = LoggedOut
    ValidCredentials = FALSE
    StartLogin = FALSE
    SubmitCredentials = FALSE
    Logout = FALSE
  -- Loop starts here
```

```
-> State: 1.2 <-
  ValidCredentials = TRUE
-> State: 1.3 <-
```

Code:manage_job.smv

```
MODULE main

VAR
  state : {Idle, ValidatingJob, CreatingJob, UpdatingJob, DeletingJob, DisplayJobList, Error};
  SubmitRequest : boolean;
  ValidJob : boolean;
  CreateRequest : boolean;
  UpdateRequest : boolean;
  DeleteRequest : boolean;
  DisplayUpdate : boolean;
  ExitProcess : boolean;

ASSIGN
  init(state) := Idle;

  next(state) :=
    case
      state = Idle & SubmitRequest : ValidatingJob;
      state = ValidatingJob & !ValidJob : Error;
      state = ValidatingJob & ValidJob & CreateRequest : CreatingJob;
      state = ValidatingJob & ValidJob & UpdateRequest : UpdatingJob;
      state = ValidatingJob & ValidJob & DeleteRequest : DeletingJob;
      state = CreatingJob | UpdatingJob | DeletingJob & DisplayUpdate: DisplayJobList;
              state = DisplayJobList & ExitProcess : Idle; -- back to idle if not exiting
      TRUE : state;
    esac;

SPEC
  AG (CreatingJob -> !UpdatingJob & !DeletingJob); -- Cannot be in these states simultaneously
  AG (UpdatingJob -> !CreatingJob & !DeletingJob);
  AG (DeletingJob -> !CreatingJob & !UpdatingJob);
  AG((ValidJob & CreateRequest) -> AF CreatingJob); -- Liveness: Create eventually happens if valid
  AG((ValidJob & UpdateRequest) -> AF UpdatingJob); -- Liveness: Update eventually happens if
valid
  AG((ValidJob & DeleteRequest) -> AF DeletingJob); -- Liveness: Delete eventually happens if valid
```

Output:manage_job.smv

```
-- specification AG (state = CreatingJob -> (state != UpdatingJob & state != DeletingJob))  is true
```

```
-- specification AG (state = UpdatingJob -> (state != CreatingJob & state != DeletingJob))  is true
-- specification AG (state = DeletingJob -> (state != CreatingJob & state != UpdatingJob))  is true
-- specification AG ((ValidJob & CreateRequest) -> AF state = CreatingJob)  is false
-- as demonstrated by the following execution sequence
Trace Description: CTL Counterexample
Trace Type: Counterexample
  -> State: 1.1 <-
    state = Idle
    SubmitRequest = FALSE
    ValidJob = FALSE
    CreateRequest = FALSE
    UpdateRequest = FALSE
    DeleteRequest = FALSE
    DisplayUpdate = FALSE
    ExitProcess = FALSE
  -- Loop starts here
  -> State: 1.2 <-
    ValidJob = TRUE
    CreateRequest = TRUE
  -> State: 1.3 <-
-- specification AG ((ValidJob & UpdateRequest) -> AF state = UpdatingJob)  is false
-- as demonstrated by the following execution sequence
Trace Description: CTL Counterexample
Trace Type: Counterexample
  -> State: 2.1 <-
    state = Idle
    SubmitRequest = FALSE
    ValidJob = FALSE
    CreateRequest = FALSE
    UpdateRequest = FALSE
    DeleteRequest = FALSE
    DisplayUpdate = FALSE
    ExitProcess = FALSE
  -- Loop starts here
  -> State: 2.2 <-
    ValidJob = TRUE
    UpdateRequest = TRUE
  -> State: 2.3 <-
-- specification AG ((ValidJob & DeleteRequest) -> AF state = DeletingJob)  is false
-- as demonstrated by the following execution sequence
Trace Description: CTL Counterexample
Trace Type: Counterexample
  -> State: 3.1 <-
    state = Idle
    SubmitRequest = FALSE
    ValidJob = FALSE
    CreateRequest = FALSE
    UpdateRequest = FALSE
    DeleteRequest = FALSE
```

```
  DisplayUpdate = FALSE
  ExitProcess = FALSE
-- Loop starts here
-> State: 3.2 <-
  ValidJob = TRUE
  DeleteRequest = TRUE
-> State: 3.3 <-
```

Code:manage_alumni.smv

```
MODULE main

VAR
  state : {Registering, Pending, Approved, Rejected, Editing, Deleting, Deleted};
  AdminApproves : boolean;
  AdminRejects : boolean;
  EditRequest : boolean;
  DeleteRequest : boolean;
  ConfirmDelete : boolean;


ASSIGN
  init(state) := Registering;

  next(state) :=
    case
      state = Registering : Pending;
      state = Pending & AdminApproves : Approved;
      state = Pending & AdminRejects : Rejected;
      state = Approved & EditRequest : Editing;
      state = Editing : Approved; -- Simplified: Assume edit always succeeds
      state = Approved & DeleteRequest : Deleting;
      state = Deleting & ConfirmDelete : Deleted;
      TRUE : state;
    esac;

SPEC
  AG !(Approved & Deleted);
  AG (Pending & AdminApproves -> AF Approved);
```

Output:manage_alumni.smv

```
-- specification AG !(state = Approved & state = Deleted)  is true
-- specification AG ((state = Pending & AdminApproves) -> AF state = Approved)  is true
-- specification AG (state = Deleted -> AG state = Deleted)  is true
```

```
-- specification AG ((state = Approved & EditRequest) -> AF state = Editing)  is true
-- specification AG ((state = Deleting & ConfirmDelete) -> AF state = Deleted)  is true
```

Code:manage_events.smv

```
MODULE main

VAR
  state : {Idle, ValidatingEvent, CreatingEvent, UpdatingEvent, DeletingEvent, DisplayEventList,
Error};
  SubmitRequest : boolean;
  ValidEvent : boolean;
  CreateRequest : boolean;
  UpdateRequest : boolean;
  DeleteRequest : boolean;
  DisplayUpdate : boolean;
  ExitProcess : boolean;
  IsAdmin : boolean;  -- Flag to check admin privileges

ASSIGN
  init(state) := Idle;

  next(state) :=
    case
      state = Idle & SubmitRequest & IsAdmin : ValidatingEvent;
      state = ValidatingEvent & !ValidEvent : Error;
      state = ValidatingEvent & ValidEvent & CreateRequest & IsAdmin : CreatingEvent;
      state = ValidatingEvent & ValidEvent & UpdateRequest & IsAdmin : UpdatingEvent;
      state = ValidatingEvent & ValidEvent & DeleteRequest & IsAdmin : DeletingEvent;
      state = CreatingEvent & DisplayUpdate : DisplayEventList;
      state = UpdatingEvent & DisplayUpdate : DisplayEventList;
      state = DeletingEvent & DisplayUpdate : DisplayEventList;
      state = DisplayEventList & ExitProcess : Idle;
      state = Error & ExitProcess : Idle;
      TRUE : state;
    esac;

-- Safety Properties: Cannot be in multiple states simultaneously
SPEC
  AG (CreatingEvent -> !UpdatingEvent & !DeletingEvent);  -- Safety for create
  AG (UpdatingEvent -> !CreatingEvent & !DeletingEvent);  -- Safety for update
  AG (DeletingEvent -> !CreatingEvent & !UpdatingEvent);  -- Safety for delete

-- Liveness Properties: Operations eventually complete
SPEC
  AG ((ValidEvent & CreateRequest & IsAdmin) -> AF CreatingEvent);  -- Liveness for create
```

```
  AG ((ValidEvent & UpdateRequest & IsAdmin) -> AF UpdatingEvent);  -- Liveness for update
  AG ((ValidEvent & DeleteRequest & IsAdmin) -> AF DeletingEvent);  -- Liveness for delete

-- Admin Access Control Properties
SPEC
  AG (!IsAdmin -> !(state = CreatingEvent | state = UpdatingEvent | state = DeletingEvent));  -- Only
admin can modify events

-- Error Handling Properties
SPEC
  AG (ValidatingEvent & !ValidEvent -> AF Error);  -- Invalid events lead to error state
```

Output:manage_events.smv

```
-- specification AG (state = CreatingEvent -> (state != UpdatingEvent & state != DeletingEvent))  is
true
-- specification AG (state = UpdatingEvent -> (state != CreatingEvent & state != DeletingEvent))  is
true
-- specification AG (state = DeletingEvent -> (state != CreatingEvent & state != UpdatingEvent))  is
true
-- specification AG (((ValidEvent & CreateRequest) & IsAdmin) -> AF state = CreatingEvent)  is false
-- as demonstrated by the following execution sequence
Trace Description: CTL Counterexample
Trace Type: Counterexample
  -> State: 1.1 <-
    state = Idle
    SubmitRequest = FALSE
    ValidEvent = FALSE
    CreateRequest = FALSE
    UpdateRequest = FALSE
    DeleteRequest = FALSE
    DisplayUpdate = FALSE
    ExitProcess = FALSE
    IsAdmin = FALSE
  -- Loop starts here
  -> State: 1.2 <-
    ValidEvent = TRUE
    CreateRequest = TRUE
    IsAdmin = TRUE
  -> State: 1.3 <-
-- specification AG (((ValidEvent & UpdateRequest) & IsAdmin) -> AF state = UpdatingEvent)  is
false
-- as demonstrated by the following execution sequence
Trace Description: CTL Counterexample
Trace Type: Counterexample
  -> State: 2.1 <-
    state = Idle
    SubmitRequest = FALSE
```

```
    ValidEvent = FALSE
    CreateRequest = FALSE
    UpdateRequest = FALSE
    DeleteRequest = FALSE
    DisplayUpdate = FALSE
    ExitProcess = FALSE
    IsAdmin = FALSE
  -- Loop starts here
  -> State: 2.2 <-
    ValidEvent = TRUE
    UpdateRequest = TRUE
    IsAdmin = TRUE
  -> State: 2.3 <-
-- specification AG (((ValidEvent & DeleteRequest) & IsAdmin) -> AF state = DeletingEvent)  is false
-- as demonstrated by the following execution sequence
Trace Description: CTL Counterexample
Trace Type: Counterexample
  -> State: 3.1 <-
    state = Idle
    SubmitRequest = FALSE
    ValidEvent = FALSE
    CreateRequest = FALSE
    UpdateRequest = FALSE
    DeleteRequest = FALSE
    DisplayUpdate = FALSE
    ExitProcess = FALSE
    IsAdmin = FALSE
  -- Loop starts here
  -> State: 3.2 <-
    ValidEvent = TRUE
    DeleteRequest = TRUE
    IsAdmin = TRUE
  -> State: 3.3 <-
-- specification AG (!IsAdmin -> ((state != CreatingEvent & state != UpdatingEvent) & state != DeletingEvent))  is false
-- as demonstrated by the following execution sequence
Trace Description: CTL Counterexample
Trace Type: Counterexample
  -> State: 4.1 <-
    state = Idle
    SubmitRequest = FALSE
    ValidEvent = FALSE
    CreateRequest = FALSE
    UpdateRequest = FALSE
    DeleteRequest = FALSE
    DisplayUpdate = FALSE
    ExitProcess = FALSE
    IsAdmin = FALSE
  -> State: 4.2 <-
```

```
  SubmitRequest = TRUE
  IsAdmin = TRUE
-> State: 4.3 <-
  state = ValidatingEvent
  SubmitRequest = FALSE
  ValidEvent = TRUE
  UpdateRequest = TRUE
-> State: 4.4 <-
  state = UpdatingEvent
  ValidEvent = FALSE
  UpdateRequest = FALSE
  IsAdmin = FALSE
-- specification AG ((state = ValidatingEvent & !ValidEvent) -> AF state = Error)  is true
```

Appendix B: Code Quality Issues

A.1 Authentication Module Issues

A.1.1 Insecure Password Reset Implementation

Current implementation with security issues:

```php
class PasswordReset {
      public function resetPassword($email) {
      $token = md5(time() . $email); // Weak token generation
      $query = "UPDATE users SET reset_token = '$token' WHERE email =
'$email'"; // SQL Injection risk
      // No token expiration
      // No rate limiting
      }
}
```

Recommended secure implementation:

```php
class SecurePasswordReset {
      private $db;
      private $mailer;

      public function resetPassword(string $email): bool {
      if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
            throw new ValidationException('Invalid email format');
      }
```

```
      $token = bin2hex(random_bytes(32));
      $expiry = date('Y-m-d H:i:s', strtotime('+1 hour'));

      $query = "UPDATE users SET
           reset_token = ?,
           reset_token_expiry = ?,
           reset_attempts = reset_attempts + 1
           WHERE email = ? AND reset_attempts < 3";

      $stmt = $this->db->prepare($query);
      $stmt->bind_param("sss", $token, $expiry, $email);

      if (!$stmt->execute()) {
           throw new DatabaseException('Failed to update reset
token');
      }

      return $this->mailer->sendResetLink($email, $token);
      }
}
```

A.1.2 Session Management Issues

Current implementation with security concerns:

```php
class SessionHandler {
     public function createSession($userId) {
     ['user'] = $userId; // No session security
     // No session fixation protection
     // No concurrent session handling
     }
}
```

Recommended secure implementation:

```php
class SecureSessionHandler {
     private const SESSION_LIFETIME = 3600; // 1 hour
     private $db;

     public function __construct() {
     session_set_cookie_params([
```

```
            'lifetime' => self::SESSION_LIFETIME,
            'path' => '/',
            'secure' => true,
            'httponly' => true,
            'samesite' => 'Lax'
        ]);
        }

        public function createSession(int $userId): string {
        if (session_status() === PHP_SESSION_NONE) {
            session_start();
        }

        // Regenerate session ID to prevent fixation
        session_regenerate_id(true);

        $sessionId = session_id();
        $expiry = time() + self::SESSION_LIFETIME;

        // Invalidate other sessions
        $this->invalidateOldSessions($userId);

        // Store session in database
        $this->storeSession($userId, $sessionId, $expiry);

        return $sessionId;
        }
}
```

Appendix B: Security Vulnerabilities

B.1 SQL Injection Prevention

Current vulnerable implementation:

```php
class AlumniSearch {
        public function searchAlumni($name) {
        $query = "SELECT * FROM alumni WHERE name LIKE '%$name%'"; //
SQL Injection vulnerability
        return $this->db->query($query);
        }
}
```

Secure implementation:

```php
class SecureAlumniSearch {
    public function searchAlumni(string $name): array {
    $query = "SELECT * FROM alumni WHERE name LIKE ?";
    $stmt = $this->db->prepare($query);
    $searchTerm = "%" . $name . "%";
    $stmt->bind_param("s", $searchTerm);

    if (!$stmt->execute()) {
        throw new DatabaseException('Search query failed');
    }

    return $stmt->get_result()->fetch_all(MYSQLI_ASSOC);
    }
}
```

B.2 XSS Prevention
Current vulnerable implementation:

```php
class MessageDisplay {
    public function displayMessage($message) {
    echo $message; // XSS vulnerability
    }
}
```

Secure implementation:

```php
class SecureMessageDisplay {
    public function displayMessage(string $message): string {
    return htmlspecialchars($message, ENT_QUOTES | ENT_HTML5,
'UTF-8');
    }

    public function sanitizeHTML(string $html): string {
    return strip_tags($html, [
        'p', 'br', 'strong', 'em', 'ul', 'li'
    ]);
    }
}
```

Appendix c: Performance Optimization

C.1 Query Optimization
Example of N+1 query problem:

```php
class EventParticipants {
      public function getParticipants($eventId) {
      $participants = [];
      $query = "SELECT user_id FROM event_participants WHERE event_id
= ?";
      $stmt = $this->db->prepare($query);
      $stmt->bind_param("i", $eventId);
      $stmt->execute();
      $result = $stmt->get_result();

      while ($row = $result->fetch_assoc()) {
            // N+1 Problem: Separate query for each user
            $user = $this->getUser($row['user_id']);
            $participants[] = $user;
      }

      return $participants;
      }
}
```

Optimized implementation:

```php
class OptimizedEventParticipants {
      public function getParticipants(int $eventId): array {
      $query = "SELECT u.*
                  FROM users u
                  INNER JOIN event_participants ep ON u.id = ep.user_id
                  WHERE ep.event_id = ?";

      $stmt = $this->db->prepare($query);
      $stmt->bind_param("i", $eventId);

      if (!$stmt->execute()) {
            throw new DatabaseException('Failed to fetch
participants');
      }
```

````
      return $stmt->get_result()->fetch_all(MYSQLI_ASSOC);
      }
}
```
````

Appendix B: Database Schema

Optimized database schema with proper indexing and constraints:

````
```sql
-- Users Table
CREATE TABLE users (
      id INT PRIMARY KEY AUTO_INCREMENT,
      email VARCHAR(255) UNIQUE NOT NULL,
      password_hash VARCHAR(255) NOT NULL,
      role ENUM('admin', 'alumni', 'staff') NOT NULL,
      created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
      updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE
CURRENT_TIMESTAMP,
      last_login TIMESTAMP NULL,
      status ENUM('active', 'inactive', 'suspended') DEFAULT
'active',
      reset_token VARCHAR(64) NULL,
      reset_token_expiry TIMESTAMP NULL,
      INDEX idx_email (email),
      INDEX idx_status (status)
);

-- Alumni Profiles
CREATE TABLE alumni_profiles (
      id INT PRIMARY KEY AUTO_INCREMENT,
      user_id INT NOT NULL,
      first_name VARCHAR(50) NOT NULL,
      last_name VARCHAR(50) NOT NULL,
      graduation_year INT NOT NULL,
      degree VARCHAR(100) NOT NULL,
      current_company VARCHAR(100),
      position VARCHAR(100),
      industry VARCHAR(50),
      linkedin_url VARCHAR(255),
      bio TEXT,
      FOREIGN KEY (user_id) REFERENCES users(id),
      INDEX idx_graduation_year (graduation_year),
      INDEX idx_industry (industry)
````

```
);
```

Appendix E: API Documentation

Example of secure API implementation with proper error handling:

```php
class AuthenticationAPI {
    public function login(Request $request): Response {
    try {
        $credentials = $this->validateLoginRequest($request);
        $token = $this->authService->authenticate($credentials);

        return new JsonResponse([
            'status' => 'success',
            'token' => $token
        ]);
    } catch (AuthenticationException $e) {
        return new JsonResponse([
            'status' => 'error',
           'message' => $e->getMessage()
        ], 401);
    }
    }
}
```

Appendix F: Testing Evidence

Example of comprehensive unit tests:

```php
class AuthenticationTest extends TestCase {
    public function testLoginWithValidCredentials() {
    $auth = new AuthenticationService($this->db, $this->config);
    $result = $auth->authenticate([
        'email' => 'test@example.com',
        'password' => 'valid_password'
    ]);

    $this->assertNotNull($result['token']);
    $this->assertEquals('success', $result['status']);
    }
```

```
    public function testLoginWithInvalidCredentials() {
    $this->expectException(AuthenticationException::class);

    $auth = new AuthenticationService($this->db, $this->config);
    $auth->authenticate([
        'email' => 'test@example.com',
        'password' => 'wrong_password'
    ]);
    }
}
```

Appendix G: Deployment Configuration

Nginx configuration for secure deployment:

```nginx
server {
    listen 80;
    server_name alumni.example.com;
    root /var/www/html/public;

    location / {
    try_files $uri $uri/ /index.php?$query_string;
    }

    location ~ \.php$ {
    fastcgi_pass unix:/var/run/php/php8.1-fpm.sock;
    fastcgi_index index.php;
    fastcgi_param SCRIPT_FILENAME
$document_root$fastcgi_script_name;
    include fastcgi_params;
    }

    # Security headers
    add_header X-Frame-Options "SAMEORIGIN";
    add_header X-XSS-Protection "1; mode=block";
    add_header X-Content-Type-Options "nosniff";
    add_header Referrer-Policy "strict-origin-when-cross-origin";
    add_header Content-Security-Policy "default-src 'self' https:;
script-src 'self' 'unsafe-inline' 'unsafe-eval'; style-src 'self'
'unsafe-inline';";
}
```